

A THEORY OF SETS

Second Edition

A. P. MORSE

A THEORY OF SETS

Second Edition

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A THEORY OF SETS

Second Edition

Anthony P. Morse

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To Barbara

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FOREWORD

BY TREVOR J. MCMINN

Here in a formal inferential system is ensconced an axiomatic logic and set theory. With rudiments and simple versatile prescriptions, ground is prepared for shaping a wide selection of formal inferential languages. Then upon this ground is fashioned a particular formal inferential language that is lean, mechanical, vigorous, and more than adequate for the purposes at hand. At the same time, within the language, on axiomatic foundations broadly and deeply laid, logic and set theory are deductively built in strikingly unified combination. The axioms are amenable to replacement of schematic expressions by almost *any* formula, guarantee a non-elemental universe, enable the set of x such that $\dots x \dots$ to be defined, and ensure the elementhood of many sets. The set-theoretic structure is substantial, with numerous interesting topics, including the most essential ones, taken up and dealt with in efficient dependent order.¹ The initial treatment of each is thoroughgoing, and, on occasion, new results are introduced.² Altogether these topics provide a firm base and house a variety of useful tools for far-reaching mathematical theories.

The system is described in spare and trenchant English which reflects the author's endeavor each time to hit the nail on the head and drive it home with one stroke. Together with a scattering of similarly phrased informal interpretive asides, suggestive headings, and stage directions, the formal language then takes over the task of elucidating mathematical ideas. A title given to a section, subsection, rule, definition, or theorem hints at subjects entertained, roles played, historical origins, or mathematical emphasis. Generally speaking an entitled section em-

¹ The Table of Contents fairly summarizes the topical sweep.

² Some, but by no means all, of these results are touched upon in the author's preface.

braces only one topic and lists at the beginning each definition first used in the section. With many results identified as numbered theorems, the reader is led with measured pace through the book. No single step is hard to take, few proofs are long, and all are perspicuous. This high degree of organization makes reference easy for one familiar with the formal language and imparts to the text a skeletal appearance somewhat belying the full-bodied mathematical treatment.

The book is the result of successive refinements of lectures given by the author over the years at the University of California at Berkeley. In the process the system has evolved from one of little formality and traditional separation of logic and set theory to the present one of complete formality and unified logic and set theory. The axioms for set theory have undergone a metamorphosis from ones like those appearing in the Appendix to J. L. Kelley's *General Topology* to the present more primitive ones. In recent years the formal language has been actively used by the author and students in course work in set theory and analysis and the axiom system has been tested with experience.

Readers at various levels of mathematical education may well profit from this elegantly handled and bold-spirited enterprise. A scholar who already largely understands the author's objectives is still apt to discover much that is original and ingenious in his way of attaining them. A student perplexed by fundamental questions stands a good chance of finding them answered. One less interested in the foundations of set theory than in its superstructure should be amply rewarded for effort spent in learning necessary preliminaries by the impetus given to his understanding of the subsequent beautiful edifice.

Because of a paucity of instructional elaboration of the book's linguistic-logical-mathematical subtleties, the less expert reader is provided in ensuing paragraphs of this foreword with some expansive, semitechnical, and advisory comments. These may help him to see better both the forest and the trees. They have been devised to give him an inkling of what the author is up to in Chapter 0, the first part of Chapter 2, and in Appendix A, to highlight some special virtues and characteristics, and to suggest points of view and ways of approach. Chapter 1 needs little comment, and the problems in understanding the latter part of Chapter 2 are almost purely mathematical. The more technical of these comments will probably be of most help when read concurrently with a study of relevant sections of the book itself.

This commentary is followed by a brief description of the axiomatic structure of some of the other set theories which have originated since the turn of the century together with comparative remarks to place the present system among them.

* * *

In that initial part of Chapter 0 which ends with the section on Indicial and Accepted Variables, the author's concern is to set forth construction rules for a written language capable of conveying mathematical meaning. The language itself consists of inscriptions called expressions which are linear arrays of certain more or less connected inscriptions called symbols. Expressions in the language are of various sorts (for example, the definor, the punctuator, schemators, definitions, axioms, constants, variables, definienda, forms, formulas, primitive constants, primitive forms, schematic forms, schematic expressions, simple expressions, etc.). Certain rudimentary expressions (for example, 'u' a schemator, 'u'xx'' a schematic form, ' $\equiv$ ' the definor and a primitive constant, ' $(x \equiv y)$ ' a primitive form, ' $(0 \equiv \wedge xx)$ ' a definition, ' $(x \leftrightarrow (0 \in x))$ ' an axiom, etc.) are specifically listed and, together with certain of their parts, are somewhat arbitrarily identified as being of one sort or another. Two very simple methods of combining expressions (replacement and schematic replacement) for the purpose of constructing further expressions are mentioned and illustrated. Prescriptions (in the form of agreements and rules) are laid down for telling us of what sort the rudimentary and constructed expressions are. Some are statements of relation between certain expressions (for example, Rule 0.2: A variable is free in a form if and only if it occurs therein less than twice). Others (for example, Rule 0.4) are descriptions of a test applied to a pair of expressions to see if one bears a certain relation (in this case freeness) to the other. Still others (for example, Rule 0.3: A is a formula if and only if some variable is free in A) tell whether an expression is of a certain sort in terms of its relations to other expressions.

We want to know what is in the language and in its various components and what relations certain members bear to others. The specific identifications, constructive replacements, and prescriptions are our sole means of finding out. Quite mechanically, one step at a time, through prescribed replacements, new expressions in the language are constructed from those that have already been constructed or from the

rudimentary ones. As these new expressions arise, they are cataloged and their relations to some of the others are noted. The expressions we are mainly interested in are constants, variables, forms, and formulas.

In order to simplify the formalization no attempt has been made to prove facts about expressions in the language. When it has seemed necessary or fruitful to use such facts, they have merely been assumed as numbered rules (for example, Rules 0.13–0.16). A rule is thus thought of in Chapter 0 as being meta-axiomatic.

Although the relations of freeness, indiciality, and acceptedness between certain expressions are categorically described by the agreements and rules, they are not explicitly defined. Thus, although a method is here prescribed for demonstrating freeness or lack of it in any given case by considering expressions bearing certain relations to others, freeness is not prescribed in the explicit form “ α is free in A if and only if”.... In the later section on Chains such prescriptions are made explicit.

The author’s second concern in Chapter 0, set forth in the section on Rules of Inference, is to establish the process of mathematical inference. Among all formulas are singled out for special attention those called theorems. At the outset all real definitions and all axioms are theorems. Starting from these, through certain prescribed changes embodied in the rules of inference, other theorems are constructed in somewhat the same mechanically iterative manner in which formulas are constructed from the rudimentary expressions. Here again tests are prescribed for telling us whether a given formula is a theorem. However, this is not to say that a test may be easily made. Hardly so. It amounts to exhibiting an ordered list of theorems, each of which results from some of its predecessors by the application of a rule of inference, terminating with the formula in question. Such a list is a formal proof. Together with definitions dispersed throughout, the whole of formal mathematical literature consists of just such an ordered list headed by the axioms.

In this section on Rules of Inference the notion of theorem, like freeness, is not explicitly defined. Unlike freeness, it is not categorically described. Thus no prescription is given here for determining that a given formula is not a theorem. However, in the section on Demonstrations the notion of theorem is made explicit.

It is to be hoped, quite naturally, on the basis of the axioms adopted in Chapters 1 and 2 that not all formulas are theorems, that is, that the resulting system is consistent.

In the foregoing nutshell description of the author's formal system none of its nicer features is really evident. Some of these deserve special mention.

The strictly linear structure of expressions that has been adhered to throughout facilitates their technical analysis and enables them to be read serially from left to right as a machine might well be made to do.

A high degree of simplicity has been achieved in the statement of rules and agreements for constructing the language by holding down the number of sorts of expressions treated. Here no distinction is made between sentential variables and nominational variables, much less between set variables and point (element) variables. Also no distinction is made between statementlike expressions and namelike expressions (elsewhere these are commonly known as formulas and terms, respectively). Together these are here simply formulas. Thus among the formulas are

$$\begin{aligned} & '(0 \in x)', \quad '(x \in 1)', \quad '(x \rightarrow x)', \quad '((0 = \bigvee x \in 1 \ x) \wedge \bigwedge x(x \rightarrow x))', \\ & '\bigvee x(0 \in x)', \quad '\{x\}', \quad '(x, x)', \quad 'Ex(0 \in x)', \quad '\bigwedge x; (0 \in x) \ x'. \end{aligned}$$

Except at the very first, each new form is introduced as a definiendum (the left side of a definition). Yet regardless of whether or not the definition is properly constructed, once the constants appearing in the form are known (see page 6), the grammar of the form, that is, a tally of which variables are free and which are indicial, is easily determined by simply counting occurrences of each variable.

The use of schematic expressions enables many definition schemas, axiom schemas, and theorem schemas to be rendered as a single formula in the system and thus amenable to treatment as such.

Once the legitimacy of a substitution or a schematic substitution has been established the substitution or schematic substitution itself is carried out by a straightforward replacement or schematic replacement which does not distinguish occurrences. The requirement that schematic expressions not only have the same schemator but be entirely the same to qualify for schematic replacement considerably simplifies the mechanics of schematic substitution.

Allowing mixed expressions in which there appear variables neither free nor bound (see A.18) provides a great deal of freedom in substitution. For example, in 2.35 is found the theorem

$$'(x \in Ex \ \underline{u}x \leftrightarrow \underline{u}x \wedge x \in U)' \tag{1}$$

in which ‘ x ’ is neither free nor bound. From it, one easily obtains as theorems

$$‘(x \in E y \underline{u} y \leftrightarrow \underline{u} x \wedge x \in U)’ \quad (2)$$

and

$$‘(x \in E y \underline{u} y \leftrightarrow \vee y (x = y \wedge \underline{u} y) \wedge x \in U)’ \quad (3)$$

An interesting aspect of (1) shared by neither (2) nor (3) is that a theorem is obtained from

$$‘(x \in E x P \leftrightarrow P \wedge x \in U)’$$

by replacing ‘ x ’ by an *arbitrary* variable and ‘ P ’ by an *arbitrary* formula.

Worthy of note is the all-embracing scope of the author’s preparatory setting. It is a novelty that his formal inferential system is not specially tailored to his logic and set theory. Without changing this system in any way, one could, by introducing appropriate mathematical definitions, equally well formally pursue radically different mathematical disciplines with quite different primitive concepts.

With the language at hand we, of course, interpret it by attaching meaning to the expressions in it. This is essentially an intellectual process for which no prescription can be given. Only the use of words appearing in definitions common also in everyday usage suggests the proper attachment of intuitive mental concepts which one already possesses. An interesting feature of the author’s development is that only a scant few basic concepts are appealed to. Once the appropriate conceptual meaning has been attached to ‘ $\equiv$ ’ (“definitionally equivalent to”), ‘ $\rightarrow$ ’ (“implies”), ‘ $\wedge$ ’ (“for each”), an adequate start has been made for logic; and once the appropriate conceptual meaning has been attached to ‘ $\in$ ’ (“is in”), and ‘ mel ’ (“choice”), an adequate start has been made for set theory.

We are specially interested in those formulas devoid of schemators in which every variable actually appearing is bound (see A.18). It is with these latter, known as sentences or names, that we associate, on the one hand, mathematical ideas or propositions, and, on the other hand, specific concrete mathematical things in another, ideal (mental) world,

the mathematical world which, of course, no one should confuse with the world of inscriptions. Thus here the formula ' $(0 \in 1)$ ' is a sentence expressing a mathematical idea, namely, that the empty set belongs to the set whose only member is the empty set, and ' 0 ' is a name for a mathematical thing, the empty set. Other formulas, like ' $(x \in y)$ ' and ' $\{x\}$ ', with free variables actually appearing in them, although neither sentences nor names are akin to them in that they can be made into sentences or names by replacement, for instance, by replacing ' x ' by ' 0 ' and ' y ' by ' 1 ' in the aforementioned to get ' $(0 \in 1)$ ' and ' $\{0\}$ ', or by quantifying, for instance, by writing

$$' \wedge x \vee y(x \in y)' \quad \text{or} \quad '\wedge x \{x\}'.$$

Each is capable, by being transformed in certain ways into sentences or names, of conveying many specific and concrete meanings, all of a more or less similar nature.

The significance of theorems lies in their interpretation. A theorem, in addition to any idea it may convey as a formula, inspires in us, at least provisionally, belief that the idea conveyed is right.

Theorems need not be sentences. For example, the formula ' $(x = x)$ ' is a theorem. We believe every specific idea conveyed as the result of replacing ' x ' in this theorem by a name as well as by universally quantifying it.

It is a peculiarity of the author's system as axiomatized in Chapter 2 that frequently in one breath a mathematical idea is expressed and a set is named. Each set is either true or false and each sentence is a name for some set. It turns out that 1 is true, that $(1, 2)$ is false, that $(0 \in 1)$ is the mathematical universe, and that $(0 \in 0)$ is the empty set. It also turns out, rather less surprisingly, that 1 is the set whose only member is the empty set, that $(1, 2)$ is an ordered pair of natural numbers, that $(0 \in 1)$ is true, and that $(0 \in 0)$ is false.

In trying to swallow this unified notion of logic and set theory one may find at first that it sticks in one's craw. A purist might argue that it is contrary to good principles of nomenclature to have a formula express an idea and name an object at the same time. In support of unification one may take the view that the idea expressed and the object named are always so different that practical confusion will not crop up over the ambiguity as to which of the two is intended. The damage to one's preconceptions turns out to be not as great as one might at first

think, and in most instances it is pretty clear whether a given formula is to be thought of as statementlike or as namelike. It turns out that

$$\forall x \in 3((0 \in x) \wedge \underline{ux})$$

if and only if

$$\text{for some } x \text{ in } 3, 0 \text{ is in } x \text{ and } \underline{ux};$$

it also turns out that

$$\forall x \in 3((0 \in x) \wedge \underline{ux})$$

equals

the union as x runs over 3 of the intersection of $(0 \in x)$ with $\underline{ux}$.

Of the two interpretations of

$$' \forall x \in 3((0 \in x) \wedge (x \neq 1)) '$$

thus suggested, the first is probably intended and we know what idea (incidentally right) is conveyed. Of the two interpretations of

$$' \forall x \in 3((0 \in x) \wedge x) '$$

likewise suggested, the second is probably intended and it can be checked that

$$(2 = \forall x \in 3((0 \in x) \wedge x)).$$

To take another instance, in Chapter 1 we are mainly interested in knowing that 0 is false, whereas in Chapter 2 we are mainly interested in the fact that 0 is the empty set.

In any event, whatever the interpretation that is made, it has no bearing on the consistency of the system considered as a game played according to rules with expressions.

Aside from the sheer elegance of this unified structure, attested to in part by the features mentioned in the author's preface, numerous technical benefits result, among which is the possibility of exploring areas of logic and set theory simultaneously. An example may help to indicate this possibility. Since $(p \rightarrow q)$ is equal to $(\sim p \vee q)$, we can say that the set $(p \rightarrow q)$ is the complement of p , union q , and, by way of an axiom (2.5.0), that $(p \rightarrow q)$ is true if and only if 0 belongs to the

complement of p or to q . Since $(\sim p \vee p)$ is the universe, and 0 belongs to the universe, we see that $(p \rightarrow p)$ is true.

To give a glimmering of other technical advantages of unification we note that in the theorem

$$'((0 \in x) = \exists y(0 \in x))'$$

the left side is more concise than the right.

The author's third concern in Chapter 0 is to formulate a practical theory of notation that will simplify formulas and leave them uniquely readable without sacrificing mathematical consistency. This is a noteworthy effort that goes beyond the mere listing of shorthand devices and rules of thumb. The language retains its machinelike precision, is eminently usable, and incorporates many of the usages of traditional mathematics. Furthermore, this effort goes beyond the adopting of conventions for a language germane only to elementary set theory. Having in mind a language with wider applicability in later branches of mathematics, the author has adopted several conventions that harmonize simplifications in this unexploited language with the herein developed language of elementary set theory.

Due to the use of constants other than introductors in forms, like '+', in ' $(x + y)$ ', a great variety of new forms can be introduced from a fixed collection of constants by varying the arrangement of them with variables and schematic expressions in a matrix. An example of this economy is given in the combinatorial sum definition

$$'((A + + B) \equiv \exists x + y (x \in A \wedge y \in B))'.$$

Other abbreviational nuances that are both interesting and useful can be gleaned from the theorems

$$\begin{aligned} & '(x, y, z \in A \leftrightarrow (x, y, z) \in A)', \\ & '(x, y, z, \in A \leftrightarrow (x, y, z), \in A \leftrightarrow x \in A \wedge y \in A \wedge z \in A)'. \end{aligned}$$

The theory of notation systematically exploits some of these possibilities.

The author's fourth concern in Chapter 0, treated in the section on Demonstrations, is to devise a framework for analyzing formal proofs. In this setting the notion of a theorem is explicitly defined in terms of the notion of a formula. Certain expressions are called demonstrations.

A demonstration, roughly speaking, is the ordered concatenation of the theorems listed in a formal proof. Certain formulas in a demonstration are identified as subformulas. A trifle more precisely, a demonstration is built up iteratively starting from definitions or axioms by juxtaposing on the right a formula obtained from subformulas of what is already at hand by inference from these subformulas. A theorem then is a subformula of some demonstration.

The author's fifth and last concern in Chapter 0, treated in the section on Chains, is to explicitly define indiciality, acceptedness, freeness, and parentheticality. One way of doing this would involve intuitive set theory somewhat in the spirit of A.29–A.38. The way adopted by the author uses concatenations of certain expressions each initiated by a symbol, the punctuator, counted among the constants but never allowed to appear in a form, and each a beginning expression or an expression obtained by certain replacement operations from preceding such expressions in the array. Such arrays are called chains. Chains are thus built up iteratively by juxtaposition on the right somewhat as demonstrations are. The expressions juxtaposed minus the punctuator are called links. Four kinds of chains are considered. In the case of a free chain each link is the concatenation of a variable with a form in which the variable is free or with an expression, obtained by replacement operations from preceding links, in which the variable is free.

In furtherance of a correct notational theory, the author has included an appendix on definitions. Since from definitions we learn what are constants, variables, and forms, and since definitions are theorems, one can easily arrive at contradictions both metamathematical and mathematical by accepting improperly constructed ones. In Appendix A rules governing the correct formulation of definitions are so made that: parentheses are constants; 'x' is a variable; the end of a definition is determined by an internal routine; each definition is a formula; the notion of a formula is categorical; no formula, which is meaningful, but not a theorem prior to the addition of a definition, becomes a theorem after the addition of the definition; every formula has a primitive translation. Also circularity is avoided since the theorem

$$'(0 \equiv (0 \wedge 1))',$$

which does not violate the last two above requirements, is, quite rightly, not allowed to be a definition.

At this point some words of reassurance should perhaps be extended to the reader impatient with precise linguistic foundations and anxious to come to grips with the mathematical content of Chapter 2. Once he has got the hang of constructing a few formulas not dependent upon the theory of notation, he will avoid, and with good reason, the laborious task of intricately checking to see if a given expression is a formula. He will probably be able to see intuitively at a glance if it is one and from which forms it was obtained and how. With the exception of certain orienting definitions like 0.0.0, 0.0.1, and 2.0.0 there are no tricks that would lead him to think, contrary to his upbringing, that expressions like

$$'(x \in', '(\rightarrow x \rightarrow)', ' \wedge \underline{ux}'$$

are formulas.

Expressions that have been simplified by taking advantage of the theory of notation may give somewhat more trouble; but since efforts have been made to make them seem at first blush reasonable, even they can be deciphered intuitively with only an occasional reference to the theory. For instance, it seems reasonable that useful meaning should be ascribed to the expressions

$$' \wedge x \in y \underline{ux}', ' \vee x; \underline{ux} \underline{yx}', '(p \wedge q \vee r)', '(x = y \wedge a \in b \subset c)'$$

and that they be included among the forms. The first and last of these can probably be guessed at here, and there is perhaps a fifty-fifty chance that the others can be too.

In reading a formula, from left to right, advantage should be taken of the fact that no formula is an initial segment of another in order to pick off one by one the formulas which have been substituted into a form. If the candidacy of a form, for such substitutions, is not clear at the start it will soon become so.

It may well comfort the reader to know that if he can contrive *one* valid reading of a formula, then *that* reading is correct. A case in point is the form

$$'(p \rightarrow q \rightarrow r)',$$

which, because of 0.37 and in keeping with the spirit of ordinary implicational proofs, means the same as

$$'((p \rightarrow q) \wedge (p \rightarrow r))'$$

and *not* the same as

$$'((p \rightarrow q) \wedge (q \rightarrow r))'.$$

It might be noticed that

$$'(p \rightarrow q \rightarrow r)'$$

can be obtained from the form

$$'(x \rightarrow y)'$$

be replacing ' x ' by ' $p \rightarrow q$ ' and ' y ' by ' r ', as well as by replacing ' x ' by ' p ' and ' y ' by ' $q \rightarrow r$ '. However as it turns out, neither of these readings is valid since neither ' $p \rightarrow q$ ' nor ' $q \rightarrow r$ ', lacking parentheses, is a formula. In this connection

$$'(p \rightarrow q \rightarrow r)', \quad '((p \rightarrow q) \rightarrow r)', \quad '(p \rightarrow (q \rightarrow r))'$$

are different from each other both in appearance and meaning.

At any rate, a thorough grasp of the Theory of Notation is far from essential for comprehending Chapters 1 and 2.

In the same vein, though some of the rules of inference may look complicated, a moment's reflection should convince the reader that they cannot all be utterly simple when stated precisely. He need examine but one application to be assured that they only accurately prescribe in general what he would almost certainly be inclined by mathematical second nature to allow in any instance. Thus by indicial substitution (0.28) we infer that since

$$'(\wedge x \underline{ux} \leftrightarrow \wedge x \underline{ux})'$$

is a theorem, then

$$'(\wedge y \underline{uy} \leftrightarrow \wedge x \underline{ux})'$$

is a theorem. This can be seen by letting

q be ' z ',

Q be ' $(z \leftrightarrow \wedge x \underline{ux})$ ',

T be ' $(\wedge x \underline{ux} \leftrightarrow \wedge x \underline{ux})$ ',

A be ' $\wedge x \underline{ux}$ ',

α be ' x ',

B be ' $\wedge y \underline{uy}$ ',

T' be ' $(\wedge y \underline{uy} \leftrightarrow \wedge x \underline{ux})$ ',

and by checking that

q is free in Q ,
 A is a form,
 α is indicial in A ,
 ‘ y ’ is accepted in A .

Having once checked in detail the validity of an application of a rule of inference, the reader will not likely do it often again. And there is no reason why he should.

In actual mathematical literature formal proofs are rarely exhibited. Ordinary proofs are considered as commentaries designed to convince one of the possibility of constructing formal ones. This treatise on set theory is no exception. However, a certain formality is retained here in that, although many gaps are permitted in the shortening of a formal proof to an ordinary one, an effort has been made, up through 2.37, to make each entry formally correct and a theorem. Thereafter in establishing implications, a somewhat relaxed but thoroughly sound method is frequently employed.

Sentences structurally incorporating formulas along with ordinary English words and phrases are punctuated in the customary way. Formulas not so incorporated stand without further punctuation.

Many readers will find it expedient to plunge directly into Chapters 1 and 2, grasping by context and preconceptions the gist of what is being said, and resorting to Chapter 0 only when necessary to pick up indispensable basic notions and notations.³ All readers are strongly urged to explore Chapter 0 up through 0.29, omitting 0.5 through 0.18, and Chapters 1 and 2 before attempting to completely master Chapter 0.

The informal passages give succinctly the necessary clues, while the formal passages encapsule much mathematical content in small compass. Because of this, the reader is well advised to read slowly and carefully and, whether he scans or delves, to take for granted that the author's words have been weighed, his formulas accurately cast, and that he means exactly what he says.

* * *

³ Use of the Index of Constants may shorten subsequent referential searches.

Set theories are largely characterized by the provisions that are made for constructing sets from properties and for relating membership in a set so constructed with satisfaction of the property, on the one hand, and, on the other, by the provisions that are made for determining what sets are capable of belonging to some set, that is, of being elements. In short, in a given system, if A is the set of x such that $\dots x \dots$, then we are interested in knowing just what are members of A and whether or not A is capable of being an element.

Since the paradox revealed by Russell in 1901 precludes the all-encompassing provision of taking as a classification axiom each formula obtained from

$$' \forall A \wedge x(x \in A \leftrightarrow P) '$$
 (4)

by replacing ' P ' by a formula in which ' A ' does not appear, other so far successful provisions have been engineered.

Those of Russell and Whitehead⁴ lie in restricting the kinds of formulas which may be considered as replacements for ' P '. Only suitably restricted ones are considered meaningful and only for one of these does (4) give rise to an axiom.

The Zermelo-Fraenkel⁵ provisions lie in accepting as a classification axiom each formula obtained from

$$' \wedge B \vee A \wedge x(x \in A \leftrightarrow P \wedge x \in B) '$$
 (5)

by replacing ' P ' by a formula in which ' A ' does not appear. Then, because of the dependence of A upon B , other axioms are adopted to assure the existence of at least one set and indeed of more than one. Thereafter (5) is used to get a great variety of sets. However, no universal set is forthcoming from (5). An important feature is the total lack of restriction placed upon the capability of a set being an element.

Stemming from von Neumann's idea that it is not so much the existence of sets constructed from arbitrary properties that induces

⁴ Russell, B. A. W., and Whitehead, A. N., *Principia Mathematica*, 3 vols., Cambridge Univ. Press, 1910, 1912, 1913.

⁵ Fraenkel, A. A., and Bar-Hillel, Y., *Foundations of Set Theory*, North-Holland Publishing Co., 1958.

paradoxes as the lack of restraint in allowing them to be elements, are the systems of von Neumann–Bernays–Gödel,⁶ Quine,⁷ and Morse.

The von Neumann–Bernays–Gödel set-theoretic axiom system is both finite and unschematic. An axiom provides for the existence of a set with a certain property, and further axioms provide for the existence of other sets dependent upon given sets. In addition an axiom provides for the existence of an element, and further axioms provide for the existence of other elements dependent upon given sets or elements. It then can be shown that a classification theorem is obtained from

$$' \forall A \wedge x(x \in A \leftrightarrow P \wedge \forall B(x \in B)) '$$
 (6)

by replacing ' P ' by such a formula φ that ' A ' does not appear in φ and quantification in φ is restricted to elements. Whether or not the set A is in turn an element must be determined from the axioms about elements. An early consequence is the existence of a unique universal set. Were it to be an element, a Russell-type paradox would ensue.

In Quine's system a classification axiom schema is described which amounts to accepting as an axiom each formula obtained from

$$' \forall A \wedge x(x \in A \leftrightarrow P \wedge \forall B(x \in B)) '$$
 (7)

by replacing ' P ' by a formula in which ' A ' does not appear. In addition there is described an elementhood axiom schema which amounts to accepting as an axiom each formula obtained from

$$' \forall A (\wedge x(x \in A \leftrightarrow P \wedge \forall B(x \in B)) \wedge \forall B(A \in B)) '$$
 (8)

by replacing ' P ' by such a formula φ that: ' A ' does not appear in φ ; φ is stratified⁸; quantification in φ is restricted to elements; the variables appearing in φ different from ' x ' and not bound therein are conjunctively restricted to elements. For example, a suitably restricted replacement for ' P ' arising from

$$' (\wedge w(w = w) \wedge (x \in y \vee x \in z)) '$$

⁶ Gödel, K., *The Consistency of the Axiom of Choice and of the Generalized Continuum Hypotheses with the Axioms of Set Theory*, Princeton Univ. Press, 1940.

⁷ Quine, W. V., *Mathematical Logic*, rev. ed., Harvard Univ. Press, 1951.

⁸ Quine, W. V., *op. cit.*, §28.

is

$$‘(\bigvee B(y \in B) \wedge \bigvee B(z \in B) \wedge \bigwedge w(\bigvee B(w \in B) \rightarrow w = w) \wedge (x \in y \vee x \in z))’.$$

The elementhood axiom schema does not preclude the elementhood of comprehensive sets. A universal set is defined which turns out to be a member of itself. One consequence of this is that the hereditary property of a subset of an element being an element does not hold.

Although the present axioms are finite in number some of them do involve schematic expressions. Among these of particular interest is 2.5.3 which enables the set of x such that $\dots x \dots$ to be defined. With a universal set U earlier defined the related subsequent useful Theorem of Classification (2.35)

$$‘(x \in Ex \ \underline{u}x \leftrightarrow \underline{u}x \wedge x \in U)’ \quad (9)$$

is analogous to theorems described by Quine⁹ as consequences of his §29 (7). Axiom 2.5.7, which, in contrast with 2.5.3, is replaceable by axioms not involving schematic expressions helps guarantee considerable elementhood. The elementhood of the universe is not guaranteed. Indeed, as with the von Neumann–Bernays–Gödel system if this were so, a contradiction would arise.

Closely related to (9) is the formula

$$‘(x \in Ex \ \underline{u}x \leftrightarrow \underline{u}x \wedge \bigvee B(x \in B))’,$$

similar to the one which appears on page xxix of the preface. It was formulated by Morse as an axiom of classification in his 1939 lectures at the University of California. Kelley¹⁰ later borrowed and put to good use this simple formula. He, in effect, accepts as an axiom each formula obtained from

$$‘(x \in Ex P \leftrightarrow P \wedge \bigvee B(x \in B))’ \quad (10)$$

by replacing ‘ P ’ by a formula. Kelley’s system, which incorporates many features of an equivalent earlier unpublished system of Morse, is *very* much the same in pure set-theoretic content as the system at hand, although it is far from clear upon glancing at the axioms that it is.

⁹ Quine, W. V., *Mathematical Logic*, Harvard Univ. Press, 1940, under *230, p. 171.

¹⁰ Kelley, J. L., *General Topology*, Van Nostrand, 1955 or Springer-Verlag, New York, 1975, Appendix.

For purposes of further comparison let us say a schema is *concise* if and only if it can be replaced by a single formula with the same effect. This formula, of course, may have schematic expressions appearing in it.

The classification schema (4), widely accepted before 1901, is made concise by

$$' \vee A \wedge x(x \in A \leftrightarrow \underline{u}x) '.$$

It is highly unlikely that the restrictions of Russell and Whitehead can be rendered into a schema that can be made concise.

The classification schema of Zermelo–Fraenkel (5) is made concise by

$$' \wedge B \vee A \wedge x(x \in A \leftrightarrow \underline{u}x \wedge x \in B) '$$

An interesting feature of the von Neumann–Bernays–Gödel system is that when translated into the present language with logic left intact the purely set-theoretic axioms are not only finite in number but devoid of schematic expressions as well. However, it is unlikely that the classification metatheorem (6) can be made concise.

The classification schema of Quine (7) is made concise by

$$' \vee A \wedge x(x \in A \leftrightarrow \underline{u}x \wedge \vee B(x \in B)) '.$$

However, it is unlikely that his elementhood schema (8) can be made concise.

The classification schema (10) used by Kelley is made concise by

$$' (x \in Ex \wedge \underline{u}x \leftrightarrow \underline{u}x \wedge \vee B(x \in B)) '.$$

Although the present system and Quine's⁷ revised system may both be consistent, they are radically different, chiefly as to provisions for elementhood.

The von Neumann–Bernays–Gödel system has been shown to be equiconsistent with the Zermelo–Fraenkel system.¹¹

The system used by Kelley is definitely stronger than the von Neumann–Bernays–Gödel system.

¹¹ Novak, I. L., (I. N. Gál), *A Construction for Models of Consistent Systems*, *Fundamenta Mathematica*, 37, 1951, pp. 87–110. (Submitted in 1948 as a Thesis to Radcliffe College, Cambridge, Massachusetts.)

Rosser, J. B., and Wang, H., *Non-standard Models for Formal Logic*, *Journal of Symbolic Logic*, 15, 1950, pp. 113–129 (Errata, p. IV).

The present system is slightly stronger than Kelley's. Their striking set-theoretic similarities though suggest that the former is consistent with the latter. I am told that Alfred Tarski and later David C. Peterson have verified this relative consistency. Thus the present unified system is just as sound as the more conventional ununified system used by Kelley.

For those interested, a more-precise account will now be given of the similarities, unearthed by the proof of relative consistency, between the present system and that of Kelley. We shall take advantage of 0.70–0.75. Let us agree here that: *P* is *statemental* if and only if *P* is either a schematic expression, or a variant of

‘ $\wedge x \underline{ux}$ ’, or ‘ $(\underline{ux} \rightarrow \underline{vx})$ ’, or ‘ $\sim \underline{ux}$ ’, or ‘ $(x \in y)$ ’;

N is *nominal* if and only if *N* is either a variable or a variant of ‘ $\text{Ex } \underline{ux}$ ’; *C* is *generated* by *T* if and only if *C* is not a subformula of *T* and there is such a subformula *B* of *T* that *C* is obtained from *B* either by replacing some free variable of *B* by some nominal formula or by schematically replacing some schematic expression of *B* by some subformula of *T*; *S* is a *statementary* if and only if *S* is a string and each subformula of *S* is either a statemental formula or a formula generated by some substring of *S*; *C* is *statementant* if and only if *C* is a subformula of some statementary; and finally *C* is *nominant* if and only if *C* is either a variable or an expression obtained from ‘ $\text{Ex}y$ ’ by replacing ‘*x*’ by some variable and ‘*y*’ by some statementant expression. A statementant expression is statementlike and a nominant expression is namelike. It turns out that if *H* is a statementant expression devoid of schemators, then *H* is a theorem in the present system if and only if it is a theorem in the translated¹² system of Kelley.

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¹² It is assumed here that Kelley's system has been translated into the present language with logic left intact. This translated system is, in effect, the more conventional system alluded to in the second paragraph of the preface which follows.

PREFACE

This book provides graduate students and professional mathematicians with a formal unified treatment of logic and set theory. The formalization can be used without change to build just about any mathematical structure on some suitable foundation of definitions and axioms. In addition to most of the topics considered standard fare for set theory several special ones are treated. It is hoped the book will be found useful as a text for a substantial one-semester course in set theory and that the student will find continuing use for the formal and highly flexible language.

The first and more fundamental part of Chapter 0 terminates with a detailed account of our rules of inference. Here the lack of distinction between formulas and terms both unifies and simplifies the formalization. Our axioms reflect this unity and, in keeping with the Remark preceding 2.33, each theorem of logic is an immediate consequence of a corresponding systematically verifiable theorem of set theory. Nevertheless without changing Chapter 0 in any way we could equally well use more conventional axioms which would preserve the usual sharp distinction between logic and set theory. The more conventional system we have in mind differs but little from the present one in the separated worlds of logic and set theory. It is somewhat more concise axiomatically. It has ' $\exists x \underline{u}x$ ' as an additional set-theoretic primitive form, and in harmony with Theorem 2.35 it uses

$$'((x \in \exists x \underline{u}x) \leftrightarrow (\underline{u}x \wedge \forall y(x \in y)))'$$

as an axiom of classification. It also fails quite naturally in our view to answer some of the simplest questions about the interplay of logic and set theory.

The last and less fundamental part of Chapter 0 is dominated by our Theory of Notation. Many notations in common use in present day

mathematics are automatically preserved. Among these are

$$'(x + y \cdot z + w)'$$

and

$$'|x + y \cdot z + w|',$$

but not among these are $'|xyz|'$ and $'|x|y|z|'$. We abandon the classical functional notation $'f(x)'$ since its use would lead to almost instant technical disaster. Mindful of such danger, von Neumann introduced the technically sound notation $'[f, x]'$ which could well be used as an alternative to our shorter notation $'\cdot fx'$. More specifically, no harm would be done by adding

$$'([f, x] \equiv \cdot fx)'$$

to our list of definitions. On the other hand, if

$$'(f(x) \equiv \cdot fx)'$$

were added, then the resulting system would, as will be pointed out in the paragraph which begins Appendix A, be inconsistent.

Our Theory of Notation is followed by a short section on demonstrations, and Chapter 0 is then concluded with a section on chains. Taken together these last two sections of Chapter 0 are independent of the Theory of Notation and make quite explicit the notion of formula and the concept of theorem.

Because of the importance we attach to definitions we formulate in Appendix A the rules we follow in making them. Earlier, S. Leśniewski worked painstakingly along these lines.

We shall employ the terminology 'point' and 'set' instead of the more usual terminology 'set' and 'class'. We feel justified in this since to us a point is a set capable of belonging to a set. We think of a set which is not a point as very large indeed.

Many problems which appear to be solvable by definitional induction may, because of the sheer size of the sets involved, not yield to ordinary definition by induction. In this connection the more-general principle 2.101 may be of real use. Specifically, we use 2.101 to

advantage in proving 2.148.4 which rounds out 2.148 as a far-reaching precursor to the Cantor–Bernstein Theorem.

In 2.115 we formulate our very powerful Maximal Principle which like that of Hausdorff does not limit the competition to points. An easy consequence of 2.115 is 2.118 of which Hausdorff's Maximal Principle 2.119 is a special case. That 2.118 is not an easy consequence of 2.119 is, we feel, brought out by 2.120. The difficulty here arises from the possibility that no point is maximal.

For purposes of further comparison, let us agree that Kuratowski's Lemma and Zorn's Lemma are, respectively, Theorems 25(d) and 25(e) given by J. L. Kelley on page 33 of his *General Topology*, Van Nostrand, Princeton, New Jersey, 1955, or Springer-Verlag, New York, 1975. From 2.118 which has no premise we learn that the conclusion in Kuratowski's Lemma is essentially independent of the premise. In 2.121 and 2.123 we have useful inductive variants of Zorn's Lemma. There is indeed a connection between Zorn's Lemma and our 2.122 in that, roughly speaking, the conclusions are the same but the premise of the former implies that of the latter.

We feel that the entitled theorems of Chapter 2 capture, short of inconsistency, the intuitive simplicity of Frege's beautiful but inconsistent system.

I am grateful to students and colleagues for comments and advice. I am doubly grateful to those students who, down through the years, have found a treatment akin to this both interesting and comprehensible.

The axioms for logic, which are easy to work with, were suggested by Alfred Tarski. Also due to him, in collaboration with Knaster, is the approach to Theorem 2.149 and the use of 2.149 in checking the Cantor–Bernstein Theorem 2.154. Beyond this I have profited greatly from many most illuminating conversations with Tarski about the fundamentals of mathematics.

Due to Trevor J. McMinn and myself are: Regularity and Choice; Appendix B.

I have received considerable help from David C. Peterson in preparing the final manuscript for publication. I owe him a special debt for valuable detailed criticisms and suggestions; in like fashion I owe a debt to R. A. Alps and R. C. Neveln.

The change in the present 0.24 is due to R. A. Alps. It is no longer possible to prove 1.4.3 without using 1.3. I am also greatly indebted to Robert A. Alps for countless other suggestions and for stintless help in preparing the present edition for publication.

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February 1984
Orinda, California

A. P. M.

LANGUAGE AND INFERENCE

INTRODUCTION

A *mark* is a more or less connected inscription. Among our marks are the Latin letters, the Greek letters, and the ten Arabic numerals. We think of a subscript, or a superscript which is not a quotation mark, as touching any inscription immediately to the left above it. Underlining is thought of as touching any inscription immediately above it. If one nonitalicized Latin letter is printed close beside another as in a word, then they are thought of as touching. In this connection the ten Arabic numerals are to be thought of as nonitalicized Latin letters.

An *expressional* is a linear array of marks. To write about expressionals some naming device is required.

It is understood here, as it often is elsewhere, that the name of an expressional is its enclosure within single quotation marks. Thus in the name of an expressional the initial mark is always the opening quotation mark and the terminal mark is always the closing quotation mark. Accordingly in written discourse

$'p \rightarrow q'$

is, because of initial understanding and the magic of names, a three mark expressional whose initial mark is a letter, whose terminal mark is a letter, and whose middle mark reminds us of an arrow. We are also convinced that

$'p \rightarrow q'$

is a three mark expressional whose initial mark is ' p ', whose terminal mark is ' q ', and whose middle mark is ' $\rightarrow$ '.

The name of an expressional is, of course, also an expressional. Since many expressionals are names of others and since each expressional has

its own name, a twinge of apprehension seems justified in setting up notational machinery describing elementary operations with expressionals.

If

$$A \text{ is } 'p \rightarrow q < r'$$

and

$$B \text{ is } 's > t',$$

then the concatenation of A and B is thought of as a writing down A and then following it on the right by writing down B to obtain

$$'p \rightarrow q < rs > t'.$$

If one tries to set up a good notation such as that introduced in 0.9.0 for the concatenation of expressionals in *general*, then a real difficulty emerges. We feel we have scotched this difficulty by insisting presently that quotation marks are *not* among our symbols.

In the present chapter we are primarily interested in formalizing mathematics. We shall soon learn that mathematical expressionals have no quotation marks in them. Our formalization of mathematics does not require the metamathematical machinery set up in 0.9.0, but we feel this machinery of concatenation now and then considerably shortens explanations.

We agree that c is a *symbol* if and only if c is a mark which is not a quotation mark.

An *expression* is a linear array of symbols.

As will be seen, quotation marks play a central role in this chapter. We do not include them among our symbols since this would lead by way of 0.9.0 into a contradiction. As will be seen in the Remark which precedes 0.10, the possibility of such a contradiction would arise primarily from the inclusion of quotation marks among our symbols rather than imagined technical defects in 0.9.0.

It helps to notice, for example, that

$$'p \leftarrow q < r + s'$$

is an expression in which, of course, no quotation marks appear; on the other hand

$$''p \leftarrow q < r + s''$$

is not only the name of an expression but is an expression in which precisely two quotation marks appear; moreover, the name of ' $p \rightarrow q$ ' is " $p \rightarrow q$ " and the name of ' x ' is " x ".

Because of theorems or rules, spawned by definitions or agreements, names arise for some of us in a quite different but rather persuasive way. Since it will turn out that

$$('x' 'x' 'x') \text{ is } 'xxx',$$

it is sometimes said that

$$('x' 'x' 'x') \text{ is a name for } 'xxx'.$$

Since it turns out in Chapter 2 that

$$(0 = \wedge xx)$$

it is sometimes said that

$$'0' \text{ is a name for } \wedge xx$$

and that

$$' \wedge xx' \text{ is a name for } 0$$

and that

$$'0' \text{ is a name for } 0.$$

Even more interestingly, it is said that

$$'(2 + 3)' \text{ is a name for } 5.$$

This pleasant manner of speaking will not reappear below.

Replacement. If A is the expression

$$'(x + y + x)',$$

then: if in A we replace ' x ' by ' 1 ', we obtain

$$'(1 + y + 1)';$$

if in A we replace ' r ' by ' x ', we obtain A ; if in A we replace ' y ' by ' x ', we obtain

$$'(x + x + x)';$$

if in A we replace ' x ' by ' t ', we obtain

$$'(t + y + t)';$$

if in A we replace ' r ' by ' t ', we obtain A ; if in A we replace ' x ' by ' r ' and ' y ' by ' s ', we obtain

$$'(r + s + r)';$$

and if in A we replace ' x ' by ' y ' by ' x ', we obtain

$$'(y + x + y)';$$

if in A we replace ' y ' by " $(p \rightarrow q)$ ", we obtain

$$'(x + '(p \rightarrow q) + x)'.$$

If x is

$$'(x + y + x)'$$

and if in x we replace ' x ' by x , we obtain

$$'((x + y + x) + y + (x + y + x))'.$$

Expressions. The inscription ' $\alpha\alpha$ ' is a two symbol inscription in which precisely one symbol appears and that symbol appears precisely twice. The inscription ' $\alpha\beta$ ' is a two symbol inscription in which precisely two symbols appear and each of these appears precisely once. ' wh ' is a two symbol expression and not a symbol whereas ' wh ' is a symbol.

Among our symbols are:

$$'+', '., ':', '...', '\alpha', '\beta', '\underline{\alpha\beta}', '., ':', '264', \\ '98.64', '38A', 'x', 'x_3', 'x_{12}'.$$

Also among our symbols are wordlike nonitalic inscriptions such as:

$$'sb', 'psb', 'inf', 'sup'.$$

If C is the expression obtained from ' $(\alpha\beta)$ ' by replacing ' α ' by A and ' β ' by B , then: if A is ' x ' and B is ' y ', then C is ' (xy) '; if A is ' x ' and B is ' y ', then C is ' (xy) '; if A is ' x ' and B is ' y ', then C is ' $(x y)$ '; if A is '34' and B is '27', then C is '(34 27)' and not '(3427)'; if A is 'sin' and B is 'cos',

then C is $(\sin \cos)$ and not $(\sin \cos)$; and if A is x and B is 2 , then C is $(x 2)$.

If C is the expression obtained from $\alpha\beta$ by replacing α by A and β by B , then: if A is wh and B is at , then C is $wh at$; if A is $\alpha\beta$ and B is $\beta\alpha$, then C is the four symbol expression $\alpha\beta\beta\alpha$; if A is wh and B is at , then C is the two symbol expression $wh at$; if A is wh and B is at , then C is the four symbol expression $what$; if A is 27 and B is 682 , then C is the two symbol expression $27 682$; and if A is x and B is y_2 then C is the two symbol expression $x y_2$.

Rudiments. Our *punctuator* is $\blacksquare$.

Our *definor* is $\equiv$.

Unlike our definor, which will appear frequently, our punctuator will never appear in the mathematical language we are trying to describe. Instead it will be used, in the final section of this chapter, to facilitate the analysis of expressions in which it does *not* appear.

Our *schemators* are

$\underline{u}$, $\underline{v}$, $\underline{w}$, $\underline{u'}$, $\underline{v'}$, $\underline{w'}$, $\underline{u''}$, $\underline{v''}$, $\underline{w''}$,
 $\underline{u'''}$, $\underline{v'''}$, $\underline{w'''}$, $\underline{u''''}$, etc.

Inference starts with *definitions* and *axioms*. Each definition and each axiom will be an expression explicitly described or explicitly introduced by an appropriate marginal label. The scope of such a marginal label will end just before the next heading, aside, or marginal notation. In addition, we have so constructed the definitions themselves that no definition can be an initial segment of a different definition. We shall now explain the precautions we have taken to achieve this somewhat limited goal.

We agree S is *framed* if and only if S is an expression in which $\blacksquare$ does not appear.

We agree S is *formative* if and only if S is a framed expression in which $\equiv$ does not appear.

We agree that: S is *parenthetic* if and only if S is $()$ or S can be obtained from (x) by replacing x by an expression in which no parenthesis appears; S *evolves* T if and only if T can be obtained from S by replacing a symbol which is not a parenthesis by a parenthetic

expression; S is *parenthetical* if and only if S is a framed expression which can be built by successive evolution from some expression in which no parenthesis appears.

Thus if A and B are parenthetical expressions and C is obtained from A by replacing a symbol which is not a parenthesis by B , then C is parenthetical.

We have so constructed our definitions that if D is one of them, then D can be obtained from ' $(x \equiv y)$ ' by replacing ' y ' by a parenthetical expression and ' x ' by a formative expression.

As is well known, parenthetical expressions have a straightforward arithmetical characterization. To '(' is assigned the value 1; to each symbol which is not a parenthesis is assigned the value 0; to ')' is assigned the value -1 . Now if S is any framed expression, then S is parenthetical if and only if the total value of S is 0 and that of each initial segment is nonnegative.

Among the parenthetical expressions are

$$'xyz', \quad 'x(y \rightarrow x)(a \rightarrow b)t \equiv z', \quad \text{and} \quad '(x \rightarrow (y \rightarrow z))'.$$

We agree that c is *fixed* by D if and only if c is a symbol and D is a definition that can be obtained from ' $(x \equiv y)$ ' by replacing ' y ' by a parenthetical expression in which c does not occur and ' x ' by a formative expression in which c does occur.

We agree that c is a *constant* if and only if either c is a symbol fixed by some definition, c is ' $\mathbf{\Gamma}$ ', c is ' $\equiv$ ', c is a schemator, or c is either a nonitalicized Latin-lettered symbol or one of the superscripted or subscripted symbols derived therefrom.

Accordingly,

$$'e', \quad 'd', \quad 'C', \quad 'cCc', \quad 'sb', \quad 'sup', \quad \text{and} \quad 'Crd'$$

are constants.

Definitions 0.0 inform us that parentheses, the impicator ' $\rightarrow$ ', the universal quantifier ' $\wedge$ ', and the semicolon are among our constants; and Definition 2.0.0 informs us that the membership sign ' ϵ ' and the Zermelo Selector ' mel ' are among our constants.

We agree that α is a *variable* if and only if α is a symbol which is not a constant.

The lightface italic Latin letters together with the superscripted and subscripted symbols derived therefrom are variables.

We agree that D *lifts* A if and only if A is a formative expression and D is a definition which can be obtained from ' $(x \equiv y)$ ' by replacing ' x ' by A and ' y ' by a parenthetical expression.

We agree that D *raises* A if and only if A is a formative expression and D is a definition which can be obtained from ' $(x \equiv y)$ ' by replacing ' x ' by A and ' y ' by a parenthetical expression different from A .

Thus our definition

$$'((p \wedge q) \equiv \sim(p \rightarrow \sim q))'$$

both lifts and raises ' $(p \wedge q)$ '. Our definition

$$'((x \rightarrow x') \equiv (x \rightarrow x'))'$$

lifts but does not raise ' $(x \rightarrow x')$ '.

We agree that A is a *definiendum* if and only if some definition lifts A .

Thus from our definition.

$$'(0 \equiv \wedge xx)'$$

we learn that ' 0 ' is both a definiendum and a constant. From our definition ' $(x \equiv x)$ ' we learn that ' x ' is a definiendum.

We agree that B is a *variant* of A if and only if B can be obtained from A by replacing variables by variables and conversely A can be obtained from B by replacing variables by variables.

We agree that A and B are *diverse* if and only if A and B are expressions and B is not a variant of A .

We agree that C is a *form* if and only if C is a variant either of ' $(x \equiv y)$ ' or of some definiendum.

From our definition

$$'((p \wedge q) \equiv \sim(p \rightarrow \sim q))'$$

we learn that ' $(p \wedge q)$ ' is a definiendum and that ' $(x \wedge y)$ ' is a form. However, ' $(x \wedge x)$ ' is not a form. From our definitions

$$'(\vee x \underline{ux} \equiv \sim \wedge x \sim \underline{ux})'$$

and

$$'(\vee x \in x' \underline{yx} \equiv \vee x; (x \in x') \underline{yx})'$$

we learn that ' $\vee y \underline{uy}$ ' and ' $\vee t \in y \underline{yt}$ ' are forms. However, ' $\vee x \underline{yx}$ ' is not a form.

Forms are to be read as a whole. The individual constants are usually incidental. The expression ' $(p \rightarrow q)$ ' has nothing to do with limits and the expression

$$'(\underline{u}x \rightarrow A \text{ as } x \rightarrow a)'$$

has nothing to do with implication.

We agree that C is *primitive* if and only if either C is a schemator, or C is ' $\equiv$ ', or C is ' $(x \equiv y)$ ', or C is such a definiendum that no definition raises a variant of C .

Our primitive constants consist of our definor and our schemators.

We agree that S is *primal* if and only if S is one of the expressions

$$\begin{aligned} &'(x \equiv y)', \\ &'x', \quad '\underline{u}x', \quad '\underline{v}x', \quad '\underline{w}x', \\ &'x'', \quad '\underline{u}xx'', \quad '\underline{v}xx'', \quad '\underline{w}xx'', \\ &'x''', \quad '\underline{u}xxx''', \quad '\underline{v}xxx''', \quad '\underline{w}xxx''', \\ &\text{etc.} \end{aligned}$$

We agree S is *schematic* if and only if S can be obtained by replacing variables by variables in some primal expression in which a schemator appears.

In an informal way we sometimes think $\underline{u}x$ if and only if x has the property $\underline{u}$. Alternatively, in an informal way we sometimes think that $\underline{u}x$ is the set corresponding to x under $\underline{u}$. In most given instances one thought is more reasonable than the other.

We agree that A is *simple* if and only if A is an expression in which no variable appears more than once.

Schematic Replacement. We agree that B is obtained from A by *schematically replacing* S by R if and only if S is a schematic expression, and there is an expression Q in which the first symbol in S does not occur and a symbol q such that A is obtained from Q by replacing q by S and B is obtained from Q by replacing q by R .

The only reason for making the first two definitions of 0.0 below is to establish certain constants. The only reason for making the remainder of the definitions is to lift certain important forms.

0.0 ORIENTING DEFINITIONS.

- .0 $((\rightarrow x) \equiv x)$
 - .1 $(\wedge ; x \equiv x)$
 - .2 $((x \rightarrow x') \equiv (x \rightarrow x'))$
 - .3 $(\wedge x \underline{u}x \equiv \wedge x \underline{u}x)$
 - .4 $(x \equiv x)$
 - .5 $(\underline{u}x \equiv \underline{u}x)$
 - .6 $(\underline{v} \equiv \underline{v}x)$
 - .7 $(\underline{w}x \equiv \underline{w}x)$
 - .8 $(x' \equiv x')$
 - .9 $(\underline{u}'xx' \equiv \underline{u}'xx')$
 - .10 $(\underline{v}'xx' \equiv \underline{v}'xx')$
 - .11 $(\underline{w}'xx' \equiv \underline{w}'xx')$
 - .12 $(x'' \equiv x'')$
 - .13 $(\underline{u}''xx'x'' \equiv \underline{u}''xx'x'')$
 - .14 $(\underline{v}''xx'x'' \equiv \underline{v}''xx'x'')$
 - .15 $(\underline{w}''xx'x'' \equiv \underline{w}''xx'x'')$
 - .16 $(x''' \equiv x''')$
 - .17 $(\underline{u}'''xx'x''x''' \equiv \underline{u}'''xx'x''x''')$
- etc.

Remark. With the help of 2.24.67 it is easy to see that all of the orienting definitions in 0.0 are theorems.

Through rules and agreements we shall try to make clear when a given variable is *free*, *indicial*, or *accepted* in an expression.¹ We shall also try to make clear just what expressions are *formulas*, and we shall give *rules of inference* for establishing *theorems*. Theorems, of course, are of particular interest to us. Our rules of inference enable us, step by step, to use theorems already known to us to discover new theorems. Formalization describes with care an explicit process for arriving at

¹ Roughly speaking, a variable is free in a formula if and only if every occurrence is a free occurrence.

theorems. Our rules are to be taken for granted, although some of them can be derived from others. Our rules are akin to, but different from, axioms and theorems. Our agreements are akin to, but different from, definitions. Mathematics is made up of statements about sets. Metamathematics is made up of statements about expressions. Axioms, theorems, and definitions belong to mathematics. Rules and agreements belong to metamathematics.

To give a rough idea of the roles played by schematic expressions, free variables, indicial variables, and accepted variables, we say that in a theorem a free variable is replaceable by a wide variety of formulas, a schematic expression is replaceable by a still wider variety of formulas, and an indicial variable, such as an index of summation or a dummy variable of integration, is replaceable by accepted variables.

FREE VARIABLES AND FORMULAS

0.1 RULE. If α is free in A , then α is a variable and A is an expression.

0.2 RULE. A variable is free in a form if and only if it occurs therein less than twice.

0.3 AGREEMENT. A is a *formula* if and only if some variable is free in A .

0.4 RULE. If A is a formula, C is a formula, B is different from A and is obtained from A either by replacing some free variable of A by C or by schematically replacing some schematic expression by C , then a variable is free in B if and only if it is free in both A and C .

The remainder of this section and the last two sections of this chapter, 0.70–0.85, shed light on the precise structure of formulas and the explicit nature of theorems. However, the reader may omit all of this material except 0.9, according to his pleasure, since the rest of this chapter, which includes the mechanics of proof, is entirely independent.

0.5 AGREEMENT. A is *strict* if and only if A is a formula and not a variable.

0.6 AGREEMENT. F is *fundamental* if and only if F is either a schematic form, or a strict formula devoid of schemators from which some form can be obtained by replacing variables by schematic expressions.

Of the eight expressions

$'(x \rightarrow t)'$, $'\underline{u}'xy'$, $'\wedge yx'$, $'x'$, $'\wedge xx'$, $'\wedge x \underline{u}x'$,
 $'\wedge x \underline{u}y'$, and $'(x \rightarrow x)'$,

the first three are fundamental formulas and the last five are formulas which are not fundamental.

0.7 AGREEMENT. α is an *introducer* if and only if α is a constant which is the initial symbol of some definiendum.

0.8 AGREEMENT. α is a *noun* if and only if α is both a constant and a definiendum.

0.9 AGREEMENTS.

.0 (AA') is the expression obtained from $'xy'$ by replacing $'x'$ by A and $'y'$ by A' .

.1 $(AA'A'')$ is $((AA')A'')$.

.2 $(AA'A''A''')$ is $((AA'A'')A''')$.
 etc.

In other words, (AB) is the concatenation of A and B . For example, ('sin' 'x') is 'sin x' .

Remark. We suppose herein that

C is ('x' 'x' 'x') .

Because of 0.9.1 and 0.9.0, it is easy to see that

C is 'xxx' .

On p. 2 we agreed that quotation marks are *not* among our symbols. If quotation marks were among our symbols, then in the light of 0.9.0 *alone* a contradiction would arise as follows.

We also suppose

A is 'x';
 A' is 'x' 'x';
 B is 'x' 'x';
 B' is 'x'.

Thus by 0.9.0 we have on the one hand

C is (AA')

and on the other

C is (BB') .

But

(AA') is 'xx' 'x'

and

(BB') is 'x' 'xx'.

However, 'xx' 'x' is not the same as 'x' 'xx', nor is either the same as 'xxx'.

0.10 AGREEMENT. A is a *prefix* if and only if A is either an introducer or an expression of the kind (αB) where α is an introducer and B is an expression devoid of introductors.

0.11 AGREEMENT. A is a *suffix* if and only if A is either a noun or an expression of the kind (αB) where α is an introducer and B is an expression.

0.12 AGREEMENT. C is *reducible* if and only if C is strict and not fundamental.

0.13 RULE. A formula is strict if and only if its initial symbol is an introducer.

0.14 RULE. A formula is fundamental if and only if it is either a simple prefix or an expression of the kind $(A\alpha)$ where A is a simple prefix and α is the initial symbol of A .

0.15 RULE. If F is a strict formula devoid of schemators, and A is a form obtained from F by replacing variables by schematic expressions, then F is a simple formula, every variable which appears in A also appears in F , α is free in F if and only if α is free in A , and A can be obtained from F by replacing variables which do not appear in A by schematic expressions.

0.16 RULE. If C is a formula, Q is an expression, α is a variable which appears precisely once in Q , M is a formula, and C is obtained from Q by replacing α by M , then:

- .0 if Q is a formula and M is a strict formula, then α is free in Q ;
- .1 if Q is fundamental and C is reducible, then either α is free in Q or M is free in Q ;
- .2 if M is a variable, then Q is a formula; and
- .3 if A is a prefix, B is an expression whose initial symbol is α , and Q is (AB) , then Q is a formula.

Thus

$$' \wedge (x \rightarrow x)y '$$

is not a formula since otherwise we could learn that ' t ' is both free and not free in ' $\wedge ty$ '.

0.17 RULE. No formula is an initial segment of a different formula.

0.18 RULE. If A is a prefix, B is a suffix, and (AB) is a reducible formula, then some initial segment of B is a formula.

Remark. In order that the foregoing rules and agreements unite in harmony, we must in making definitions take some technical precautions such as, for example, those outlined in A.0–A.8 of Appendix A.

INDICIAL AND ACCEPTED VARIABLES

0.19 RULE. If α is indicial in A , then α is a variable and A is a formula; if α is accepted in A , then α is a variable and A is a formula.

0.20 RULE. A variable is indicial in a form if and only if it occurs therein more than once.

0.21 RULE. A variable is accepted in a form if and only if it occurs therein less than twice.

Thus a variable is accepted in a form, as opposed to a formula, if and only if the variable is free in the form. On the other hand, a variable is indicial in a form if and only if it is not free therein.

0.22 RULE. If A , B , and C are formulas with A different from B and B different from C , and if B can be obtained from A by replacing a free and accepted variable of A by C , then α is accepted in B if and only if α is accepted in A , and α is indicial in B if and only if α is indicial in A and does not appear in C .

0.23 RULE. If A , B , and C are formulas with A different from B and B different from C , S is a schematic expression, some variable in S is indicial in A , and if B is obtained from A by schematically replacing S by C , then α is indicial in B if and only if α is indicial in A , and α is accepted in B if and only if α is accepted in A and does not appear in C .

If some variable is indicial in a form, then it is quite reasonable to assign special significance to the positions occupied by the free variables and the schematic expressions, and to the positions outside of schematic expressions occupied by individual variables. For example, in

$$' \forall x \in A \ \underline{u}x '$$

we might say ' x ' in its first appearance is in an indicial position, ' A ' is in a free position, and ' $\underline{u}x$ ' is in a position subservient to the indicial.

Indicial and accepted variables can be looked at, less mechanically, in another way. Suppose F is a form which is neither a schematic form nor

a variable. Now if F' is obtained from F by simultaneously replacing free variables which appear in F by formulas and schematically replacing schematic expressions which appear in F by formulas, then a variable is indicial in F' if and only if it is indicial in F and does not appear in any of the formulas replacing free variables, and a variable is accepted in F' if and only if it is accepted in F and does not appear in any of the formulas schematically replacing schematic expressions.

Remarks. A variable which does not occur in a formula is free, accepted, but not indicial therein. In the expression

$$' \wedge x(x \rightarrow y) '$$

' x ' is indicial and not free, whereas ' y ' is free but neither indicial nor accepted. Because ' x ' is free in ' $\wedge y \underline{y}$ ' and in ' $(x \in t)$ ', it follows that ' x ' is free in

$$' \wedge y(x \in t) '.$$

In the expression ' $\wedge x \wedge yy$ ', ' x ' is indicial and not free while ' y ' is neither free nor indicial nor accepted. In the expressions

$$' \wedge y \wedge yy ', ' \forall y \in \wedge xx y ', \text{ and } ' \forall y \in x y ',$$

' y ' is indicial and ' x ' is accepted. In the expression ' $\forall x \in x x$ ', ' x ' is neither free nor indicial nor accepted. In the expression

$$' (\wedge xx \rightarrow \wedge xx) '$$

' x ' is accepted but neither free nor indicial.

If A is any formula exhibited previously, then a variable is free in A if and only if in A it never immediately follows ' $\wedge$ ' or ' $\forall$ '. A variable b is indicial in A provided A is obtained from ' $\wedge xz$ ' or ' $\forall xz$ ' or ' $\forall x \in y z$ ' by replacing ' x ' by b , ' y ' by a formula in which b does not appear, and ' z ' by a formula. Furthermore, if A is obtained from ' $\wedge xz$ ' or ' $\forall xz$ ' or ' $\forall x \in y z$ ' by replacing ' x ' by a variable α , ' y ' by a formula, and ' z ' by a formula C , then a variable is accepted in A if and only if it differs from α and does not appear in C .

Our Rule of Inference 0.28 is, of course, to be understood in the light of 0.19-0.23. There is a natural temptation to simplify 0.28, by abandoning acceptedness, using a simpler notion of indiciality, and, in

0.28, replacing the words ‘is accepted’ by ‘does not appear’. To assess the consequences of this proposed simplification let us adopt the unordered summation notation

$$‘\sum_{x \in A} \underline{u}x’.$$

It is then natural to expect that

$$(\sum_{n \in 4} n = 6 = \sum_{m \in 4} m).$$

Also it seems clear that

$$‘\sum_{x \in y} x’$$

is a formula. Inasmuch as

$$‘(y \rightarrow \wedge xy)’$$

is a formula in which ‘y’ is free, it seems inescapable that

$$‘(x \rightarrow \wedge xx)’$$

is a formula, but not, we hope, a theorem. Thus, since ‘y’ is free in

$$‘\sum_{x \in y} x’,$$

we feel compelled to admit that

$$‘\sum_{x \in x} x’$$

is also a formula. If 0.28 were simplified in the way we momentarily have in mind, then we would be unable to interpret the formula

$$‘\sum_{x \in x} x’.$$

But with 0.28 and 0.19–0.23, as they stand, at our disposal we notice that

$$(\sum_{y \in x} y = \sum_{y \in x} y)$$

and use 0.28 to infer that

$$(\sum_{x \in x} x = \sum_{y \in x} y).$$

RULES OF INFERENCE; THEOREMS

We agree D is *real* if and only if D is a formula which raises some definiendum.

0.24 INITIATION. Every real definition is a theorem; every axiom that is a formula is also a theorem.

It should not be assumed that a formula is a definition just because it looks like one. A variant of a definition is seldom a definition. In particular, the formula

$$(0 \equiv \wedge yy)',$$

which is a variant of Definition 1.0.3, is not a definition.

Definitions are especially important for three reasons. Definitions generate forms; real definitions are more than mere shorthand devices; real definitions are accepted as theorems. Accordingly, definitions should be made with care.

Presumably, only by mistake would someone fashion an axiom which is not a formula.

0.25 DETACHMENT. If a theorem is obtained from ' $(p \rightarrow q)$ ' by replacing ' p ' by a theorem and ' q ' by a formula T , then T is a theorem.

0.26 SUBSTITUTION. If T is a theorem in which b is free and A is such a formula that each variable in it is free in T , then the expression obtained from T by replacing b by A is also a theorem.

0.27 SCHEMATIC SUBSTITUTION. If T is a theorem, S is a schematic expression, and A is such a formula that each variable in it is either free in T or occurs explicitly in S , and T' is a formula obtained from T by schematically replacing S by A , then T' is a theorem.

0.28 INDICIAL SUBSTITUTION. If q is free in Q , T is a theorem obtained from Q by replacing q by a formula A in which α is indicial, B is obtained from A by replacing α by a variable which is

accepted in A , and finally T' is obtained from Q by replacing q by B , then T' is a theorem.

0.29 UNIVERSALIZATION. If T is a formula obtained from ' $\wedge xy$ ' by replacing ' x ' by a variable and ' y ' by a theorem, then T is a theorem.

We shall eventually categorically describe theorems in Rule 0.75. This rule is independent of the intervening Theory of Notation.

Examples. By detachment we learn that if

$$'(\wedge x(x \rightarrow x) \rightarrow (x \rightarrow x))'$$

is a theorem and if

$$' \wedge x(x \rightarrow x)'$$

is a theorem, then ' $(x \rightarrow x)$ ' is a theorem.

By substitution we learn that if

$$'((y \rightarrow t) \rightarrow \wedge x(y \rightarrow t))'$$

is a theorem, then

$$'((y \rightarrow y) \rightarrow \wedge x(y \rightarrow y))'$$

is a theorem. However, replacing ' t ' or ' y ' by ' x ' is not allowed by substitution.

By schematic substitution we learn that if

$$'(\wedge x \underline{u}x \rightarrow \underline{u}x)'$$

is a theorem, then

$$'(\wedge x(x \rightarrow x) \rightarrow (x \rightarrow x))'$$

is also a theorem. Note, however, that from the supposition that

$$'(\wedge xy \rightarrow y)'$$

is a theorem we cannot employ either substitution or schematic substitution to infer that

$$'(\wedge x(x \rightarrow x) \rightarrow (x \rightarrow x))'$$

is a theorem.

By indicial substitution we learn that if

$$‘(\forall x \in y \ x \rightarrow \forall x \in y \ x)’,$$

$$‘(\wedge x \ \underline{u}x \rightarrow \underline{u}x)’,$$

and

$$‘(\wedge x(x \rightarrow x) \rightarrow (x \rightarrow x))’$$

are theorems, then

$$‘(\forall x \in y \ x \rightarrow \forall y \in y \ y)’,$$

$$‘(\wedge y \ \underline{u}y \rightarrow \underline{u}x)’,$$

and

$$‘(\wedge y(y \rightarrow y) \rightarrow (x \rightarrow x))’$$

are theorems. However, from the assumption that

$$‘(\wedge y \ \underline{u}y \rightarrow \underline{u}x)’$$

is a theorem we cannot directly employ schematic substitution to learn that

$$‘(\wedge y(y \rightarrow y) \rightarrow (x \rightarrow x))’$$

is a theorem. This is because ‘ $\underline{u}y$ ’ and ‘ $\underline{u}x$ ’ are not the same.

From universalization it follows that if ‘ $(x \rightarrow x)$ ’ is a theorem, then

$$‘\wedge x(x \rightarrow x)’$$

is a theorem.

We henceforth try to bear these foregoing rules in mind.

THEORY OF NOTATION

The reader may find some of our notations different from those to which he has become accustomed. We find ourselves a little reluctant to introduce nonlinear notations and somewhat more reluctant to introduce notations which make it very easy to reach a contradiction.

As we have indicated before, reluctance of the latter sort caused us to use the functional notation ' $.fx$ ' in place of the customary, and incidentally more cumbersome, notation ' $f(x)$ '.

Although most of the formulas we use can be deciphered intuitively, we nevertheless suggest a somewhat cursory persual of and occasional reference to this section. We formulate herein a general and flexible theory of notation which permits useful simplification of a vast number of complicated expressions and justifies many of the informal conventions of present-day mathematics. Attention paid to the examples should make considerably easier the reader's understanding of the rudiments of the theory.

We shall make no real use of 0.50–0.64 until we reach 2.57.

In 0.30 we are interested in superficial metaconformity with Appendix A.

0.30 AGREEMENT. α is of type n if and only if

.2 $n = 2$ and α is

' $\rightarrow$ ';

.4 $n = 4$ and α is

' $\leftrightarrow$ ';

.5 $n = 5$ and α is one of

' $\wedge$ ', ' $\vee$ ';

.6 $n = 6$ and α is one of

' $\in$ ', ' $\ni$ ', ' $\subset$ ', ' $\supset$ ', ' $=$ ', ' $\neq$ ',
'wellorders', 'orders', 'eq', 'topologizes',
' $<$ ', ' $>$ ', ' $\leq$ ', ' $\geq$ ',
'metrizes', 'simplymetrizes', 'measures';

.7 $n = 7$ and α is

' $,$ ';

.8 $n = 8$ and α is one of

' $\setminus$ ', ' $\cup$ ';

.9 $n = 9$ and α is one of

' $\oplus$ ', ' $+$ ', ' $\oplus_1$ ', ' $\oplus_2$ ', ' $\oplus_3$ ',
' $\oplus_4$ ', ' $\oplus_5$ ', ' $\oplus_6$ ', ' $\oplus_7$ ';

.11 $n = 11$ and α is one of

‘ $-$ ’, ‘ $\ominus_3$ ’, ‘ $\ominus_6$ ’;

.13 $n = 13$ and α is

‘ $/$ ’;

.15 $n = 15$ and α is one of

‘ $\cap$ ’, ‘ $\cup$ ’, ‘ $:$ ’, ‘ $::$ ’, ‘ $\odot$ ’, ‘ $\cdot$ ’,
‘ $\odot_1$ ’, ‘ $\odot_2$ ’, ‘ $\odot_3$ ’, ‘ $\odot_4$ ’, ‘ $\odot_5$ ’,
‘ $\odot_6$ ’, ‘ $\odot_7$ ’, ‘ $\bullet$ ’, ‘ $*$ ’, ‘ $\underline{\cap}$ ’; or

.17 $n = 17$ and α is

‘ $\sim$ ’.

0.31 AGREEMENTS.

.0 A symbol is a *binarian* if and only if it is a symbol of some type.

.1 A symbol is a *binariate* if and only if it is either ‘ x ’ or one of the primed symbols derived therefrom.

0.32 AGREEMENT. c is a *nexus* if and only if c is an expression in which each symbol is a binarian.

In 0.33 we use 0.9.

0.33 AGREEMENT. A is a *parade* if and only if A is such an expression in which some binarian appears that A can be obtained from one of the expressions

‘ (xx') ’, ‘ $(xx'x'')$ ’, ‘ $(xx'x''x''')$ ’, ...

by replacing each binariate α which is different from ‘ x ’ by some expression which either is α itself or is of the kind $(c\alpha)$ where c is a nexus.

A not unusual sort of parade is

‘ $(x \subset x' \subset x'')$ ’.

A less common sort is

‘ $(x \cup x' \cap \cap x'' \rightarrow x''' \subset x'''' \ni x''''')$ ’.

Our theory of notation and subsequent mathematical definitions will make possible a unique interpretation of the two parades just mentioned as well as a host of others.

0.34 AGREEMENT. A is of power n if and only if A is a nexus in which some symbol of type n appears and no symbol of type less than n appears.

For example,

$$'<', '<<', '<\cap', '\subset\cdot\epsilon'$$

are of power 6.

0.35 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained by replacing ' $\wedge$ ' by an expression of odd power in any one of the expressions:

$$\begin{aligned} '((x \wedge x' \wedge x'') \equiv ((x \wedge x') \wedge x''))', \\ '((x \wedge x' \wedge x'' \wedge x''') \equiv ((x \wedge x' \wedge x'') \wedge x'''))', \end{aligned}$$

etc.

0.36 DEFINITION. $((x) \equiv (x))$

0.37 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained by replacing ' $\rightarrow$ ' by an expression of power 2 in any one of the expressions:

$$\begin{aligned} '((x \rightarrow x' \rightarrow x'') \equiv ((x \rightarrow x') \wedge (x \rightarrow x'')))', \\ '((x \rightarrow x' \rightarrow x'' \rightarrow x''') \equiv ((x \rightarrow x') \wedge (x \rightarrow x'') \wedge (x \rightarrow x''')))', \end{aligned}$$

etc.

In each of these expressions note well that the sixth symbol from the end is ' x ' and not one of the primed symbols derived therefrom. More usual and very similar to each other are 0.38 and 0.39 below.

0.38 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained by replacing ' $\leftrightarrow$ ' by an expression of power 4 in any one of the expressions

$$\begin{aligned} '((x \leftrightarrow x' \leftrightarrow x'') \equiv ((x \leftrightarrow x') \wedge (x' \leftrightarrow x'')))', \\ '((x \leftrightarrow x' \leftrightarrow x'' \leftrightarrow x''') \equiv ((x \leftrightarrow x' \leftrightarrow x'') \wedge (x'' \leftrightarrow x''')))', \end{aligned}$$

etc.

0.39 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained by replacing '=' and each primed symbol derived therefrom by expressions of power 6 in any one of the expressions

$$\begin{aligned} &'((x = x' = 'x'') \equiv ((x = x') \wedge (x' = 'x'')))', \\ &'((x = x' = 'x' = ''x''') \equiv ((x = x' = 'x'') \wedge (x'' = ''x''')))', \\ &\text{etc.} \end{aligned}$$

0.40 AGREEMENT. α is a *biniate* if and only if α is a binariate or a parenthesis.

0.41 AGREEMENTS.

.0 B in A is a *leftfence* for C if and only if C is a binariate and either B is a biniate for which (BC) is a segment of A , or B is such a nexus that for some binariate α , (αBC) is a segment of A .

.1 D in A is a *rightfence* for C if and only if C is a binariate and either D is a biniate for which (CD) is a segment of A or D is such a nexus that for some binariate β , $(CD\beta)$ is a segment of A .

.2 C occurs in A *betwixt* B and D if and only if A is a parade and either C is a binariate, B in A is a leftfence for C , D in A is a rightfence for C , or C is a nexus, B is a binariate, D is a binariate, (BCD) is a segment of A .

If A is the expression

$$'(x \cap x' \rightarrow \cup x'' + \leftrightarrow \cdot x''' x''')',$$

then in A

' x ' is betwixt '(', and ' $\cap$ ',
 ' $\cap$ ' is betwixt ' x ', and ' x' ',
 ' x' ' is betwixt ' $\cap$ ' and ' $\rightarrow \cup$ ',
 ' $\rightarrow \cup$ ' is betwixt ' x' ' and ' x'' ',
 ' x'' ' is betwixt ' $\rightarrow \cup$ ' and ' $+ \leftrightarrow \cdot$ ',
 ' $+ \leftrightarrow \cdot$ ' is betwixt ' x'' ' and ' x''' ',
 ' x''' ' is betwixt ' $+ \leftrightarrow \cdot$ ' and ' x'''' ', and
 ' x'''' ' is betwixt ' x'''' ' and ')'.
 .

0.42 AGREEMENT. B is a *bisegment* of A if and only if B is a nexus which occurs in A betwixt two binariates.

0.43 AGREEMENT. We agree that B is *minimal* in A if and only if B is a bisegment of A and no bisegment of A is of lower power than that of B .

0.44 AGREEMENT. We agree B is of *prime importance* in A if and only if either B is a parenthesis, or B is of even power and is minimal in A , or B is of odd power, is minimal in A , and, among those expressions which are minimal in A , B is the expression whose first appearance in A betwixt two binariates is deferred the longest.

0.45 AGREEMENTS.

.0 We agree that α is *left* in A if and only if α is a binariate and there are such expressions B and C that α occurs in A betwixt B and C , B is of prime importance in A , and C is not of prime importance in A .

.1 We agree that α is *right* in A if and only if α is a binariate and there are such expressions B and C that α occurs in A betwixt B and C , B is not of prime importance in A , and C is of prime importance in A .

0.46 AGREEMENTS.

.0 *Leftdistend* α is $(\cdot(\cdot\alpha))$.

.1 *Rightdistend* β is $(\beta^{\cdot})^{\cdot}$.

Thus *Leftdistend* ' x ' is ' $(x$ ' and *Rightdistend* ' x ' is ' $x)$ '.

0.47 AGREEMENT. *Complicate* A is the expression obtained from A by first replacing each binariate α which is left in A by *Leftdistend* α and then in this result replacing each binariate β which is right in A by *Rightdistend* β .

0.48 DEFINITIONAL SCHEMA. If A is a parade, then we accept as a definition the expression obtained from ' $(x \equiv y)$ ' by replacing ' x ' by A and ' y ' by *Complicate* A .

From 0.39 we learn that

$$'((x \subset x' = x'' \ni x''') \equiv ((x \subset x' = x'') \wedge (x'' \ni x'''))))'$$

is a definition.

If A is the rather weird expression

$$'(x \rightarrow x'x'' \rightarrow \cdot x''' < \cap \cup x''')',$$

then the bisegments of A are ' $\rightarrow$ ', ' $\rightarrow \cdot$ ', and ' $< \cap \cup$ '; ' $\rightarrow$ ', ' $\rightarrow \cdot$ ', and the parentheses are of prime importance in A ; ' x ' and ' x''' ' are left in A ; ' x'' ' and ' x'''' ' are right in A ; and Complicate A is

$$'(x \rightarrow (x'x'') \rightarrow \cdot (x''' < \cap \cup x'''))'.$$

If A is the expression

$$'(x \cup x' \cap x'' \cup x''')',$$

then ' $\cap$ ' is of prime importance in A , ' x ' and ' x''' ' are left in A , ' x' ' and ' x'' ' are right in A , and Complicate A is

$$'((x \cup x') \cap (x'' \cup x'''))'.$$

However, if A is the expression

$$'(x \cap x' \cup x'' \cap x''')',$$

then ' $\cup$ ' is of prime importance in A and Complicate A is

$$'((x \cap x') \cup (x'' \cap x'''))'.$$

If A is the expression

$$'(x \wedge < x' \wedge << x'' \wedge < x''' \wedge << x''')',$$

then ' $\wedge < <$ ' is of prime importance in A and Complicate A is

$$'((x \wedge < x') \wedge << (x'' \wedge < x''') \wedge << x''')'.$$

If A is the expression

$$'(x \in x' \cap x'')',$$

then ' $\in$ ' is of prime importance in A and Complicate A is

$$'(x \in (x' \cap x''))'.$$

On the other hand, if A is the expression

$$'(x \in x' \wedge x'')',$$

then ' $\wedge$ ' is of prime importance in A and Complicate A is

$$'((x \in x') \wedge x'')'.$$

If A is the expression

$$'(x \rightarrow x' = x'' \in x''' \rightarrow x'''' \vee x''''')',$$

then Complicate A is

$$'(x \rightarrow (x' = x'' \in x''') \rightarrow (x'''' \vee x'''''))'.$$

If A is the expression

$$'(x + x' \cdot x'' + x''')',$$

then Complicate A is

$$'(x + (x' \cdot x'') + x''')'$$

and we know that

$$'((x + x' \cdot x'' + x''') \equiv (x + (x' \cdot x'') + x'''))'$$

is among our definitions.

If A is the expression

$$'(x = x' \in x'' x''' \ni x'''' \cup x''''' x''''')',$$

then Complicate A is

$$'(x = x' \in (x'' x''') \ni (x'''' \cup x''''' x'''''))'.$$

If A is the expression

$$'(x \cup \cup \cup x')'$$

then Complicate A is A .

A convenient negating device is

0.49 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$'((x \smile \in y) \equiv ((x \in y) \rightarrow \wedge zz))'$$

by replacing ' $\in$ ' by a nexus different from ' $\smile$ '.

As we have said before, we shall make no real use of 0.50–0.64 until we reach 2.57.

0.50 AGREEMENT. α is of class n if and only if

- .0 $n = 0$ and α is one of
‘E’, ‘One’, ‘The’;
- .1 $n = 1$ and α is one of
‘ $\wedge$ ’, ‘ λ ’, ‘far R ’, ‘large’, ‘small’, ‘big’,
‘alm $\neq M\varphi$ ’, ‘alm φ ’, ‘Alm φ ’, ‘Al φB ’;
- .2 $n = 2$ and α is one of
‘V’, ‘Ad’, ‘ $\sum$ ’;
- .3 $n = 3$ and α is one of
‘sup’, ‘inf’, ‘ad’, ‘osc’;
- .4 $n = 4$ and α is of class 0 or 1 or 2 or 3;
- .5 $n = 5$ and α is
‘|’; or
- .6 $n = 6$ and α is
‘Al’.

0.51 AGREEMENT. A symbol is a *notarian* if and only if it is the initial symbol of some expression of some class.

0.52 AGREEMENT. A is a *march* if and only if some parade in which no binariate is adjacent to a binariate can be obtained from ‘(p)’ by replacing ‘ p ’ by A .

Thus of the three expressions

$$‘xx' \in x''’, ‘x' \ni x’, \text{ and } ‘x \in x' \subset \subset x'' \ni x'''’$$

only the third is a march.

It will turn out that a march is never a formula.

0.53 AGREEMENTS.

- .0 α is a *verb* if and only if α is a binarian of type 2 or 4 or 6.
- .1 C is *verbal* if and only if C is an expression in which some verb appears.

.2 C is *verbless* if and only if C is an expression in which no verb appears.

0.54 AGREEMENT. s is a *subject* of A if and only if s is such a verbless expression, whose terminal symbol is a binariate, that either A is s or A can be obtained from ' xyz ' by replacing ' x ' by s , ' y ' by a verbal nexus, and ' z ' by an expression.

Clearly an expression can have at most one subject.

0.55 AGREEMENT. C is of *order* n if and only if C has a subject in which precisely n binariates appear.

0.56 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained by replacing ' E ' by an expression of class 0, ' $\wedge$ ' by an expression of class 1, ' $\vee$ ' by an expression of class 2, and ' $\sup$ ' by an expression of class 3 in any one of the following expressions:

$$\begin{aligned} & '(Ex; \underline{u}x \underline{v}x \equiv Ex(\underline{u}x \wedge \underline{v}x))' \\ & '(\wedge x; \underline{u}x \underline{v}x \equiv \wedge x(0 \in \underline{u}x \rightarrow \underline{v}x))' \\ & '(\vee x; \underline{u}x \underline{v}x \equiv \vee x(0 \in \underline{u}x \wedge \underline{v}x))' \\ & '(\sup x \underline{v}x \equiv \sup x; (x = x) \underline{v}x)' \end{aligned}$$

We suggest that the semicolon in any form whose initial symbol is of class 4 be read "subject to the condition that".

0.57 DEFINITIONS.

.0 $(st \ zx \ \underline{u}x \equiv \vee x(z = x \wedge \underline{u}x))$

.1 $(\text{substitute } z \text{ for } x \text{ in } \underline{u}x \equiv st \ zx \ \underline{u}x)$

0.58 DEFINITIONAL SCHEMAS.

.0 We accept as a definition each expression which can be obtained by replacing ' p ' by a march whose terminal symbol is ' x ' in

$$'(st \ zp \ \underline{u}'xx' \equiv \vee x \vee x'(z = (p) \wedge \underline{u}'xx'))'$$

.1 We accept as a definition each expression which can be obtained by replacing ' p ' by a march whose terminal symbol is ' x ' in

$$(\text{st } zp \underline{u}''xx'x'' \equiv \forall x \forall x' \forall x''(z = (p) \wedge \underline{u}''xx'x''))'.$$

etc.

Thus among our theorems are:

$$(\text{st } z \, x \, , \, y \, \underline{u}'xy \equiv \forall x \forall y(z = (x \, , \, y) \wedge \underline{u}'xy))',$$

$$(\text{st } z \, x + y \, \underline{u}'xy \equiv \forall x \forall y(z = (x + y) \wedge \underline{u}'xy))'.$$

0.59 AGREEMENTS.

.0 A is a 1 stencil if and only if A can be obtained by replacing ' E ' by an expression of class 0, ' $\vee$ ' by an expression of class 1 or class 2 or class 3, and ' $\wedge$ ' by an expression of class 4 in any one of the expressions

$$(\exists p ; qr \equiv \exists z \text{ st } zp (q \wedge r))',$$

$$(\vee p ; qr \equiv \vee z ; \text{st } ztq \text{ st } ztr)',$$

$$(\wedge pr \equiv \wedge p ; (x = x) r)'.$$

.1 A is a 2 stencil if and only if A can be obtained by replacing ' $\wedge$ ' by an expression of class 4 in any one of the expressions

$$(\wedge p ; qr \equiv \wedge s ; ((p) \wedge q) r)',$$

$$(\wedge pr \equiv \wedge s ; (p) r)'.$$

0.60 DEFINITION SCHEMAS.

.0 We accept as a definition each expression which can be obtained from a 1 stencil by replacing ' p ' by a verbless march of order 2, ' t ' by ' x, x' ', ' q ' by ' $\underline{u}'xx'$ ', and ' r ' by ' $\underline{v}'xx'$ '.

.1 We accept as a definition each expression which can be obtained from a 1 stencil by replacing ' p ' by a verbless march of order 3, ' t ' by ' x, x', x'' ', ' q ' by ' $\underline{u}''xx'x''$ ', and ' r ' by ' $\underline{v}''xx'x''$ '.

etc.

0.61 DEFINITIONAL SCHEMAS.

.0 We accept as a definition each expression which can be obtained from a 2 stencil by replacing ' p ' by a verbal march M of order 1, ' s ' by a subject of M , ' q ' by ' $\underline{u}x$ ', and ' r ' by ' $\underline{v}x$ '.

.1 We accept as a definition each expression which can be obtained from a 2 stencil by replacing 'p' by a verbal march M of order 2, 's' by a subject of M , 'q' by ' $\underline{u'xx'}$ ', and 'r' by ' $\underline{v'xx'}$ '.

etc.

Thus, among our theorems are

$$\begin{aligned}
 & '(Ex, y; \underline{u'xy} \underline{v'xy} \equiv Ez \text{ st } z \text{ x}, y (\underline{u'xy} \wedge \underline{v'xy}))', \\
 & '(Ex, y \underline{v'xy} \equiv Ex, y; (x = x) \underline{v'xy})', \\
 & '(Ex + y; \underline{u'xy} \underline{v'xy} \equiv Ez \text{ st } z \text{ x} + y (\underline{u'xy} \wedge \underline{v'xy}))', \\
 & '(\wedge x + y; \underline{u'xy} \underline{v'xy} \equiv \wedge z; \text{st } z \text{ x}, y \underline{u'xy} \text{ st } z \text{ x}, y \underline{v'xy})', \\
 & '(\wedge x, y; \underline{u'xy} \underline{v'xy} \equiv \wedge z; \text{st } z \text{ x}, y \underline{u'xy} \text{ st } z \text{ x}, y \underline{v'xy})', \\
 & '(\wedge x \in A \cup B \underline{ux} \equiv \wedge x; (x \in A \cup B) \underline{ux})', \\
 & '(\wedge x \cup y \in A \underline{ux} \equiv \wedge x \cup y; (x \cup y \in A) \underline{ux})', \\
 & '(\wedge x \cup y; \underline{u'xy} \underline{v'xy} \equiv \wedge z; \text{st } z \text{ x}, y \underline{u'xy} \text{ st } z \text{ x}, y \underline{v'xy})', \\
 & '(\wedge x \subset A; (xyz \subset B) \underline{ux} \equiv \wedge x; ((x \subset A) \wedge (xyz \subset B)) \underline{ux})'.
 \end{aligned}$$

A convenient combinatorial device is

0.62 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$'((A \cap \cap B) \equiv Ex \cap y (x \in A \wedge y \in B))'$$

by replacing ' $\cap$ ' by a verbless binarian.

Remark. In 0.30–0.48 we were primarily interested in parenthetical simplification. In 0.49–0.62 we have been primarily interested in notational uniformity as well as brevity. In 0.63–0.64 we shall again be interested primarily only in parenthetical simplification.

0.63 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$'(|p| \equiv |(p)|)'$$

by replacing 'p' by a march and '|' by an expression of class 5.

For example,

$$'(|x + x' + x''| \equiv |(x + x' + x'')|)'$$

is among our definitions.

0.64 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(Al \varphi By \underline{u}y \equiv Al \varphi(B)y \underline{u}y)’$$

by replacing ‘ B ’ by a march and ‘ Al ’ by an expression of class 6.

In trying to make sure that our definitions conform to Appendix A we now pick up some loose ends.

0.65 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(\wedge ; x \equiv x)’$$

by replacing ‘ $\wedge$ ’ by a notarian not of class 5.

0.66 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(| \rightarrow x | \equiv x)’$$

by replacing ‘ $\rightarrow$ ’ by a binarian and ‘ $|$ ’ by a symbol of class 5.

0.67 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(\wedge x \underline{u}x \equiv \wedge x \underline{u}x)’$$

by replacing ‘ $\wedge$ ’ by an expression of either class 0 or class 1 or class 2.

0.68 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(\sup x ; \underline{u}x \underline{v}x \equiv \sup x ; \underline{u}x \underline{v}x)’$$

by replacing ‘ $\sup$ ’ by an expression of class 3.

0.69 DEFINITIONAL SCHEMA. We accept as a definition each expression which can be obtained from

$$‘(|x| \equiv |x|)’$$

by replacing ‘ $|$ ’ by an expression of class 5.

Remark. By added effort we could in the spirit of the foregoing give a much more sweeping theory of notation than the one herein given. We could, for example, so arrange things that expressions like

$$\left(\int \underline{u}x \cdot \underline{v}x \, dx \equiv \int (\underline{u}x \cdot \underline{v}x) \, dx \right),$$

would automatically become theorems.

Remark. We urge the reader to take for granted that c is a constant if and only if c is either our definor, or our punctuator, or a semicolon, or a schemator, or a parenthesis, or a binarian,² or a notarian,² or a symbol fixed by 0.57.0, or a symbol fixed by 0.57.1, or a symbol fixed by the first formula in which it appears among those listed in the chapters which follow, or either a nonitalicized latin-lettered symbol or one of the superscripted or subscripted symbols derived therefrom. In this connection we should like to point out that no symbol is ever fixed by an expression in which ‘ $\equiv$ ’ does not appear.

DEMONSTRATIONS

0.70 AGREEMENT. C is entailed by A and by B if and only if A , B , and C are formulas for which there are such a formula Q , such a variable t , and such a schematic expression S that either

- .0 B is obtained from ‘ $(p \rightarrow q)$ ’ by replacing ‘ p ’ by A and ‘ q ’ by C ; or
- .1 t is free in A , each variable in Q is free in A , and C is obtained from A by replacing t by Q ; or
- .2 each variable in Q is either free in A or occurs explicitly in S , and C is obtained from A by schematically replacing S by Q ; or
- .3 there are such formulas A' and B' and such variables α and β that t is free in Q , α is indicial in A' , β is accepted in A' , A is obtained from Q by

² To illustrate the flexibility of our theory of notation we have included among our binarians and notarians a good many mathematical symbols seldom used in elementary set theory.

replacing t by A' , B' is obtained from A' by replacing α by β , and C is obtained from Q by replacing t by B' ; or

.4 C is obtained from ' $\wedge xy$ ' by replacing ' x ' by t and ' y ' by A .

0.71 AGREEMENT. S is a *string* if and only if S can be obtained from one of the expressions

$'x', 'xx', 'xx'x'', \dots$

by replacing variables by formulas.

In other words, a string is a linear arrangement or concatenation of formulas.

With the help of 0.9 it can be seen that if S is a string and T is an expression, then (ST) is a string if and only if T is a string.

Also if S is a string which is not a formula, then there are a unique string T and a unique formula A such that S is (TA) . In this connection let us examine some specific expressions. If

R is ' $||x| \rightarrow y|$ ',

T is ' $||x$ ', and

A is ' $| \rightarrow y|$ ',

then because of 0.63, 0.66, and 0.69, R and A are formulas for which R is (TA) ; the expression T , however, is not a string. Accordingly if

S is ' $x||x| \rightarrow y|$ ',

T is ' x ',

A is ' $||x| \rightarrow y|$ ',

T' is ' $x||x$ ', and

A' is ' $| \rightarrow y|$ ',

then

S is a string which is not a formula,

S is (TA) , S is $(T'A')$,

A is a formula, A' is a formula,

T is a formula and therefore a string, but

T' is an expression which is not even a string.

Nevertheless, if S is a string which is not a formula, then there are a unique formula A and a unique expression T for which S is (AT) .

0.72 AGREEMENTS.

.0 S is a *substring* of T if and only if T is a string and S is a string which is an initial segment of T .

.1 T *terminates* with A if and only if A is a formula and T is either A or an expression of the kind (SA) where S is a string.

.2 A is a *subformula* of T if and only if some substring of T terminates with A .

0.73 AGREEMENT. C is *enlisted* by S if and only if there are such A and B that A and B are subformulas of S , C is not a subformula of S , and C is entailed by A and by B .

0.74 AGREEMENT. S is a *demonstration* if and only if S is a string and each subformula of S is either a definition or an axiom or a formula enlisted by some substring of S .

The next rule categorically determines just what expressions are theorems. If we had accepted it earlier as an agreement, then we could have *derived* our rules of inference. It is of interest in this connection that: if A and B are demonstrations, then (AB) is also a demonstration; if S is a demonstration which enlists C , then (SC) is a demonstration.

0.75 RULE. T is a theorem if and only if there is a demonstration of which T is a subformula.

CHAINS

We now take the trouble to make a number of earlier notions quite explicit. It is to be noted that 0.79, 0.81, 0.83, and 0.85, have, in reverse order, the force of a sequence of agreements.

0.76 AGREEMENTS.

0.1 C is a *chain* if and only if C is an expression whose initial symbol is 'I'.

- .1 C' is a *subchain* of C if and only if C is a chain, C' is a chain, and either C' is C or there is such a chain C'' that C is $(C'C'')$.
- .2 C ends with A if and only if A is framed and C is of the kind $(C' A)$ where C' is a chain whose terminal symbol is '■'.
- .3 A is a *link* of C if and only if some subchain of C ends with A .

0.77 AGREEMENTS.

- .0 $(A::B)$ is the expression obtained from ' ixy ' by replacing ' x ' by A and ' y ' by B .
- .1 $(A:B)$ is the expression obtained from ' axy ' by replacing ' x ' by A and ' y ' by B .

We shall use 0.77 in 0.80 and 0.81.

0.78 AGREEMENTS.

- .0 A is a *string-link* of C if and only if A is a link of C and there is such a subchain C' of C that A is not a link of C' and A is of the kind (FB) where F is a formula and B is a link of C' .
- .1 C is a *string-chain* if and only if each link of C is either a formula or a string-link of C .

0.79 RULE. S is a string if and only if there is a string-chain of which S is a link.

0.80 AGREEMENTS.

- .0 A is an *indicial-schematic-link* of C if and only if A is a link of C and A is of the kind $(\alpha::B)$ where α is such a variable and B is such a formula that there are a subchain C' of C , a variable a , a schematic expression S , and a formula B' for which

A is not a link of C' ,
 a appears in S ,
 $(a::B')$ is a link of C' ,
 $(\alpha::B')$ is a link of C' ,

and B can be obtained from B' by schematically replacing S by a formula.

.1 A is an *accepted-schematic-link* of C if and only if A is a link of C and A is of the kind $(\alpha:B)$ where α is such a variable and B is such a formula that there are a subchain C' of C , a variable a , a schematic expression S , and a formula B' for which

A is not a link of C' ,
 a appears in S ,
 $(a::B')$ is a link of C' ,
 $(\alpha:B')$ is a link of C' ,

B can be obtained from B' by schematically replacing S by a formula in which α does not appear.

.2 A is an *indicial-free-link* of C if and only if A is a link of C and A is of the kind $(\alpha::B)$ where α is such a variable and B is such a formula that there are a subchain C' of C , a variable β , and a formula B' for which

A is not a link of C' ,
 $(\alpha::B')$ is a link of C' ,
 β is free in B' ,
 $(\beta:B')$ is a link of C' ,

B can be obtained from B' by replacing β by a formula in which α does not appear.

.3 A is an *accepted-free-link* of C if and only if A is a link of C and A is of the kind $(\alpha:B)$ where α is such a variable and B is such a formula that there are a subchain C' of C , a variable β , and a strict formula B' for which

A is not a link of C' ,
 $(\alpha:B')$ is a link of C' ,
 β is free in B' ,
 $(\beta:B')$ is a link of C' ,

B can be obtained from B' by replacing β by a formula.

.4 A is an *indicial-start* if and only if A is of the kind $(\alpha::F)$ where F is a form and α is a variable which occurs in F more than once.

.5 A is an *accepted-start* if and only if A is of the kind $(\alpha:F)$ where F is a form and α is a variable which occurs in F less than twice.

.6 C is a *double-chain* if and only if C is a chain and each link of C is either an indicial-start, or an accepted-start, or an indicial-schematic-

link of C , or an accepted-schematic-link of C , or an indicial-free-link of C , or an accepted-free-link of C .

0.81 RULES.

.0 α is indicial in A if and only if α is a variable, A is a formula, and $(\alpha::A)$ is a link of some double-chain.

.1 α is accepted in A if and only if α is a variable, A is a formula, and $(\alpha:A)$ is a link of some double-chain.

0.82 AGREEMENTS.

A is a *free-link* of C if and only if A is a link of C and A is of the kind (αB) where α is such a variable and B is such an expression that there are a subchain C' of C , a variable β , an expression B' , and an expression B'' for which

A is not a link of C' ,

$(\beta B')$ is a link of C' ,

$(\alpha B')$ is a link of C' ,

$(\alpha B'')$ is a link of C' ,

and B is obtained from B' either by replacing β by B'' or by schematically replacing some schematic expression by B'' .

.1 A is a *free-start* if and only if A is of the kind (αF) where F is a form and α is a variable which occurs in F less than twice.

Thus if α is a variable and F is a form, then (αF) is a free-start if and only if $(\alpha:F)$ is an accepted-start.

.2 C is a *free-chain* if and only if C is a chain and each link of C is either a free-start or a free-link of C .

0.83 RULE. α is free in A if and only if α is a variable, A is an expression, and (αA) is a link of some free-chain.

0.84 AGREEMENTS.

.0 A is a *parenthetical-link* of C if and only if A is a link of C and there are such a subchain C' of C and such a link B of C' that

A is not a link of C'

and A can be obtained from B by replacing a symbol which is not a parenthesis by a parenthetic expression.

.1 C is a *parenthetical-chain* if and only if C is a chain and each link of C either is an expression devoid of parentheses, or is a parenthetical-link of C .

0.85 RULE. S is parenthetical if and only if there is a parenthetical-chain of which S is a link.

CHAPTER 1

LOGIC

1.0 DEFINITIONS.

- .0 (If p then $q \equiv (p \rightarrow q)$)
- .1 ($(x \text{ implies } y) \equiv (x \rightarrow y)$)
- .2 (For each x , $\underline{u}x \equiv \wedge x \underline{u}x$)
- .3 ($0 \equiv \wedge xx$)
- .4 ($\sim p \equiv (p \rightarrow 0)$)
- .5 (Not $p \equiv \sim p$)
- .6 ($U \equiv \sim 0$)
- .7 ($((p \wedge q) \equiv \sim (p \rightarrow \sim q))$)
- .8 ($((p \text{ and } q) \equiv (p \wedge q))$)
- .9 ($((p \leftrightarrow q) \equiv ((p \rightarrow q) \wedge (q \rightarrow p))$)
- .10 ($((p \text{ if and only if } q) \equiv (p \leftrightarrow q))$)
- .11 ($((x \text{ is equivalent to } y) \equiv (x \leftrightarrow y))$)
- .12 ($((p \vee q) \equiv \sim (\sim p \wedge \sim q))$)
- .13 ($((p \text{ or } q) \equiv (p \vee q))$)
- .14 ($((p \sim q) \equiv (p \wedge \sim q))$)
- .15 ($((p \text{ not } q) \equiv (p \sim q))$)
- .16 ($(\vee x \underline{u}x \equiv \sim \wedge x \sim \underline{u}x)$)
- .17 (For some x , $\underline{u}x \equiv \vee x \underline{u}x$)

With Remark A.28 in mind we make

1.1 DEFINITIONAL AXIOMS FOR LOGIC.

- .0 ($0 \equiv \wedge xx$)
- .1 ($\sim p \equiv (p \rightarrow 0)$)

1.2 AXIOMS OF DEFINITION FOR LOGIC.

$$.0 \quad ((x \equiv y) \rightarrow (x \rightarrow y))$$

$$.1 \quad ((x \equiv y) \rightarrow (y \rightarrow x))$$

1.3 AXIOMS FOR LOGIC.

$$.0 \quad (\wedge x \underline{u}x \rightarrow \underline{u}x)$$

$$.1 \quad (y \rightarrow \wedge xy)$$

$$.2 \quad (\wedge x(\underline{u}x \rightarrow \underline{v}x) \rightarrow (\wedge x \underline{u}x \rightarrow \wedge x \underline{v}x))$$

$$.3 \quad ((p \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (p \rightarrow r)))$$

$$.4 \quad ((p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r)))$$

$$.5 \quad ((\sim p \rightarrow p) \rightarrow p)$$

1.4 THEOREMS.

$$.0 \quad (0 \rightarrow p)$$

Proof. Because of 1.1.0 and initiation

$$(0 \equiv \wedge xx).$$

Initiating 1.2.0, we learn

$$((x \equiv y) \rightarrow (x \rightarrow y))$$

and from this we learn successively by substitution that

$$((0 \equiv y) \rightarrow (0 \rightarrow y))$$

and

$$((0 \equiv \wedge xx) \rightarrow (0 \rightarrow \wedge xx)).$$

By detachment

$$(0 \rightarrow \wedge xx).$$

Initiating 1.3.0 we learn

$$(\wedge x \underline{u}x \rightarrow \underline{u}x)$$

and from this we learn by schematic substitution that

$$(\wedge xx \rightarrow x).$$

Initiating 1.3.3 we learn

$$((p \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (p \rightarrow r)))$$

and from this we learn successively by substitution that

$$((0 \rightarrow q) \rightarrow ((q \rightarrow r) \rightarrow (0 \rightarrow r))),$$

$$((0 \rightarrow q) \rightarrow ((q \rightarrow x) \rightarrow (0 \rightarrow x))),$$

and

$$((0 \rightarrow \wedge xx) \rightarrow ((\wedge xx \rightarrow x) \rightarrow (0 \rightarrow x))).$$

By detachment

$$((\wedge xx \rightarrow x) \rightarrow (0 \rightarrow x)),$$

and again by detachment

$$(0 \rightarrow x).$$

By substitution we conclude

$$(0 \rightarrow p).$$

We shall never again be so detailed in our proofs.

.1 $(\sim p \rightarrow (p \rightarrow q))$

Proof. Because of 1.1.1, 1.2.0, and 1.3.3,

$$(\sim p \rightarrow (p \rightarrow 0))$$

and

$$((p \rightarrow 0) \rightarrow ((0 \rightarrow q) \rightarrow (p \rightarrow q))).$$

Substituting in 1.3.3 and detaching twice, we see

$$(\sim p \rightarrow ((0 \rightarrow q) \rightarrow (p \rightarrow q))).$$

Consequently, because of 1.3.4,

$$((0 \rightarrow q) \rightarrow (\sim p \rightarrow (p \rightarrow q))).$$

Now employ .0 and detach the desired result.

.2 $(p \rightarrow (\sim p \rightarrow q))$

Proof. Use .1 and 1.3.4.

A remarkable discovery by Łukasiewicz is that all ordinary sentential theorems which can be written down in terms of parentheses, variables, ' $\sim$ ', and ' $\rightarrow$ ' can be inferred from 1.3.3, 1.4.2, and 1.3.5 by using only substitution and detachment.

.3 $(p \rightarrow p)$

Proof. Using .2, we see

$$(p \rightarrow (\sim p \rightarrow p)).$$

Initiating 1.3.5, we see

$$((\sim p \rightarrow p) \rightarrow p).$$

Substituting in 1.3.3 and detaching twice, we complete the proof.

.4 ~ 0

Proof. Use 1.2.1, 1.1.1, and .0.

.5 U

.6 $(p \rightarrow ((p \rightarrow q) \rightarrow q))$

Proof. Substituting in .3, we see

$$((p \rightarrow q) \rightarrow (p \rightarrow q)).$$

Substituting in 1.3.4, we see

$$(((p \rightarrow q) \rightarrow (p \rightarrow q)) \rightarrow (p \rightarrow ((p \rightarrow q) \rightarrow q))).$$

Detachment in the above completes the proof.

.7 $((q \rightarrow r) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)))$

Hint. Apply 1.3.4 to 1.3.3.

Remark. Because of 1.3.3, 1.3.4, and 1.4.7, we now feel free to use the first four supplementary rules of inference found in 1.11.

.8 $(\sim \sim p \rightarrow p)$

Proof. Use .1 to see

$$\begin{aligned} &(\sim \sim p \rightarrow (\sim p \rightarrow q)) \\ &(\sim \sim p \rightarrow (\sim p \rightarrow p)). \end{aligned}$$

The above, 1.3.5, and sorites (1.11.0) complete the proof.

.9 $((p \rightarrow \sim p) \rightarrow \sim p)$

Proof. Because of .8 and inflow (1.11.2),

$$((p \rightarrow \sim p) \rightarrow (\sim \sim p \rightarrow \sim p)).$$

Substituting in 1.3.5, we see

$$((\sim \sim p \rightarrow \sim p) \rightarrow \sim p).$$

Use of sorites completes the proof.

.10 $(p \rightarrow \sim \sim p)$

Hint. Use .2 to see

$$(p \rightarrow (\sim p \rightarrow 0)).$$

.11 $((p \rightarrow q) \rightarrow (\sim q \rightarrow \sim p))$

Proof. The proof culminates in Step 3.

Step 0. $((p \rightarrow q) \rightarrow ((q \rightarrow 0) \rightarrow (p \rightarrow 0)))$

Step 1. $((((q \rightarrow 0) \rightarrow (p \rightarrow 0)) \rightarrow (\sim q \rightarrow (p \rightarrow 0))))$

Proof. Because of 1.0.4 and 1.2.0,

$$(\sim q \rightarrow (q \rightarrow 0))$$

and the desired conclusion follows by inflow.

Step 2. $((\sim q \rightarrow (p \rightarrow 0)) \rightarrow (\sim q \rightarrow \sim p))$

Proof. Because of 1.0.4 and 1.2.1,

$$((p \rightarrow 0) \rightarrow \sim p)$$

and the desired conclusion follows by outflow (1.11.3).

Step 0–Step 2 telescope soritically to yield

$$\text{Step 3. } ((p \rightarrow q) \rightarrow (\sim q \rightarrow \sim p))$$

$$\text{.12 } ((\sim p \rightarrow \sim q) \rightarrow (q \rightarrow p))$$

Proof. The proof culminates in Step 3.

$$\text{Step 0. } ((\sim p \rightarrow \sim q) \rightarrow (\sim \sim q \rightarrow \sim \sim p))$$

Proof. Use .11.

$$\text{Step 1. } ((\sim \sim q \rightarrow \sim \sim p) \rightarrow (\sim \sim q \rightarrow p))$$

Proof. Use .8 and outflow.

$$\text{Step 2. } ((\sim \sim q \rightarrow p) \rightarrow (q \rightarrow p))$$

Proof. Use .10 and inflow.

Step 0–Step 2 telescope soritically to yield

$$\text{Step 3. } ((\sim p \rightarrow \sim q) \rightarrow (q \rightarrow p))$$

$$\text{.13 } (p \rightarrow (q \rightarrow p))$$

Proof. Use .2, .12, and sorites.

$$\text{.14 } (p \rightarrow (q \rightarrow (p \wedge q)))$$

Proof. The proof culminates in Step 4.

$$\text{Step 0. } (p \rightarrow ((p \rightarrow \sim q) \rightarrow \sim q))$$

Proof. Use .6.

Step 1. $((p \rightarrow \sim q) \rightarrow \sim q) \rightarrow (\sim \sim q \rightarrow \sim(p \rightarrow \sim q))$

Proof. Use .11.

Step 2. $((\sim \sim q \rightarrow \sim(p \rightarrow \sim q)) \rightarrow (q \rightarrow \sim(p \rightarrow \sim q)))$

Proof. Use .10 and inflow.

Step 3. $((q \rightarrow \sim(p \rightarrow \sim q)) \rightarrow (q \rightarrow (p \wedge q)))$

Proof. Because of 1.0.7 and 1.2.1, we see

$$(\sim(p \rightarrow \sim q) \rightarrow (p \wedge q)).$$

Outflow yields the desired conclusion.

Step 0–Step 3 telescope soritically to yield

Step 4. $(p \rightarrow (q \rightarrow (p \wedge q)))$

Remark. Because of .13 and .14, we now feel free to use the first six supplementary rules of inference found in 1.11.

.15 $((p \wedge q) \rightarrow p)$

Proof. The proof culminates in Step 3.

Step 0. $((p \wedge q) \rightarrow \sim(p \rightarrow \sim q))$

Proof. Use 1.0.7 and 1.2.0.

Step 1. $(\sim p \rightarrow (p \rightarrow \sim q))$

Proof. Use .1.

Step 2. $(\sim(p \rightarrow \sim q) \rightarrow \sim \sim p)$

Proof. Use Step 1 and .11 and detachment.

Step 0, Step 2, and .8 telescope soritically to yield

Step 3. $((p \wedge q) \rightarrow p)$

.16 $((p \wedge q) \rightarrow q)$

Proof. The proof culminates in Step 3. As in the proof of .15 we find again

Step 0. $((p \wedge q) \rightarrow \sim(p \rightarrow \sim q))$

Step 1. $(\sim q \rightarrow (p \rightarrow \sim q))$

Proof. Use .13.

Step 2. $(\sim(p \rightarrow \sim q) \rightarrow \sim \sim q)$

Proof. Use Step 1 and .11 and detachment.

Step 0, Step 2, and .8 telescope soritically to yield

Step 3. $((p \wedge q) \rightarrow q)$

Remark. It will become increasingly important to *correctly* understand 0.37 especially if our approach here to Set Theory is to be carried forward into aspects of Elementary Analysis, Metrics, Measure, Linear Measure and Total Variation, Integration, Covering and Differentiation. In particular note that

$$((p \rightarrow q \rightarrow r \rightarrow s) \equiv ((p \rightarrow q) \wedge (p \rightarrow r) \wedge (p \rightarrow s))).$$

We do *not* assert, in the spirit of sorites, that

$$((p \rightarrow q \rightarrow r \rightarrow s) \equiv ((p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow s))),$$

although it is of real interest that

$$(((p \rightarrow q) \wedge (q \rightarrow r) \wedge (r \rightarrow s)) \rightarrow (p \rightarrow q \rightarrow r \rightarrow s)).$$

In connection with soritic reasoning we easily see that

$$((s \rightarrow t) \rightarrow ((p \rightarrow q \rightarrow r \rightarrow s) \rightarrow (p \rightarrow q \rightarrow r \rightarrow s \rightarrow t))),$$

but a much weaker antecedent implies the same consequent. In fact, because of forthcoming .19,

$$((p \rightarrow (s \rightarrow t)) \rightarrow ((p \rightarrow q \rightarrow r \rightarrow s) \rightarrow (p \rightarrow q \rightarrow r \rightarrow s \rightarrow t))).$$

Remark. In trying to prove .17 below we write down a single formula which, when properly understood, should make us believe .17 is a theorem. That the formula written in proof is a theorem follows by intuitive induction from 1.3.3 and .14. That this formula yields .17 is a consequence of .16.

.17 $((p \rightarrow (p \rightarrow q)) \rightarrow (p \rightarrow q))$

Proof. In the reasoning below use is made of .11 and outflow, 1.3.4, .9 and outflow, .12.

$$\begin{aligned} & ((p \rightarrow (p \rightarrow q)) \rightarrow (p \rightarrow (\sim q \rightarrow \sim p))) \\ & \rightarrow (\sim q \rightarrow (p \rightarrow \sim p)) \\ & \rightarrow (\sim q \rightarrow \sim p) \\ & \rightarrow (p \rightarrow q) \end{aligned}$$

.18 $(p \rightarrow (p \wedge p))$

Hint. Use .14 and .17.

.19 $((p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)))$

Proof. In the reasoning below use is made of 1.3.4, .7, .17, and outflow.

$$\begin{aligned} & ((p \rightarrow (q \rightarrow r)) \rightarrow (q \rightarrow (p \rightarrow r))) \\ & \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow (p \rightarrow r))) \\ & \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r)) \end{aligned}$$

.20 $((p \rightarrow (q \rightarrow r)) \rightarrow ((p \wedge q) \rightarrow r))$

Proof. In the reasoning below use is made of .15 and inflow, 1.3.4, .16 and inflow, .17.

$$\begin{aligned} & ((p \rightarrow (q \rightarrow r)) \rightarrow ((p \wedge q) \rightarrow (q \rightarrow r))) \\ & \rightarrow (q \rightarrow ((p \wedge q) \rightarrow r)) \\ & \rightarrow ((p \wedge q) \rightarrow ((p \wedge q) \rightarrow r)) \\ & \rightarrow ((p \wedge q) \rightarrow r) \end{aligned}$$

.21 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Hint. Use .20 and 1.3.3.

.22 $(p \rightarrow (q \rightarrow (q \wedge p)))$

Hint. Use 1.3.4 and .14.

$$\textbf{.23 } ((p \wedge q) \rightarrow (q \wedge p))$$

Hint. Use .20 and .22.

$$\textbf{.24 } ((p \rightarrow (q \rightarrow (r \rightarrow s))) \rightarrow ((p \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (p \rightarrow s))))$$

Proof. Because of .19 and then because of .19 and outflow,

$$((p \rightarrow (q \rightarrow (r \rightarrow s))) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow (r \rightarrow s))) \rightarrow ((p \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (p \rightarrow s)))).$$

$$\textbf{.25 } ((p \rightarrow q) \rightarrow ((p \rightarrow r) \rightarrow (p \rightarrow (q \wedge r))))$$

Hint. In .24 replace 's' by '(q ∧ r)'; then use .14 and consequence (1.11.4) and detachment.

$$\textbf{.26 } ((p \rightarrow p') \rightarrow ((q \rightarrow q') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))))$$

Proof. The proof culminates in Step 7.

$$\textbf{Step 0. } ((p \rightarrow p') \rightarrow ((p \wedge q) \rightarrow p'))$$

Proof. Use .15 and inflow.

$$\textbf{Step 1. } ((q \rightarrow q') \rightarrow ((p \wedge q) \rightarrow q'))$$

Proof. Use .16 and inflow.

Substitution in .25 yields:

$$\textbf{Step 2. } ((p \rightarrow p') \rightarrow ((p \rightarrow q') \rightarrow (p \rightarrow (p' \wedge q'))))$$

Substitution in Step 2 yields:

$$\textbf{Step 3. } (((p \wedge q) \rightarrow p') \rightarrow (((p \wedge q) \rightarrow q') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))))$$

Step 0 and Step 3 and sorites yield:

Step 4. $((p \rightarrow p') \rightarrow (((p \wedge q) \rightarrow q') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))))$

Step 4 and commutation (1.11.1) yield:

Step 5. $((((p \wedge q) \rightarrow q') \rightarrow ((p \rightarrow p') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q')))))$

Step 1 and Step 5 and sorites yield:

Step 6. $((q \rightarrow q') \rightarrow ((p \rightarrow p') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))))$

Step 6 and commutation yield:

Step 7. $((p \rightarrow p') \rightarrow ((q \rightarrow q') \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))))$

Because of .26, .20 yields very useful:

.27 $((p \rightarrow p') \wedge (q \rightarrow q')) \rightarrow ((p \wedge q) \rightarrow (p' \wedge q'))$

.28 $((p \rightarrow q) \wedge (p \rightarrow r)) \rightarrow (p \rightarrow (q \wedge r))$

Proof. In the reasoning below use is made of .27, .18, and inflow.

$$\begin{aligned} ((p \rightarrow q) \wedge (p \rightarrow r)) &\rightarrow ((p \wedge p) \rightarrow (q \wedge r)) \\ &\rightarrow (p \rightarrow (q \wedge r)) \end{aligned}$$

Remark. Because of .27 and .28, we now feel free to use the first eight of the supplementary rules of inference found in 1.11.

.29 $((p \rightarrow (q \wedge r)) \rightarrow ((p \rightarrow q) \wedge (p \rightarrow r)))$

Proof. Because of .15 and outflow,

$$((p \rightarrow (q \wedge r)) \rightarrow (p \rightarrow q)).$$

Because of .16 and outflow,

$$((p \rightarrow (q \wedge r)) \rightarrow (p \rightarrow r)).$$

Because of semiconjuncture (1.11.7), the desired result is at hand.

1.5 THEOREMS.

$$.0 \quad ((p \leftrightarrow q) \rightarrow ((p \rightarrow q) \wedge (q \rightarrow p)))$$

Hint. Use 1.0.9 and 1.2.0.

$$.1 \quad ((p \leftrightarrow q) \rightarrow (p \rightarrow q))$$

$$.2 \quad ((p \leftrightarrow q) \rightarrow (q \rightarrow p))$$

$$.3 \quad (((p \rightarrow q) \wedge (q \rightarrow p)) \rightarrow (p \leftrightarrow q))$$

Hint. Use 1.0.9 and 1.2.1.

Remark. Because .1, .2, and .3, we now feel free to use the first ten of the rules of inference found in 1.11.

$$.4 \quad ((p \equiv q) \rightarrow (p \leftrightarrow q))$$

Proof. Because of 1.2.0, 1.2.1, and semiconjuncture,

$$\begin{aligned} &((p \equiv q) \rightarrow (p \rightarrow q)) \\ &((p \equiv q) \rightarrow (q \rightarrow p)) \\ &((p \equiv q) \rightarrow ((p \rightarrow q) \wedge (q \rightarrow p))). \end{aligned}$$

The above result and .3 telescope soritically to yield the desired result.

Because of .0, .3, and equivalence (1.11.9) or .4, 1.0.9 and detachment

$$((p \leftrightarrow q) \leftrightarrow ((p \rightarrow q) \wedge (q \rightarrow p))).$$

This, because of 0.48, simplifies into

$$.5 \quad ((p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p))$$

The following theorems of 1.4 can similarly be simplified: .14, .15, .16, .18, .20, .22, .23, .25, .26, .27, .28, and .29. Because of 0.48 and sometimes sorites, inflow, or outflow, these simplifications are also theorems. We shall henceforth feel free to use simplifications engendered by 0.48 without comment.

$$.6 \quad ((p \rightarrow (q \rightarrow r)) \leftrightarrow (q \rightarrow (p \rightarrow r)))$$

Hint. Use 1.3.4, a variant of 1.3.4, and equivalence.

Unlike 1.3.4, 1.3.5 is used but thrice previously and now for the last time in checking

.7 $((\sim p \rightarrow p) \leftrightarrow p)$

Hint. Use 1.3.5 and 1.4.2.

.8 $(p \leftrightarrow p)$

Hint. Use 1.4.3.

.9 $((p \rightarrow \sim p) \leftrightarrow \sim p)$

Hint. Use 1.4.9. and 1.4.1.

.10 $(p \leftrightarrow \sim \sim p)$

Hint. Use 1.4.10. and 1.4.8.

.11 $((p \rightarrow (p \rightarrow q)) \leftrightarrow (p \rightarrow q))$

Hint. Use 1.4.17. and 1.4.13.

.12 $((p \rightarrow q) \leftrightarrow (\sim q \rightarrow \sim p))$

Hint. Use 1.4.11. and 1.4.12.

.13 $(p \leftrightarrow p \wedge p)$

Hint. Use 1.4.18. and 1.4.15.

.14 $(p \wedge q \leftrightarrow q \wedge p)$

Hint. Use 1.4.23.

.15 $((p \rightarrow q) \wedge (p \rightarrow r) \leftrightarrow (p \rightarrow q \wedge r))$

Hint. Use 1.4.28. and 1.4.29.

.16 $(p \rightarrow (p \rightarrow r) \leftrightarrow r)$

Hint. Use 1.4.6, 1.4.13, consequence, semiconjuncture, and .3.

.17 $((p \leftrightarrow q) \leftrightarrow (q \leftrightarrow p))$

Proof.

$$\begin{aligned}(p \leftrightarrow q &\rightarrow (p \rightarrow q) \wedge (q \rightarrow p)) \\ (p \leftrightarrow q &\rightarrow (q \rightarrow p) \wedge (p \rightarrow q)) \\ (p \leftrightarrow q &\rightarrow q \leftrightarrow p)\end{aligned}$$

From this, a variant, and equivalence, we arrive at the desired result.

$$.18 \ ((p \leftrightarrow q) \wedge (q \leftrightarrow r) \rightarrow p \leftrightarrow r)$$

Proof. By .1, conjuncture, 1.4.21, .2, semiconjuncture, and .3,

$$\begin{aligned}((p \leftrightarrow q) \wedge (q \leftrightarrow r) &\rightarrow (p \rightarrow r)) \\ ((p \leftrightarrow q) \wedge (q \leftrightarrow r) &\rightarrow (r \rightarrow p)) \\ ((p \leftrightarrow q) \wedge (q \leftrightarrow r) &\rightarrow p \leftrightarrow r).\end{aligned}$$

Remark. Because of .18, we now feel free to use the first eleven rules of inference found in 1.11.

The next theorem follows from 0.32.

$$.19 \ ((p \leftrightarrow q \leftrightarrow r) \equiv ((p \leftrightarrow q) \wedge (q \leftrightarrow r)))$$

$$.20 \ (p \leftrightarrow q \leftrightarrow r \rightarrow p \leftrightarrow r)$$

$$.21 \ ((p \leftrightarrow q) \leftrightarrow (\sim p \leftrightarrow \sim q))$$

Proof. By 1.4.11, a variant of 1.4.11, and conjuncture (1.11.6),

$$((p \rightarrow q) \wedge (q \rightarrow p) \rightarrow (\sim q \rightarrow \sim p) \wedge (\sim p \rightarrow \sim q)).$$

Similarly, by 1.4.12,

$$((\sim q \rightarrow \sim p) \wedge (\sim p \rightarrow \sim q) \rightarrow (p \rightarrow q) \wedge (q \rightarrow p)).$$

By equivalence,

$$((p \rightarrow q) \wedge (q \rightarrow p) \leftrightarrow (\sim q \rightarrow \sim p) \wedge (\sim p \rightarrow \sim q)).$$

By .5, equivalence-transitivity (1.11.10), .5 again, and equivalence-transitivity again, the desired result is obtained.

$$.22 \ (p \leftrightarrow p' \rightarrow (p \rightarrow r) \leftrightarrow (p' \rightarrow r))$$

Hint. Use two variants of 1.3.3, conjuncture, 1.4.23, .5, and .3.

$$.23 \ (q \leftrightarrow q' \rightarrow (r \rightarrow q) \leftrightarrow (r \rightarrow q'))$$

Hint. Use 1.4.7.

$$\mathbf{.24} \quad ((p \rightarrow (q \rightarrow r)) \leftrightarrow (p \wedge q \rightarrow r))$$

Proof. Use .12, .23, .6, equivalence-transitivity, .17, .10, 1.0.7, .4, and .22 in checking

$$\begin{aligned} ((p \rightarrow (q \rightarrow r)) &\leftrightarrow (p \rightarrow (\sim r \rightarrow \sim q))) \\ &\leftrightarrow (\sim r \rightarrow (p \rightarrow \sim q)) \\ &\leftrightarrow (\sim (p \rightarrow \sim q) \rightarrow \sim \sim r) \\ &\leftrightarrow (\sim (p \rightarrow \sim q) \rightarrow r) \\ &\leftrightarrow (p \wedge q \rightarrow r)). \end{aligned}$$

$$\mathbf{.25} \quad ((p \leftrightarrow p') \wedge (q \leftrightarrow q') \rightarrow (p \rightarrow q) \leftrightarrow (p' \rightarrow q'))$$

Proof. Use .22, .23, conjuncture, and .18 to get

$$\begin{aligned} (p \leftrightarrow p' \rightarrow (p \rightarrow q) &\leftrightarrow (p' \rightarrow q)) \\ (q \leftrightarrow q' \rightarrow (p' \rightarrow q) &\leftrightarrow (p' \rightarrow q')) \\ ((p \leftrightarrow p') \wedge (q \leftrightarrow q') &\rightarrow (p \rightarrow q) \leftrightarrow (p' \rightarrow q')). \end{aligned}$$

$$\mathbf{.26} \quad ((p \leftrightarrow p') \wedge (q \leftrightarrow q') \rightarrow p \wedge q \leftrightarrow p' \wedge q').$$

Hint. Use .1, .2, conjuncture, 1.4.27, semiconjuncture, and .3 or use .21, .25, and 1.0.7.

$$\mathbf{.27} \quad ((p \leftrightarrow p') \wedge (q \leftrightarrow q') \rightarrow (p \leftrightarrow q) \leftrightarrow (p' \leftrightarrow q'))$$

Hint. Use .25, .14, semiconjuncture, .26, .5, and .18.

$$\mathbf{.28} \quad (p \wedge (q \wedge r) \leftrightarrow (p \wedge q) \wedge r \leftrightarrow p \wedge q \wedge r)$$

Hint. Use 1.0.7 directly or use .24.

1.6 THEOREMS.

$$\mathbf{.0} \quad (p \vee q \leftrightarrow \sim (\sim p \wedge \sim q))$$

$$\mathbf{.1} \quad (\sim (p \vee q) \leftrightarrow \sim p \wedge \sim q)$$

$$\mathbf{.2} \quad (\sim (p \wedge q) \leftrightarrow \sim p \vee \sim q)$$

$$\mathbf{.3} \quad (p \vee q \leftrightarrow (\sim p \rightarrow q))$$

$$\mathbf{.4} \quad ((p \rightarrow q) \leftrightarrow \sim p \vee q)$$

- .5 $(p \vee q \leftrightarrow q \vee p)$
 .6 $(p \vee (q \vee r) \leftrightarrow (p \vee q) \vee r \leftrightarrow p \vee q \vee r)$
 .7 $(p \rightarrow p \vee q)$
 .8 $(p \leftrightarrow p \vee p)$
 .9 $((p \wedge q) \vee r) \leftrightarrow p \wedge q \vee r \leftrightarrow p \vee r \wedge q \vee r)$

Hint. Use .5, .3, 1.5.15, and 1.5.26.

- .10 $((p \vee q) \wedge r) \leftrightarrow p \vee q \wedge r \leftrightarrow p \wedge r \vee q \wedge r)$
 .11 $((p \rightarrow q) \wedge (p' \rightarrow q') \rightarrow (p \vee p' \rightarrow q \vee q'))$

Hint. Use 1.4.27.

- .12 $((p \leftrightarrow q) \wedge (p' \leftrightarrow q') \rightarrow p \vee p' \leftrightarrow q \vee q')$

Hint. Make a proof like that of 1.5.26 using .3.

- .13 $((p \rightarrow r) \wedge (q \rightarrow r) \leftrightarrow (p \vee q \rightarrow r))$

Hint. Use .11 and 1.4.29 or base an equivalence-transitive proof on 1.5.15.

1.7 THEOREMS.

- .0 $(p \sim q \leftrightarrow p \wedge \sim q)$
 .1 $((p \rightarrow q \sim r) \leftrightarrow (p \rightarrow (q \sim r)))$
 .2 $(p \wedge q \sim r \leftrightarrow p \wedge (q \sim r))$
 .3 $((p \leftrightarrow q \sim r) \leftrightarrow (p \leftrightarrow (q \sim r)))$
 .4 $(p \vee q \sim r \leftrightarrow p \vee (q \sim r))$
 .5 $(p \sim q \sim r \leftrightarrow p \wedge \sim q \wedge \sim r)$
 .6 $(\sim(p \sim q) \leftrightarrow (p \rightarrow q))$
 .7 $(p \sim \sim q \leftrightarrow p \wedge q)$
 .8 $(\sim(p \sim q) \sim (q \sim p) \leftrightarrow (p \leftrightarrow q))$
 .9 $(\sim(\sim p \sim q) \leftrightarrow p \vee q)$
 .10 $((p \leftrightarrow p') \wedge (q \leftrightarrow q') \rightarrow p \sim q \leftrightarrow p' \sim q')$

The somewhat detailed proofs in 1.8 illustrate the use of all of the substitution rules. Inference by universalization is first encountered in

1.8.4. In 1.8 and 1.9 we shall feel free to use a distribution rule (1.11.11) after a theorem on which it is based has been established.

1.8 THEOREMS.

.0 $(\wedge y \underline{v}y \rightarrow \underline{v}s)$

Proof.

$$(\wedge x \underline{u}x \rightarrow \underline{u}x)$$

$$(\wedge x \underline{v}x \rightarrow \underline{v}x)$$

$$(\wedge y \underline{v}y \rightarrow \underline{v}x)$$

$$(\wedge y \underline{v}y \rightarrow \underline{v}s)$$

.1 $(\wedge x \underline{v}x \rightarrow \underline{v}s)$

.2 $(\wedge s \underline{v}s \rightarrow \underline{v}s)$

.3 $(\wedge xy \leftrightarrow y)$

.4 $(\wedge x(\underline{u}x \wedge \underline{v}x) \leftrightarrow \wedge x \underline{u}x \wedge \wedge x \underline{v}x)$

Proof. By successive substitution starting with 1.4.16,

$$(\underline{u}x \wedge \underline{v}x \rightarrow \underline{v}x).$$

By universalization, schematic substitution in 1.3.2, and detachment,

$$(\wedge x(\underline{u}x \wedge \underline{v}x) \rightarrow \wedge x \underline{v}x).$$

Hence, because of a similarly proved theorem and semiconjuncture,

$$(\wedge x(\underline{u}x \wedge \underline{v}x) \rightarrow \wedge x \underline{u}x \wedge \wedge x \underline{v}x).$$

By 1.4.14 and substitution twice,

$$(\underline{u}x \rightarrow (\underline{v}x \rightarrow \underline{u}x \wedge \underline{v}x)).$$

By universal distribution (1.11.11.0),

$$(\wedge x \underline{u}x \rightarrow \wedge x(\underline{v}x \rightarrow \underline{u}x \wedge \underline{v}x)).$$

By schematic substitution in a version of 1.3.2 and sorites,

$$(\wedge x \underline{u}x \rightarrow (\wedge x \underline{v}x \rightarrow \wedge x(\underline{u}x \wedge \underline{v}x))).$$

Hence, by specialization of 1.4.20 and detachment,

$$(\wedge x \underline{u}x \wedge \wedge x \underline{v}x \rightarrow \wedge x(\underline{u}x \wedge \underline{v}x)).$$

Finally, by equivalence,

$$(\wedge x(\underline{u}x \wedge \underline{v}x) \leftrightarrow \wedge x \underline{u}x \wedge \wedge x \underline{v}x).$$

.5 $(\wedge x \wedge y \underline{u}'xy \leftrightarrow \wedge y \wedge x \underline{u}'xy)$

Proof.

$$\begin{aligned} & (\wedge s \underline{u}'xs \rightarrow \underline{u}'xs) \\ & (\wedge s \wedge x \wedge s \underline{u}'xs \rightarrow \wedge s \wedge x \underline{u}'xs) \\ & (\wedge s \wedge x \wedge y \underline{u}'xy \rightarrow \wedge y \wedge x \underline{u}'xy) \\ & (\wedge x \wedge y \underline{u}'xy \rightarrow \wedge s \wedge x \wedge y \underline{u}'xy \\ & \quad \rightarrow \wedge y \wedge x \underline{u}'xy) \\ & (\wedge x \wedge y \underline{u}'xy \rightarrow \wedge y \wedge x \underline{u}'xy) \\ & (\wedge x \wedge s \underline{u}'xs \rightarrow \wedge y \wedge s \underline{u}'sy) \\ & (\wedge y \wedge x \underline{u}'yx \rightarrow \wedge x \wedge y \underline{u}'yx) \\ & (\wedge y \wedge x \underline{u}'xy \rightarrow \wedge x \wedge y \underline{u}'xy) \\ & (\wedge x \wedge y \underline{u}'xy \leftrightarrow \wedge y \wedge x \underline{u}'xy) \end{aligned}$$

.6 $(\wedge x(\underline{u}x \leftrightarrow \underline{v}x) \rightarrow \wedge x \underline{u}x \leftrightarrow \wedge x \underline{v}x)$

Hint. Use a version of 1.5.0, 1.11.11.0, a specialization of .4, 1.3.2, a version of 1.3.2, conjuncture, a specialization of 1.5.3, and sorites thrice.

1.9 THEOREMS.

.0 $(\forall x \underline{u}x \leftrightarrow \sim \wedge x \sim \underline{u}x)$

.1 $(\underline{u}x \rightarrow \forall x \underline{u}x)$

.2 $(\wedge x(\underline{u}x \rightarrow \underline{v}x) \rightarrow (\forall x \underline{u}x \rightarrow \forall x \underline{v}x))$

Hint. Use 1.4.11, 1.11.11.0, 1.3.2, 1.4.11 again, and .0.

.3 $(\wedge x(\underline{u}x \vee \underline{v}x) \rightarrow \forall x \underline{u}x \vee \wedge x \underline{v}x)$

.4 $(\wedge x(\underline{u}x \leftrightarrow \underline{v}x) \rightarrow \forall x \underline{u}x \leftrightarrow \forall x \underline{v}x)$

Hint. Use 1.11.11.0, 1.8.4, 1.3.2 twice, and .0 or use 1.11.11.0, 1.8.4 and .2 twice.

.5 $(\wedge x \underline{u}x \leftrightarrow \sim \forall x \sim \underline{u}x)$

Hint. Use .0 and 1.8.6.

.6 $(\forall x(\underline{u}x \vee \underline{v}x) \leftrightarrow \forall x \underline{u}x \vee \forall x \underline{v}x)$

Hint. Use 1.8.4, 1.11.11.2, and .0.

.7 $(\wedge x \underline{u}x \rightarrow \forall x \underline{u}x)$

Hint. Use 1.3.0 and .1.

.8 $(\forall xy \leftrightarrow y)$

Hint. Use 1.3.1, .0, and .1.

.9 $(\wedge x \underline{u}x \wedge \forall x \underline{v}x \rightarrow \forall x(\underline{u}x \wedge \underline{v}x))$

Hint. Use 1.4.14, 1.11.11.0, .2, and 1.4.20.

.10 $(\forall x \forall y \underline{u}'xy \leftrightarrow \forall y \forall x \underline{u}'xy)$

Hint. Use 1.8.5, .0, and .4.

.11 $(\forall x(\underline{u}x \rightarrow \underline{v}x) \leftrightarrow (\wedge x \underline{u}x \rightarrow \forall x \underline{v}x))$

Hint. Use 1.11.11.3, .6, and .5.

.12 $(\forall x(\underline{u}x \wedge \underline{v}x) \rightarrow \forall x \underline{u}x \wedge \forall x \underline{v}x)$

Hint. Use .1, conjuncture, 1.11.11.1, and .8.

.13 $(\forall x \forall y \underline{u}'xy \rightarrow \wedge y \forall x \underline{u}'xy)$

.14 $(\wedge x(\underline{u}x \frown \underline{v}x) \leftrightarrow \wedge x \underline{u}x \frown \forall x \underline{v}x)$

.15 $(\forall x(\underline{u}x \frown \underline{v}x) \rightarrow \forall x \underline{u}x \frown \wedge x \underline{v}x)$

1.10 THEOREMS.

.0 $(\wedge x(y \wedge \underline{u}x) \leftrightarrow y \wedge \forall x \underline{u}x)$

Hint. Use 1.8.4 and 1.8.3.

.1 $(\forall x(y \vee \underline{u}x) \leftrightarrow y \vee \forall x \underline{u}x)$

.2 $(\forall x(y \wedge \underline{u}x) \leftrightarrow y \wedge \forall x \underline{u}x)$

Hint. 1.9.12, 1.9.8, 1.9.9, and 1.3.1.

- .3 $(\wedge x(y \vee \underline{ux}) \leftrightarrow y \vee \wedge x \underline{ux})$
 .4 $(\wedge x(y \rightarrow \underline{ux}) \leftrightarrow (y \rightarrow \wedge x \underline{ux}))$
 .5 $(\wedge x(\underline{ux} \rightarrow y) \leftrightarrow (\vee x \underline{ux} \rightarrow y))$
 .6 $(\vee x(y \rightarrow \underline{ux}) \leftrightarrow (y \rightarrow \vee x \underline{ux}))$
 .7 $(\vee x(\underline{ux} \rightarrow y) \leftrightarrow (\vee x \underline{ux} \rightarrow y))$
 .8 $(\wedge x \vee s((y \rightarrow \underline{ux}) \wedge (\underline{us} \rightarrow y)) \leftrightarrow (y \leftrightarrow \wedge x \underline{ux}))$
 .9 $(\vee x \wedge s((y \rightarrow \underline{ux}) \wedge (\underline{us} \rightarrow y)) \leftrightarrow (y \leftrightarrow \vee x \underline{ux}))$
 .10 $(\vee x(y \sim \underline{ux}) \leftrightarrow y \sim \wedge x \underline{ux})$
 .11 $(\wedge x(y \sim \underline{ux}) \leftrightarrow y \sim \vee x \underline{ux})$
 .12 $(\wedge x(\underline{ux} \sim y) \leftrightarrow \wedge x \underline{ux} \sim y)$
 .13 $(\vee x(\underline{ux} \sim y) \leftrightarrow \vee x \underline{ux} \sim y)$

Remark. Given a formula A of logic³ each of whose subformulas initiated by a schemator is devoid of quantifiers (' $\wedge$ ' and ' $\vee$ '), then by resorting to theorems of logic devoid of quantifiers and to only certain theorems of 1.8, 1.9, and 1.10, namely, 1.9.0, 1.8.6, 1.9.4, and 1.10.4 through 1.10.7, one can mechanically construct such a formula A' of logic that each constant of A' which precedes a quantifier of A' is a quantifier and a theorem is obtained from ' $(\alpha' \leftrightarrow \alpha)$ ' by replacing ' α ' by A and ' α' ' by A' .

1.11 SUPPLEMENTARY RULES OF INFERENCE.

Based on 1.3.3, substitution, and detachment is

.0 *Sorites.* If A , B , and C are formulas and theorems are obtained from

$$'(\alpha \rightarrow \beta)'$$
 and $'(\beta \rightarrow \gamma)'$

by replacing ' α ', ' β ', and ' γ ' respectively by A , B , and C , then a theorem is obtained from

$$'(\alpha \rightarrow \gamma)'$$

by replacing ' α ' and ' γ ' respectively by A and C .

³ A "formula" of logic is more or less any formula generated from forms in this chapter. More precisely, using A.30, Q is a formula of logic if and only if Q is a member of (formulas G) or a simplified version of such a member obtained by means of 0.48, where G consists of ' $\wedge x \underline{ux}$ ' and ' $(x \rightarrow x)$ ' and all of the definienda of 1.0.

Based on 1.3.4 and 0.24–0.26 is

.1 Commutation. If A , B , and C are formulas and a theorem is obtained from

$$‘(\alpha \rightarrow (\beta \rightarrow \gamma))’$$

by replacing ‘ α ’, ‘ β ’, and ‘ γ ’ respectively by A , B , and C , then a theorem is obtained from

$$‘(\beta \rightarrow (\alpha \rightarrow \gamma))’$$

by making the same replacements.

Based on 1.3.3, substitution, and detachment is

.2 Inflow. If A , A' , and B are formulas and a theorem is obtained from

$$‘(\alpha' \rightarrow \alpha)’$$

by replacing ‘ α' ’ and ‘ α ’ respectively by A' and A , then a theorem is obtained from

$$‘((\alpha \rightarrow \beta) \rightarrow (\alpha' \rightarrow \beta))’$$

by replacing ‘ α ’, ‘ β ’, and ‘ α' ’ respectively by A , B , and A' .

Based on 1.4.7, substitution, and detachment is

.3 Outflow. If A , B , and B' are formulas and a theorem is obtained from

$$‘(\beta \rightarrow \beta')’$$

by replacing ‘ β ’ and ‘ β' ’ respectively by B and B' , then a theorem is obtained from

$$‘((\alpha \rightarrow \beta) \rightarrow (\alpha \rightarrow \beta'))’$$

by replacing ‘ α ’, ‘ β ’, and ‘ β' ’ respectively by A , B , and B' .

Based on 1.4.13, substitution, and detachment is

.4 Consequention. If A is a theorem and B is a formula, then a theorem is obtained from

$$‘(\beta \rightarrow \alpha)’$$

by replacing ‘ β ’ and ‘ α ’ respectively by B and A .

Based on 1.4.14, substitution, and detachment is

.5 Conjunction. If A and B are theorems, then a theorem is obtained from

$$‘(\alpha \wedge \beta)’$$

by replacing ‘ α ’ and ‘ β ’ respectively by A and B .

Based on 1.4.26, substitution, and detachment is

.6 Conjuncture. If A , A' , B , and B' are formulas and theorems are obtained from

$$‘(\alpha \rightarrow \alpha’)’ \quad \text{and} \quad ‘(\beta \rightarrow \beta’)’$$

by replacing ‘ α ’, ‘ α' ’, ‘ β ’, and ‘ β' ’ respectively by A , A' , B , and B' , then a theorem is obtained from

$$‘((\alpha \wedge \beta) \rightarrow (\alpha' \wedge \beta'))’$$

by making the same replacements.

Based on 1.4.28, substitution, and detachment is

.7 Semiconjuncture. If A , B , and C are formulas and theorems are obtained from

$$‘(\alpha \rightarrow \beta)’ \quad \text{and} \quad ‘(\alpha \rightarrow \gamma)’$$

by replacing ‘ α ’, ‘ β ’, and ‘ γ ’ respectively by A , B , and C , then a theorem is obtained from

$$‘(\alpha \rightarrow (\beta \wedge \gamma))’$$

by making the same replacements.

Based on 1.5.1, 1.5.2, substitution, and detachment is

.8 Equivalence-detachment. If A and B are formulas and a theorem is obtained from

$$‘(\alpha \leftrightarrow \beta)’$$

by replacing ‘ α ’ and ‘ β ’ respectively by A and B , then

.0 if A is a theorem, then B is a theorem, and

.1 if B is a theorem, then A is a theorem.

Based on 1.5.3, conjunction, substitution, and detachment is

.9 Equivalence. If A and B are formulas and theorems are obtained from

$$'(\alpha \rightarrow \beta)' \quad \text{and} \quad '(\beta \rightarrow \alpha)'$$

by replacing ' α ' and ' β ' respectively by A and B , then a theorem is obtained from

$$'(\alpha \leftrightarrow \beta)'$$

by making the same replacements.

Based on 1.5.18, conjunction, and detachment is

.10 Equivalence-transitivity. If A , B , and C are formulas and theorems are obtained from

$$'(\alpha \leftrightarrow \beta)' \quad \text{and} \quad '(\beta \leftrightarrow \gamma)'$$

by replacing ' α ', ' β ', and ' γ ' respectively by A , B , and C , then a theorem is obtained from

$$'(\alpha \leftrightarrow \gamma)'$$

by replacing ' α ' and ' γ ' respectively by A and C .

Based on 1.3.2, 1.8.6, 1.9.2, 1.9.4, universalization, indicial substitution, schematic substitution, and detachment is

.11 Distribution. If A and B are formulas, v is a variable, c is either ' $\rightarrow$ ' or ' $\leftrightarrow$ ', q is either ' $\wedge$ ' or ' $\vee$ ', and a theorem is obtained from

$$'(\alpha \gamma \beta)'$$

by replacing ' α ', ' γ ' and ' β ' respectively by A , c , and B , then a theorem is obtained from

$$'(\xi \delta \alpha \gamma \xi \delta \beta)'$$

by replacing ' ξ ', ' δ ', ' α ', ' γ ', and ' β ' respectively by q , v , A , c , and B .

The distribution is

- .0 *universal* if c is ' $\rightarrow$ ' and q is ' $\wedge$ ',
- .1 *existential* if c is ' $\rightarrow$ ' and q is ' $\vee$ ',
- .2 *equivalence-universal* if c is ' $\leftrightarrow$ ' and q is ' $\wedge$ ',
- .3 *equivalence-existential* if c is ' $\leftrightarrow$ ' and q is ' $\vee$ '.

Based on 1.58, 1.5.21, 1.5.25, 1.5.26, 1.5.27, 1.6.12, 1.7.10, 0.48, detachment, substitution, conjunction, equivalence-universal distribution, equivalence-existential distribution, and induction (on the number of symbols in a formula of logic, say) is

.12 Equivalence substitution. If Q is such a formula of logic³ and v such a variable that v is free in Q and v occurs in no subformula of Q begun by schemator, B and B' are formulas, A and A' are respectively obtained from Q by replacing v by B and v by B' , and a theorem is obtained from

$$(\beta \leftrightarrow \beta')$$

by replacing ' β ' and ' β ' respectively by B and B' , then a theorem is obtained from

$$(\alpha \leftrightarrow \alpha')$$

by replacing ' α ' and ' α ' respectively by A and A' .

CHAPTER 2

SET THEORY

Any set theory formalized in the spirit of Chapter 0 naturally suggests certain simple provocative questions.

When is x ? What is $(x \in y)$? What is $(p \rightarrow q)$? What is $\bigwedge x \bigvee x$?

We shall commit ourselves to a set theory unorthodox in the sense that these questions are answered. In fact, our Axioms 2.5.0–2.5.3 were conceived for this express purpose. We were led to these four axioms somewhat as follows.

We believe every (mathematical) thing is a set. We believe there is no difference between the conjunction of two or more things and their intersection. We believe there is no difference between the disjunction of two things and their union. We believe there is no difference between the negation of a thing and its complement. We have come to believe a thing if and only if the empty set is a member of the thing. We believe $(x \in y)$ if and only if x is a member of y . We believe $(x \in y)$ if and only if $(x \in y)$ is the universe. We disbelieve $(x \in y)$ if and only if $(x \in y)$ is the empty set.

PRELIMINARIES

2.0 ORIENTING DEFINITIONS.

- .0 $(\text{mel} \in x \equiv x)$
- .1 $((x \in x') \equiv (x \in x'))$
- .2 $(\text{mel } x \equiv \text{mel } x)$

2.1 DEFINITIONS.

- .0 $((x \text{ is a set}) \equiv (x \rightarrow x))$
- .1 $(\text{The empty set} \equiv 0)$

- .2** (The universe $\equiv U$)
- .3** (Complement $x \equiv \sim x$)
- .4** ((x is a member of y) $\equiv (x \in y)$)
- .5** ((x is in y) $\equiv (x \in y)$)
- .6** ((x is a point) $\equiv \forall y(x \in y)$)
- .7** ((x is true) $\equiv (0 \in x)$)
- .8** ((x is false) $\equiv \sim(0 \in x)$)
- .9** (($x \ni y$) $\equiv (y \in x)$)
- .10** ((x holds y) $\equiv (x \ni y)$)
- .11** (($x \subset y$) $\equiv \wedge t((t \in x) \rightarrow (t \in y))$)
- .12** ((x is included in y) $\equiv (x \subset y)$)
- .13** ((x is a subset of y) $\equiv (x \subset y)$)
- .14** (($x \supset y$) $\equiv (y \subset x)$)
- .15** ((x includes y) $\equiv (x \supset y)$)
- .16** ((x is a superset of y) $\equiv (x \supset y)$)
- .17** (($x = y$) $\equiv ((x \subset y) \wedge (y \subset x))$)
- .18** ((x equals y) $\equiv (x = y)$)
- .19** (($x \neq y$) $\equiv \sim(x = y)$)
- .20** (($x \subset \cdot y$) $\equiv (x \subset y \neq x)$)
- .21** ((x is a proper subset of y) $\equiv (x \subset \cdot y)$)
- .22** (($x \supset \cdot y$) $\equiv (y \subset \cdot x)$)
- .23** ((x is a proper superset of y) $\equiv (x \supset \cdot y)$)
- .24** (The intersection as x runs , of $\underline{ux} \equiv \wedge x \underline{ux}$)
- .25** (The union as x runs , of $\underline{ux} \equiv \vee x \underline{ux}$)
- .26** ($\prod A \equiv \wedge y(y \in A \rightarrow y)$)
- .27** (The intersection of $A \equiv \prod A$)
- .28** ($\nabla A \equiv \vee y(y \in A \wedge y)$)
- .29** (The union of $A \equiv \nabla A$)
- .30** (($x \cap y$) $\equiv (x \wedge y)$)
- .31** ((x intersect y) $\equiv (x \cap y)$)
- .32** (($x \cup y$) $\equiv (x \vee y)$)
- .33** ((x union y) $\equiv (x \cup y)$)
- .34** ((x complement y) $\equiv (x \sim y)$)

- .35 $(\text{sng } x \equiv \bigwedge y(y \rightarrow (x \in y)))$
- .36 $(\text{singleton } x \equiv \text{sng } x)$
- .37 $(\text{The set whose sole member is } x \equiv \text{sng } x)$

In formulating 2.2 and 2.3 we have Remark A.28 in mind.

2.2 LOGICAL DEFINITIONAL AXIOMS FOR SET THEORY.

- .0 $(U \equiv \sim 0)$
- .1 $((p \wedge q) \equiv \sim(p \rightarrow \sim q))$
- .2 $((p \leftrightarrow q) \equiv ((p \rightarrow q) \wedge (q \rightarrow p)))$
- .3 $(\bigvee x \underline{u}x \equiv \sim \bigwedge x \sim \underline{u}x)$

2.3 SET-THEORETIC DEFINITIONAL AXIOMS FOR SET THEORY.

- .0 $((x \subset y) \equiv \bigwedge t((t \in x) \rightarrow (t \in y)))$
- .1 $((x = y) \equiv ((x \subset y) \wedge (y \subset x)))$
- .2 $(\text{sng } x \equiv \bigwedge y(y \rightarrow (x \in y)))$

2.4 AXIOM OF DEFINITION FOR SET THEORY.

$$((x \equiv y) = (x = y))$$

2.5 AXIOMS FOR SET THEORY.

- .0 $(x \leftrightarrow (0 \in x))$
- .1 $((t \in U) \rightarrow ((t \in (x \in y)) \leftrightarrow (x \in y)))$
- .2 $((t \in a) \rightarrow ((t \in (p \rightarrow q)) \leftrightarrow ((t \in p) \rightarrow (t \in q))))$
- .3 $((t \in \bigwedge x \underline{u}x) \leftrightarrow \bigwedge x(t \in \underline{u}x))$
- .4 $((x \in U) \rightarrow ((x = y) \leftrightarrow \bigwedge t((x \in t) \rightarrow (y \in t))))$
- .5 $((x = y) \rightarrow (\underline{u}x = \underline{u}y))$
- .6 $(U = \bigvee c \bigwedge x((c \wedge (c \in U)) \wedge ((x \in c) \rightarrow (\text{sng } x \in c))))$
- .7 $(\bigvee x(((A \in U) \wedge (x \subset A)) \wedge ((\underline{u}x \in U) \wedge \underline{u}x)) \in U)$
- .8 $((z \in A) \rightarrow (\text{mel } A \in A))$
- .9 $(\sim \bigvee y((y \in \sim A) \wedge (y \subset A)) \rightarrow (A = U))$

Remark. Axiom 2.5.9, which supplants the more customary axiom of regularity, is our axiom of size. Without the use of 2.5.9 it will be shown that 2.5.9 is equivalent to plausible 2.105.1 which assures us that if the set of subsets of A is included in A , then A must be the universe.

Axiom 2.5.8 is our version of the Zermelo principle of choice. It can without the loss of 2.108 be replaced by the longer but less-demanding formula

$$'(((z \in A) \wedge (A \in U)) \rightarrow (\text{mel } A \in A))'.$$

Axiom 2.5.7 is our axiom of replacement. From it we learn that quite a few sets are points. It can, at some cost, be replaced by axioms in which no schemator appears. Axiom 2.5.6 is our axiom of infinity. Its purpose is to assure that

$$\forall c \wedge x((c \wedge (c \in U)) \wedge ((x \in c) \rightarrow (\text{sng } x \in c)))$$

and that

$$(\text{sng } x \in U).$$

Thus at small cost we could replace 2.5.6 by two simpler axioms. Axiom 2.5.5 is our axiom of equality. It is principally of value in establishing certain intuitively reasonable general formulas. In order to stress its initial unimportance we do not use it prior to the proof of 2.31. Axiom 2.5.4 is our version of the law of Leibniz. Axiom 2.5.3 is our axiom of intersection; Axiom 2.5.2 is our axiom of distribution; Axiom 2.5.1 is our axiom of value; and Axiom 2.5.0 is our axiom of truth.

Let us agree here that our *revised* axioms are those obtained by striking out 2.5.9, 2.5.8, 2.5.5, and then replacing 2.5.7 by the following axiom of heredity:

$$.7' \quad (((A \subset B) \wedge (B \in U)) \rightarrow (A \in U))$$

In Appendix B we shall show that 2.5.9 is consistent with our revised axioms. Nevertheless we are not at all sure that Gödel's approach exploited so successfully elsewhere⁴ can be used here to show 2.5.8 is consistent with our revised axioms.

In Appendix C we easily check that our 1965 edition is axiomatically equivalent to our present edition.

⁴ Gödel, K., *The Consistency of the Continuum Hypothesis* (Ann. of Math. Studies No. 3), Princeton Univ. Press, Princeton, New Jersey, 1940.

2.6 THEOREM. $(x \text{ is a set})$

2.7 THEOREM. $(0 \in U)$

Proof. Use 1.4.5 and 2.5.0.

2.8 THEOREM. $(x \subset y \rightarrow (x \rightarrow y))$

Proof. Helped by 2.5.0 we infer

$$\begin{aligned} (x \subset y &\rightarrow (t \in x \rightarrow t \in y)) \\ &\rightarrow (0 \in x \rightarrow 0 \in y) \\ &\rightarrow (x \rightarrow y)). \end{aligned}$$

2.9 THEOREM. $(x = y \rightarrow x \leftrightarrow y)$

2.10 THEOREM. $((x \equiv y) \leftrightarrow x = y)$

2.11 THE THEOREM OF EXTENT.

$$(a = b \leftrightarrow \wedge x(x \in a \leftrightarrow x \in b))$$

2.12 LEMMA. $(x \in a \rightarrow x \in U)$

Proof. Because of 2.5.2, 1.0.4, and 1.0.6, we know

$$\begin{aligned} (x \in a &\rightarrow ((x \in 0 \rightarrow x \in 0) \rightarrow x \in (0 \rightarrow 0))), \\ ((x \in 0 &\rightarrow x \in 0) \rightarrow (x \in a \rightarrow x \in (0 \rightarrow 0))), \end{aligned}$$

and

$$(x \in a \rightarrow x \in (0 \rightarrow 0) \rightarrow x \in \sim 0 \rightarrow x \in U).$$

2.13 THEOREM. $(x \in y \rightarrow (x \text{ is a point}))$

2.14 THEOREM. $((x \text{ is a point}) \leftrightarrow x \in U)$

2.15 LEMMA. $(x \in 0 \rightarrow x \in a)$

Proof. $(t \in 0 \rightarrow t \in \wedge x x \rightarrow \wedge x(t \in x) \rightarrow t \in a)$

2.16 THEOREM. $(0 \subset a \subset U)$

Taking advantage of 2.12, we infer

2.17 LEMMAS.

$$.0 \quad ((t \in a \leftrightarrow y \wedge t \in U) \leftrightarrow (t \in U \rightarrow t \in a \leftrightarrow y))$$

$$.1 \quad ((t \in a \leftrightarrow t \in b) \leftrightarrow (t \in U \rightarrow t \in a \leftrightarrow t \in b))$$

From 2.5.1, 2.5.2, and 2.17.0 we deduce

2.18 THEOREMS.

$$.0 \quad (t \in (x \in y) \leftrightarrow x \in y \wedge t \in U)$$

$$.1 \quad (t \in (p \rightarrow q) \leftrightarrow (t \in p \rightarrow t \in q) \wedge t \in U)$$

2.19 LEMMAS.

$$.0 \quad (x \in 0 \rightarrow 0)$$

Proof. Using 2.15, 2.18.0, and 2.5.0 we infer

$$(x \in 0 \rightarrow x \in (0 \in 0) \rightarrow 0 \in 0 \rightarrow 0).$$

$$.1 \quad (x \in 0 \leftrightarrow 0)$$

$$.2 \quad \sim(x \in 0)$$

2.20 THEOREMS.

$$.0 \quad (t \in U \rightarrow t \in \sim a \leftrightarrow t \in a)$$

Proof. Recalling 0.49 and using 2.19.1 we deduce

$$\begin{aligned} (t \in U \rightarrow t \in \sim a &\leftrightarrow t \in (a \rightarrow 0) \\ &\leftrightarrow (t \in a \rightarrow t \in 0) \\ &\leftrightarrow (t \in a \rightarrow 0) \\ &\leftrightarrow t \in a). \end{aligned}$$

$$.1 \quad (t \in \sim a \leftrightarrow t \in a \wedge t \in U)$$

$$.2 \quad (t \in (a \wedge b) \leftrightarrow t \in a \wedge t \in b)$$

Proof.

$$\begin{aligned}
 (t \in U \rightarrow t \in (a \wedge b)) &\leftrightarrow t \in \sim (a \rightarrow \sim b) \\
 &\leftrightarrow \sim (t \in (a \rightarrow \sim b)) \\
 &\leftrightarrow \sim (t \in a \rightarrow t \in \sim b) \\
 &\leftrightarrow \sim (t \in a \rightarrow t \notin b) \\
 &\leftrightarrow t \in a \wedge t \in b
 \end{aligned}$$

- .3 $(t \in (a \vee b) \leftrightarrow t \in a \vee t \in b)$
- .4 $(t \in U \rightarrow t \in (p \leftrightarrow q) \leftrightarrow (t \in p \leftrightarrow t \in q))$
- .5 $(t \in (p \leftrightarrow q) \leftrightarrow (t \in p \leftrightarrow t \in q) \wedge t \in U)$
- .6 $(t \in \bigvee x \underline{u}x \leftrightarrow \bigvee x (t \in \underline{u}x))$
- .7 $(t \in U \rightarrow t \in (x \subset y) \leftrightarrow x \subset y)$
- .8 $(t \in (x \subset y) \leftrightarrow x \subset y \wedge t \in U)$
- .9 $(t \in U \rightarrow t \in (x = y) \leftrightarrow x = y)$
- .10 $(t \in (x = y) \leftrightarrow x = y \wedge t \in U)$

2.21 THEOREM. $(a = 0 \leftrightarrow \bigwedge x \sim (x \in a) \leftrightarrow \sim \bigvee x (x \in a))$

2.22 LEMMAS.

- .0 $(p \wedge \bigwedge t (t \in p \leftrightarrow t \in U \wedge p) \rightarrow p = U)$
- .1 $(\sim p \wedge \bigwedge t (t \in p \leftrightarrow t \in U \wedge p) \rightarrow p = 0)$
- .2 $(\bigwedge t (t \in p \leftrightarrow t \in U \wedge p) \rightarrow p = U \vee p = 0)$
- .3 $(a = 0 \wedge b = U \wedge (a \leftrightarrow b) \rightarrow 0)$

2.23 LEMMA.

$$(a = 0 \vee a = U \wedge b = 0 \vee b = U \rightarrow (a \leftrightarrow b) \leftrightarrow a = b)$$

2.24 EXERCISES.

- .0 $(a \subset a)$
- .1 $(a \subset b \subset c \rightarrow a \subset c)$
- .2 $(x = x)$
- .3 $(x = y \rightarrow y = x)$

- .4** $(x = y = z \rightarrow x = z)$
- .5** $((p \rightarrow q) = (\sim p \vee q) = \sim p \cup q)$
- .6** $(a \cap U = a)$
- .7** $(a \cup 0 = a)$
- .8** $(a \cap 0 = 0)$
- .9** $(a \cup U = U)$
- .10** $(a \cap a = a)$
- .11** $(a \cup a = a)$
- .12** $(a \cap b = b \cap a)$
- .13** $(a \cup b = b \cup a)$
- .14** $(a \cap (b \cap c) = (a \cap b) \cap c = a \cap b \cap c)$
- .15** $(a \cup (b \cup c) = (a \cup b) \cup c = a \cup b \cup c)$
- .16** $(a \cup b \cap c = a \cap c \cup b \cap c)$
- .17** $(a \cap b \cup c = a \cup c \cap b \cup c)$
- .18** $(\sim \sim a = a)$
- .19** $(\sim (a \cap b) = \sim a \cup \sim b)$
- .20** $(\sim (a \cup b) = \sim a \cap \sim b)$
- .21** $(a \cap \sim a = 0)$
- .22** $(a \cup \sim a = U)$
- .23** $(\sim \wedge x \underline{u}x = \vee x \sim \underline{u}x)$
- .24** $(\sim \vee x \underline{u}x = \wedge x \sim \underline{u}x)$
- .25** $(a \subset b \wedge c \subset d \rightarrow a \cap c \subset b \cap d)$
- .26** $(a \subset b \wedge c \subset d \rightarrow a \cup c \subset b \cup d)$
- .27** $(a \cap b \subset a)$
- .28** $(a \subset a \cup b)$
- .29** $(a \subset b \leftrightarrow a = a \cap b)$
- .30** $(a \subset b \leftrightarrow b = a \cup b)$
- .31** $(y \in a \rightarrow \top \top a \subset y)$
- .32** $(y \in a \rightarrow y \subset \nabla a)$
- .33** $(y \in a \rightarrow \top \top a \subset y \subset \nabla a)$
- .34** $(a \neq 0 \rightarrow \top \top a \subset \nabla a)$
- .35** $(\top \top (a \cup b) = \top \top a \cap \top \top b)$

- .36 $(\nabla(a \cup b) = \nabla a \cup \nabla b)$
- .37 $(\prod \nabla x \underline{u}x = \wedge x \prod \underline{u}x)$
- .38 $(\nabla \nabla x \underline{u}x = \nabla x \nabla \underline{u}x)$
- .39 $(a \subset b \rightarrow \prod b \subset \prod a)$
- .40 $(a \subset b \rightarrow \nabla a \subset \nabla b)$
- .41 $(\prod U = 0)$
- .42 $(\prod 0 = U)$
- .43 $(\nabla 0 = 0)$
- .44 $(a \subset b \rightarrow a \subset \cdot b \vee a = b)$
- .45 $(a \subset \cdot b \subset c \rightarrow a \subset \cdot c)$
- .46 $(a \subset b \subset \cdot c \rightarrow a \subset \cdot c)$
- .47 $(\wedge xy = y)$
- .48 $(\vee xy = y)$
- .49 $((a \rightarrow b) = U \leftrightarrow a \subset b)$
- .50 $(x \in y \rightarrow (x \in y) = U)$
- .51 $(x \sim \in y \rightarrow (x \in y) = 0)$
- .52 $(x \subset y \rightarrow (x \subset y) = U)$
- .53 $(x \sim \subset y \rightarrow (x \subset y) = 0)$
- .54 $(x = y \rightarrow (x = y) = U)$
- .55 $(x \neq y \rightarrow (x = y) = 0)$
- .56 $(\wedge x \wedge y \underline{u}'xy = \wedge y \wedge x \underline{u}'xy)$
- .57 $(\vee x \vee y \underline{u}'xy = \vee y \vee x \underline{u}'xy)$
- .58 $(\wedge x(\underline{u}x \cap \underline{v}x) = \wedge x \underline{u}x \cap \wedge x \underline{v}x)$
- .59 $(\vee x(\underline{u}x \cup \underline{v}x) = \vee x \underline{u}x \cup \vee x \underline{v}x)$
- .60 $(\wedge x(\underline{u}x \subset \underline{v}x) \rightarrow \wedge x \underline{u}x \subset \wedge x \underline{v}x \wedge \vee x \underline{u}x \subset \vee x \underline{v}x)$
- .61 $(\wedge x(y \cup \underline{u}x) = y \cup \wedge x \underline{u}x)$
- .62 $(\vee x(y \cap \underline{u}x) = y \cap \vee x \underline{u}x)$
- .63 $(\wedge x \underline{u}x \subset \underline{u}x \subset \vee x \underline{u}x)$
- .64 $(\wedge y(y \in a \rightarrow b \subset y) \leftrightarrow b \subset \prod a)$
- .65 $(\wedge y(y \in a \rightarrow y \subset b) \leftrightarrow \nabla a \subset b)$
- .66 $(\vee x \wedge y \underline{u}'xy \subset \wedge y \vee x \underline{u}'xy)$
- .67 $(x \equiv x)$

SOME ASPECTS OF EQUALITY

2.25 THEOREM. $(q = r \rightarrow (p \rightarrow q) = (p \rightarrow r))$

Proof.

$$\begin{aligned}
 (q = r \rightarrow (t \in U \rightarrow t \in q \leftrightarrow t \in r \\
 \rightarrow (t \in p \rightarrow t \in q) \leftrightarrow (t \in p \rightarrow t \in r) \\
 \rightarrow t \in (p \rightarrow q) \leftrightarrow t \in (p \rightarrow r))) \\
 (q = r \rightarrow (t \in U \rightarrow t \in (p \rightarrow q) \leftrightarrow t \in (p \rightarrow r)))
 \end{aligned}$$

Now use 2.17.1 to infer

$$(q = r \rightarrow t \in (p \rightarrow q) \leftrightarrow t \in (p \rightarrow r));$$

conclude

$$(q = r \rightarrow \wedge t(t \in (p \rightarrow q) \leftrightarrow t \in (p \rightarrow r)))$$

and

$$(q = r \rightarrow (p \rightarrow q) = (p \rightarrow r)).$$

2.26 THEOREM. $(p = q \rightarrow (p \rightarrow r) = (q \rightarrow r))$

2.27 THEOREM. $(\wedge x(\underline{u}x = \underline{v}x) \rightarrow \wedge x \underline{u}x = \wedge x \underline{v}x)$

2.28 THEOREM. $(y = z \rightarrow (x \in y) = (x \in z))$

So far we have used Axioms 2.5.0 through 2.5.3. We now use 2.5.4.

2.29 LEMMAS.

.0 $(x = y \rightarrow (x \in t \rightarrow y \in t))$

Proof.

$$\begin{aligned}
 (x \in t \rightarrow x \in U) \\
 (x \in U \rightarrow (x = y \rightarrow (x \in t \rightarrow y \in t))) \\
 (x \in t \rightarrow (x = y \rightarrow (x \in t \rightarrow y \in t))) \\
 (x = y \rightarrow (x \in t \rightarrow (x \in t \rightarrow y \in t))) \\
 (x = y \rightarrow (x \in t \rightarrow y \in t))
 \end{aligned}$$

.1 $(x = y \rightarrow (x \in t \leftrightarrow y \in t))$

2.30 THEOREM. $(x = y \rightarrow (x \in t) = (y \in t))$

We now use 2.5.5 for the first time.

2.31 THEOREM. $(x = y \rightarrow \underline{v}x = \underline{v}y)$

Proof.

$$\begin{aligned}
 &(x = y \rightarrow \underline{u}x = \underline{u}y) \\
 &(\wedge x(\underline{u}x = \underline{v}x) \rightarrow \underline{u}x = \underline{v}x) \\
 &(\wedge x(\underline{u}x = \underline{v}x) \rightarrow \underline{u}y = \underline{v}y) \\
 &(x = y \wedge \wedge x(\underline{u}x = \underline{v}x) \rightarrow \underline{u}x = \underline{u}y \wedge \underline{u}x = \underline{v}x \wedge \underline{u}y = \underline{v}y \\
 &\quad \rightarrow \underline{v}x = \underline{v}y) \\
 &(\wedge x(\underline{u}x = \underline{v}x) \rightarrow (x = y \rightarrow \underline{v}x = \underline{v}y))
 \end{aligned}$$

In this last formula schematically replace ' $\underline{u}x$ ' by ' $\underline{v}x$ ' and detach.

The next four theorems, as well as many earlier ones, can be proved by similar devices.

2.32 THEOREMS.

- .0** $(x = y \rightarrow \text{mel } x = \text{mel } y)$
- .1** $(x = y \rightarrow \underline{u}'xz = \underline{u}'yz)$
- .2** $(x = y \rightarrow \underline{u}'zx = \underline{u}'zy)$
- .3** $(x = s \wedge y = t \rightarrow \underline{u}'xy = \underline{u}'st)$

Remark. Because of 2.3.1, 2.4, 2.5.4, 2.5.5, 2.10, 2.11, 2.24.2, 2.24.3, 2.24.4, 2.25, 2.26, 2.27, 2.28, 2.30, 2.31, and 2.32, it seems that equality might well possess those intuitive attributes ascribed to it. Hereafter we shall accept formulas like those in 2.32 without question.

Remark. It is now easy to check and useful to bear in mind that

$$(a \subset b \leftrightarrow \sim b \cap a = 0).$$

Remark. Logic helps us investigate set theory. Conversely, set theory can be used to establish theorems in logic.

We agree that a formula A is *universal* if and only if a theorem is obtained from ' $(x = U)$ ' by replacing ' x ' by A . Now all theorems which can be inferred from universal axioms are themselves universal. An examination reveals that all our axioms for logic are universal and that all but Axiom 2.5.0 of our axioms for set theory are universal. Since every universal formula is obviously a theorem, we now have at hand a systematic set-theoretic method for exploring logic.

CLASSIFICATION

2.33 DEFINITIONS.

- .0 $(\text{Ex } \underline{u}x \equiv \forall x(0 \in \underline{u}x \wedge \text{sng } x))$
- .1 (The set of points x such that $\underline{u}x \equiv \text{Ex } \underline{u}x$)
- .2 $(\{x : \underline{u}x\} \equiv \text{Ex } \underline{u}x)$
- .3 $(\text{sb } A \equiv \text{Ex}(x \subset A))$
- .4 $(\text{psb } A \equiv \text{Ex}(x \subset \cdot A))$
- .5 $(\text{sp } A \equiv \text{Ex}(x \supset A))$
- .6 $(\text{psp } A \equiv \text{Ex}(x \supset \cdot A))$

Unused in the present section but used in the next is 2.34.4 below.

2.34 LEMMAS.

- .0 $(y \in U \rightarrow y \in \text{sng } x \leftrightarrow y = x)$

Proof. Recalling 2.5.4 we see

$$\begin{aligned}
 (y \in U \rightarrow y \in \text{sng } x &\leftrightarrow y \in \wedge t(t \rightarrow x \in t) \\
 &\leftrightarrow \wedge t(y \in t \rightarrow x \in t) \\
 &\leftrightarrow y = x).
 \end{aligned}$$

- .1 $(y \in \text{sng } x \leftrightarrow y = x \wedge y \in U)$
- .2 $(x \in U \rightarrow x \in \text{sng } x)$

$$.3 \quad (x \in U \rightarrow x \in A \leftrightarrow \text{sng } x \subset A)$$

$$.4 \quad (x \in U \wedge a = \text{sng } x \rightarrow x = \prod a = \nabla a)$$

Proof. From .0 we infer

$$(y \in a \rightarrow y \subset x \wedge x \subset y).$$

Because of this, 2.24.65, and 2.24.64, we have on the one hand

$$(\nabla a \subset x \subset \prod a).$$

Because of .2 and 2.24.33 we have on the other hand

$$(\prod a \subset x \subset \nabla a).$$

The only use of 2.5.6 until 2.139 is in proving

$$.5 \quad (x \in U \rightarrow \text{sng } x \in U)$$

Proof. First

$$\begin{aligned} & (x \in (c \wedge (x \in c \rightarrow \text{sng } x \in c)) \\ & \rightarrow x \in c \wedge x \in (x \in c \rightarrow \text{sng } x \in c) \\ & \rightarrow U \wedge x \in (U \rightarrow \text{sng } x \in c) \\ & \rightarrow x \in (\text{sng } x \in c) \rightarrow \text{sng } x \in c \rightarrow \text{sng } x \in U). \end{aligned}$$

Accordingly,

$$(x \in (c \wedge (x \in c \rightarrow \text{sng } x \in c)) \rightarrow \text{sng } x \in U)$$

Because of this and because 2.5.6 makes clear

$$(U = \bigvee c(c \wedge (x \in c \rightarrow \text{sng } x \in c)))$$

it follows that

$$\begin{aligned} & (x \in U \rightarrow \bigvee c(x \in (c \wedge (x \in c \rightarrow \text{sng } x \in c))) \\ & \rightarrow \bigvee c(\text{sng } x \in U) \rightarrow \text{sng } x \in U). \end{aligned}$$

$$.6 \quad (A = \bigvee x(x \in A \wedge \text{sng } x))$$

We use 2.5.5 in proving the following very useful theorem.

2.35 THE THEOREM OF CLASSIFICATION.

$$(x \in \text{Ex } \underline{u}x \leftrightarrow \underline{u}x \wedge x \in U)$$

Proof. Using 2.5.0, 2.34.1, 2.5.5, and 2.9 for the fifth, sixth, and seventh equivalences we see

$$\begin{aligned} (x \in \text{Ex } \underline{u}x &\leftrightarrow x \in \bigvee x(0 \in \underline{u}x \wedge \text{sng } x) \\ &\leftrightarrow \bigvee t(x \in (0 \in \underline{u}t \wedge \text{sng } t)) \\ &\leftrightarrow \bigvee t(x \in (0 \in \underline{u}t) \wedge x \in \text{sng } t) \\ &\leftrightarrow \bigvee t(0 \in \underline{u}t \wedge x \in \text{sng } t) \\ &\leftrightarrow \bigvee t(\underline{u}t \wedge x \in \text{sng } t) \\ &\leftrightarrow \bigvee t(\underline{u}t \wedge x \in U \wedge x = t) \\ &\leftrightarrow \bigvee t(\underline{u}x \wedge x \in U \wedge x = t) \\ &\leftrightarrow \underline{u}x \wedge x \in U \wedge \bigvee t(x = t) \\ &\leftrightarrow \underline{u}x \wedge x \in U). \end{aligned}$$

2.36 THEOREM. $(x \in U \rightarrow x \in \text{Ex } \underline{u}x \leftrightarrow \underline{u}x)$

2.37 THEOREMS.

- .0** $(\bigwedge x(\underline{u}x \rightarrow \underline{v}x) \rightarrow \text{Ex } \underline{u}x \subset \text{Ex } \underline{v}x)$
- .1** $(\bigwedge x(\underline{u}x \leftrightarrow \underline{v}x) \rightarrow \text{Ex } \underline{u}x = \text{Ex } \underline{v}x)$
- .2** $(y = \text{Ex}(x \in y) = \bigvee x(x \in y \wedge \text{sng } x))$
- .3** $(\text{Ex}(\underline{u}x \rightarrow \underline{v}x) = (\text{Ex } \underline{u}x \rightarrow \text{Ex } \underline{v}x))$
- .4** $(\text{Ex } \sim \underline{u}x = \sim \text{Ex } \underline{u}x)$
- .5** $(\text{Ex}(\underline{u}x \wedge \underline{v}x) = \text{Ex } \underline{u}x \cap \text{Ex } \underline{v}x)$
- .6** $(\text{Ex}(x \in U \wedge \underline{u}x) = \text{Ex } \underline{u}x)$
- .7** $(\text{Ex}(\underline{u}x \vee \underline{v}x) = \text{Ex } \underline{u}x \cup \text{Ex } \underline{v}x)$
- .8** $(\text{Ex}(\underline{u}x \leftrightarrow \underline{v}x) = (\text{Ex } \underline{u}x \leftrightarrow \text{Ex } \underline{v}x))$
- .9** $(\text{Ex } \bigwedge y \underline{u}'xy = \bigwedge y \text{Ex } \underline{u}'xy)$
- .10** $(\text{Ex } \bigvee y \underline{u}'xy = \bigvee y \text{Ex } \underline{u}'xy)$
- .11** $(\text{Ex } \underline{u}x = U \leftrightarrow \bigwedge x(x \in U \rightarrow \underline{u}x))$
- .12** $(p = 0 \vee p = U \rightarrow p = \text{Etp})$
- .13** $(x \subset y \rightarrow \text{sb } x \subset \text{sb } y \wedge \text{psb } x \subset \text{psb } y \subset \text{sb } y)$
- .14** $(\text{sb } A \cap \text{sb } B = \text{sb } (A \cap B))$

$$.15 \text{ (sb } A \cup \text{sb } B \subset \text{sb } (A \cup B))$$

$$.16 \text{ (}\nabla \text{ sb } A = A \subset \text{sb } \nabla A)$$

$$.17 \text{ (sb } U = U)$$

$$.18 \text{ (}\nabla U = U)$$

$$.19 \text{ (sng } x = \text{E}y(y = x))$$

$$.20 \text{ (}0 \in \text{sb } 0 = \text{sng } 0 \subset \text{sb } A)$$

$$.21 \text{ (}\nabla A \subset B \leftrightarrow A \subset \text{sb } B)$$

$$.22 \text{ (}y \subset \bigwedge x \underline{u}x \leftrightarrow \bigwedge x(y \subset \underline{u}x))$$

$$.23 \text{ (sb } \bigwedge x \underline{u}x = \bigwedge x \text{sb } \underline{u}x)$$

In establishing implications we shall frequently, as in the proof of 2.38 below, write down formulas some of which may not be theorems but each of which is implied by the premise of the implication to be established.

2.38 THEOREM. $(\alpha = \text{E}x \sim (x \in x) \rightarrow \alpha \sim \in U)$

Proof. From 2.36 we learn

$$(t \in U \rightarrow t \in \text{E}t \underline{u}t \leftrightarrow \underline{u}t).$$

Hence

$$\begin{aligned} (t \in U \rightarrow t \in \alpha \leftrightarrow t \sim \in t), \\ (\alpha \in U \rightarrow \alpha \in \alpha \leftrightarrow \alpha \sim \in \alpha), \end{aligned}$$

and the desired conclusion is at hand.

Remark. Our proof of 2.38 is closely patterned after the reasoning used by Bertrand Russell in reaching his famous paradox. Of course if by misfortune

$$'(x \in \text{E}x \sim (x \in x) \leftrightarrow x \sim \in x)'$$

is a theorem, then the Russell paradox is at hand.

THE ROLE OF REPLACEMENT

Usually of only local interest are definitions introduced in the spirit of

2.39 DEFINITION.

$$(2A \ A \equiv \forall x(A \in U \wedge x \subset A \wedge \nabla x \in U \wedge \nabla x))$$

2.40 LEMMAS.

$$.0 \quad (2A \ A \in U)$$

Proof. See 2.5.7.

$$.1 \quad (x \in A \in U \rightarrow x \subset 2A \ A)$$

Proof. $((A \in U \wedge t \subset A \wedge \nabla t \in U \wedge \nabla t) \subset 2A \ A)$

because of 2.24.63. Substituting, we see that

$$((A \in U \wedge \text{sng } x \subset A \wedge \nabla \text{sng } x \in U \wedge \nabla \text{sng } x) \subset 2A \ A).$$

From this, 2.34.3, and 2.34.4 we infer

$$\begin{aligned} (x \in A \in U \rightarrow x &= (U \wedge U \wedge U \wedge x)) \\ &= (A \in U \wedge x \in A \wedge x \in U \wedge x) \\ &= (A \in U \wedge \text{sng } x \subset A \wedge \nabla \text{sng } x \in U \wedge \nabla \text{sng } x) \\ &\subset 2A \ A). \end{aligned}$$

$$.2 \quad ((x \subset 2A \ A \wedge A \in U \wedge x \in A) = (A \in U \wedge x \in A))$$

Proof. Use .1.

$$.3 \quad ((x \in A \wedge \text{sng } x \in U) = (x \in A))$$

Proof. Use 2.34.5.

$$.4 \quad (((x \in y \wedge t) \in U \wedge x \in y) = (x \in y \wedge t \in U))$$

Proof. $((x \in y) = 0 \vee (x \in y) = U)$

2.41 THE THEOREM OF REPLACEMENT.

$$(\forall x(A \in U \wedge x \in A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U)$$

Proof. From 2.5.7 we learn

$$(\forall x(2A \ A \in U \wedge x \subset 2A \ A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

In the light of 2.40.0 we see

$$(\forall x(x \subset 2A \ A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

Schematic substitution reveals that

$$(\forall x(x \subset 2A \ A \wedge (A \in U \wedge \underline{u}x) \in U \wedge A \in U \wedge \underline{u}x) \in U).$$

Employing 2.40.4, we find

$$(\forall x(x \subset 2A \ A \wedge A \in U \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

Another schematic substitution now reveals that

$$(\forall x(x \subset 2A \ A \wedge A \in U \wedge (x \in A \wedge \underline{u}x) \in U \wedge x \in A \wedge \underline{u}x) \in U).$$

Employing 2.40.4 again, we find

$$(\forall x(x \subset 2A \ A \wedge A \in U \wedge x \in A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

From 2.40.2 we consequently conclude

$$(\forall x(A \in U \wedge x \in A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

2.42 THEOREM. $(A \in U \rightarrow B \cap A \in U)$

Proof. According to 2.41,

$$(\forall x(x \in A \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

Schematic substitution yields

$$(\forall x(x \in A \wedge (x \in B \wedge \underline{u}x) \in U \wedge x \in B \wedge \underline{u}x) \in U).$$

According to 2.40.4,

$$(\forall x(x \in A \wedge x \in B \wedge \underline{u}x \in U \wedge \underline{u}x) \in U).$$

Hence

$$(\forall x(x \in A \cap B \wedge \underline{u}x \in U \wedge \underline{u}x) \in U)$$

and

$$(\forall x(x \in A \cap B \wedge \text{sng } x \in U \wedge \text{sng } x) \in U).$$

Helped by 2.34.6 and 2.40.3, we conclude

$$\begin{aligned} (A \cap B &= \forall x(x \in A \cap B \wedge \text{sng } x) \\ &= \forall x(x \in A \cap B \wedge \text{sng } x \in U \wedge \text{sng } x) \in U). \end{aligned}$$

2.43 THE THEOREM OF HEREDITY.

$$(B \subset A \in U \rightarrow B \in U)$$

2.44 THE THEOREM OF SUBSETS.

$$(A \in U \leftrightarrow \text{sb } A \in U)$$

Proof. Schematic substitution in 2.5.7 assures us

$$(A \in U \rightarrow \forall x(x \subset A \wedge \text{sng } x \in U \wedge \text{sng } x) \in U).$$

Helped by 2.34.6, the Theorem of Classification, 2.40.3, and the Theorem of Heredity we infer

$$\begin{aligned} (A \in U \rightarrow \text{sb } A &= \forall x(x \in \text{sb } A \wedge \text{sng } x) \\ &= \forall x(x \subset A \wedge x \in U \wedge \text{sng } x) \\ &= \forall x(x \subset A \wedge x \in U \wedge \text{sng } x \in U \wedge \text{sng } x) \\ &= \forall x(x \subset A \wedge \text{sng } x \in U \wedge \text{sng } x) \in U). \end{aligned}$$

Thus

$$.0 \quad (A \in U \rightarrow \text{sb } A \in U).$$

Because of the Theorem of Replacement

$$\begin{aligned} (A \in U \rightarrow \nabla A &= \forall x(x \in A \wedge x) \\ &= \forall x(x \in A \wedge x \in U \wedge x) \in U) \end{aligned}$$

and hence

$$.1 \quad (A \in U \rightarrow \nabla A \in U).$$

Because of .1 and 2.37.16

$$.2 \quad (\text{sb } A \in U \rightarrow A = \nabla \text{sb } A \in U).$$

The desired conclusion now follows from .0 and .2.

From 2.44.1 on the one hand, and from 2.44.0, 2.37.16, and the Theorem of Heredity on the other, we infer

2.45 THE THEOREM OF AMALGAMATION.

$$(A \in U \leftrightarrow \nabla A \in U)$$

2.46 DEFINITION. $(2B \text{ } abx \equiv (a \wedge x = 0 \vee b \wedge x = \text{sng } 0))$

2.47 LEMMAS.

- .0 $(0 \in U \wedge \text{sng } 0 \in U \wedge \text{sb sng } 0 \in U \wedge 0 \in \text{sng } 0 \wedge 0 \neq \text{sng } 0)$
- .1 $(2B \text{ } ab0 = a)$
- .2 $(2B \text{ } ab \text{ sng } 0 = b)$
- .3 $(2B \text{ } abx \neq 0 \rightarrow x = 0 \vee x = \text{sng } 0)$
- .4 $(0 \in \text{sb sng } 0 \wedge \text{sng } 0 \in \text{sb sng } 0)$
- .5 $(2B \text{ } abx = (x \in \text{sb sng } 0 \wedge 2B \text{ } abx))$
- .6 $(a \in U \wedge b \in U \rightarrow 2B \text{ } abx \in U)$

2.48 THE THEOREM OF UNIONS.

$$(a \in U \wedge b \in U \leftrightarrow a \cup b \in U)$$

This follows from the Theorem of Heredity and:

Lemma. $(a \in U \wedge b \in U \rightarrow a \cup b \in U)$

Proof. Using 2.47.5 we see

$$\begin{aligned} (\forall x \text{ } 2B \text{ } abx &= \forall x(x \in \text{sb sng } 0 \wedge 2B \text{ } abx) \\ &= \forall x(\text{sb sng } 0 \in U \wedge x \in \text{sb sng } 0 \wedge \\ &\quad 2B \text{ } abx \in U \wedge 2B \text{ } abx \in U). \end{aligned}$$

But 2.47.1 and 2.47.2 tell us

$$(a \cup b = 2B \text{ } ab0 \cup 2B \text{ } ab \text{ sng } 0 \subset \forall x \text{ } 2B \text{ } abx \in U).$$

2.49 THEOREM. $\sim(U \in U)$

Proof. From the Theorem of Heredity we have

$$(U \in U \rightarrow x \subset U \in U \rightarrow x \in U).$$

Hence

$$(x \sim \in U \rightarrow U \sim \in U).$$

Accordingly, because of 2.38,

$$(\alpha = Ex \sim (x \in x) \rightarrow \alpha \sim \in U \rightarrow U \sim \in U),$$

and the desired conclusion is at hand.

SINGLETONS

For the sake of completeness we restate in 2.51 and 2.54.10 results known earlier in 2.34 and 2.37.19. We shall use 2.34 without further reference.

The inverted word order in 2.50.1, for example, obviates parentheses.

2.50 DEFINITIONS.

- .0 $(\text{sngl } x \equiv \wedge y(x \in y \rightarrow y))$
- .1 $(\text{singleton is } a \equiv (\nabla a = \text{TT}a))$

2.51 THEOREMS.

- .0 $(y \in U \rightarrow y \in \text{sng } x \leftrightarrow y = x)$
- .1 $(x \in U \rightarrow y \in \text{sngl } x \leftrightarrow y = x)$
- .2 $(y \in \text{sng } x \leftrightarrow y = x \wedge y \in U)$

2.52 LEMMAS.

- .0 $(\text{singleton is } a \rightarrow a \neq 0)$
- .1 $(\text{singleton is } a \wedge y \in a \rightarrow y = \text{TT}a = \nabla a)$

Proof. See 2.24.33.

2.53 LEMMA. $(x \in U \rightarrow \text{singleton is sng } x)$

Theorem 2.54.8 is included for the sake of completeness, not utility.

In checking 2.54.6 we use 2.52 and the Theorem of Amalgamation. In connection with 2.54.7 we can infer, for example, that

$$(a = \text{sngl } \top\top a \rightarrow \top\top a \in U)$$

by noticing that

$$(a = \text{sngl } \top\top a \rightarrow \top\top a = \top\top \text{sngl } \top\top a)$$

and then using 2.54.3.

2.54 THEOREMS.

- .0** $(x \cup \in U \rightarrow \text{sng } x = 0 \wedge \text{sngl } x = U)$
- .1** $(x \in U \leftrightarrow \text{sngl } x \in U \leftrightarrow \text{sng } x = \text{sngl } x)$
- .2** $(x \in U \leftrightarrow x \in \text{sng } x \leftrightarrow x \in \text{sngl } x)$
- .3** $(x \in U \leftrightarrow x = \nabla \text{sng } x \leftrightarrow x = \top\top \text{sngl } x)$
- .4** $(\text{sng } x \in U)$
- .5** $(\text{sngl } x \neq 0)$
- .6** $(\text{singleton is } a \leftrightarrow \nabla a = \top\top a \in a \in U)$
- .7** $(\text{singleton is } a \leftrightarrow a = \text{sng } \nabla a \leftrightarrow a = \text{sngl } \top\top a)$
- .8** $(\text{singleton is } a \leftrightarrow \nabla y \wedge x(x \in a \leftrightarrow x = y) \leftrightarrow a \neq 0 \wedge \wedge x \wedge y(x \in a \wedge y \in a \rightarrow x = y))$
- .9** $(x \in U \leftrightarrow \text{singleton is sng } x \leftrightarrow \text{singleton is sngl } x)$
- .10** $(\text{sng } x = \text{E}y(y = x))$
- .11** $(0 \neq a \subset \text{sng } x \rightarrow a = \text{sng } x)$

ORDERED PAIRS

In connection with 2.57.1 recall 0.58, 0.60, 0.62, and the relevant examples.

The preliminary ordered pair,

$$(x, y),$$

is described by 2.56.0 and is due to N. Wiener. It naturally gives rise to a preliminary Cartesian product,

$$(a \text{ ,, } b),$$

which behaves beautifully under nonvacuous intersection and nesting union, and falters as an ordered pair only when one of the coordinates is 0. This defect can be remedied by one-to-one correlating with each set a nonvacuous set. Depending on the correlation chosen, the new ordered pair may have much or little in common with the preliminary Cartesian product from which it is fashioned. In 1949 J. W. Weihe hit upon the splendid idea of correlating with each set a the nonvacuous set $sb\ a$. The ordered pair of Weihe,

$$(sb\ a \text{ ,, } sb\ b),$$

behaves smoothly under nonvacuous intersection but erratically under nonvacuous nesting union. D. C. Peterson and I hit upon the idea of correlating with each set a the nonvacuous set $ss\ a$, described by 2.57.0, and of reaching (a, b) by way of 2.57.1. Because of 2.60 and especially 2.60.2, the correlation we use is somewhat more of a homomorphism than that of Weihe, and our ordered pair behaves smoothly under both nonvacuous intersection and nonvacuous union.

We are trying here to fashion an ordered pair which will stand up under strain. As an ordered pair,

$$((sng\ 0 \text{ ,, } a) \cup (sng\ sng\ 0 \text{ ,, } b))$$

qualifies under 2.61 and 2.62 but fails under 2.63. As a rather attractive ordered pair,

$$((ss\ a \text{ ,, } ss\ 0) \cup (ss\ 0 \text{ ,, } ss\ b))$$

qualifies under 2.61, 2.62, and 2.63 but fails in some respects as a 2-tuple. We have so arranged things that (a, b) qualifies as a 2-tuple and we shall later so arrange things that (a, b, c) qualifies as a 3-tuple.

Theorems 2.63 facilitate constructions.

The reader uninterested in 2.62 is advised to ignore 2.59.10–2.59.15 and 2.60.2–2.60.4.

2.55 DEFINITIONS.

- .0 $(\{x\} \equiv \text{sngl } x)$
- .1 $(\{xx'\} \equiv (\text{sngl } x \cup \text{sngl } x'))$
- .2 $(\{xx'x''\} \equiv (\text{sngl } x \cup \text{sngl } x' \cup \text{sngl } x''))$
- etc.

2.56 DEFINITIONS.

- .0 $((x, y) \equiv \{\{x\}\{xy\}\})$
- .1 $(\text{basicorderedpair } xy \equiv (x, y))$
- .2 $(\text{basicorderedpair is } p \equiv \forall x \forall y (p = x, y \in U))$
- .3 $(\text{basicrelation is } R \equiv \wedge p (p \in R \rightarrow \text{basicorderedpair is } p))$
- .4 $(\text{bsvs } Rx \equiv \exists y (x, y \in R))$
- .5 $(\text{The basic vertical section of } R \text{ at } x \equiv \text{bsvs } Rx)$

2.57 DEFINITIONS.

- .0 $(ss \ a \equiv (\text{sng } 0 \cup \forall x (x \in a \wedge \text{sng } \text{sng } x)))$
- .1 $((a, b) \equiv ((\text{sng } 0, ss \ a) \cup (\text{sng } \text{sng } 0, ss \ b)))$
- .2 $(\text{orderedpair } ab \equiv (a, b))$
- .3 $(\text{orderedpair is } p \equiv \forall a \forall b (p = a, b))$
- .4 $(\text{crd}' \ p \equiv \nabla \text{bsvs } p \ 0)$
- .5 $(\text{The first coordinate of } p \equiv \text{crd}' \ p)$
- .6 $(\text{crd}'' \ p \equiv \nabla \text{bsvs } p \ \text{sng } 0)$
- .7 $(\text{The second coordinate of } p \equiv \text{crd}'' \ p)$

2.58 LEMMAS.

- .0 $(x, y \in U \leftrightarrow x \in U \wedge y \in U)$
- .1 $(p = x, y \in U \rightarrow \nabla p = \{xy\} \wedge \prod p = \{x\} \wedge \nabla \nabla p = x \cup y \wedge \nabla \prod p = x \wedge \prod \nabla p = x \cap y \wedge \prod \prod p = x)$
- .2 $(p = x, y \in U \rightarrow x = \prod \prod p \wedge y = \nabla \nabla p \sim \nabla \prod p \cup \prod \nabla p)$
- .3 $(x, y = u, v \in U \leftrightarrow x = u \in U \wedge y = v \in U)$
- .4 $(x, y \neq 0)$

2.59 LEMMAS.

$$.0 \quad (x, y \in E x, y \underline{u'}xy \leftrightarrow \underline{u'}xy \wedge x, y \in U)$$

Proof. Again recall 0.58, 0.60, and the relevant examples.

$$\begin{aligned}
 & (z \in E s, t \underline{u'}st) \\
 & \leftrightarrow z \in Ez \vee s \vee t(z = s, t \wedge \underline{u'}st) \\
 & \leftrightarrow \vee s \vee t(z = s, t \wedge \underline{u'}st) \wedge z \in U \\
 & \leftrightarrow \vee s \vee t(\underline{u'}st \wedge z = s, t \in U) \\
 \\
 & (x, y \in E s, t \underline{u'}st \\
 & \leftrightarrow \vee s \vee t(\underline{u'}st \wedge x, y = s, t \in U) \\
 & \leftrightarrow \vee s \vee t(\underline{u'}st \wedge x = s \in U \wedge y = t \in U) \\
 & \leftrightarrow \vee s \vee t(\underline{u'}xy \wedge x \in U \wedge y \in U \wedge x = s \wedge y = t) \\
 & \leftrightarrow \underline{u'}xy \wedge x \in U \wedge y \in U \wedge \vee s \vee t(x = s \wedge y = t) \\
 & \leftrightarrow \underline{u'}xy \wedge x, y \in U) \\
 \\
 & (x, y \in E s, t \underline{u'}st \leftrightarrow \underline{u'}xy \wedge x, y \in U) \\
 & (x, y \in E x, y \underline{u'}xy \leftrightarrow \underline{u'}xy \wedge x, y \in U)
 \end{aligned}$$

$$.1 \quad (A \,, B = \vee x \vee y(x \in A \wedge y \in B \wedge \text{sng}(x, y)))$$

Proof.

$$\begin{aligned}
 (A \,, B &= E x, y(x \in A \wedge y \in B) \\
 &= Ez \vee x \vee y(z = x, y \wedge x \in A \wedge y \in B) \\
 &= \vee x \vee y Ez(z = x, y \wedge x \in A \wedge y \in B) \\
 &= \vee x \vee y(x \in A \wedge y \in B \wedge Ez(z = x, y)) \\
 &= \vee x \vee y(x \in A \wedge y \in B \wedge \text{sng}(x, y)))
 \end{aligned}$$

$$.2 \quad (x, y \in A \,, B \leftrightarrow x \in A \wedge y \in B)$$

$$.3 \quad (\text{bsvs}(p \cup q)x = \text{bsvs } px \cup \text{bsvs } qx)$$

$$.4 \quad (x \in A \rightarrow \text{bsvs}(A \,, B)x = B)$$

$$.5 \quad (x \sim \in A \rightarrow \text{bsvs}(A \,, B)x = 0)$$

$$.6 \quad (R \subset E x, y \underline{u'}xy \rightarrow \text{basicrelation is } R)$$

Hint. Glance at the first formula in the proof of .0.

$$.7 \quad \sim(0 \in A \,, B)$$

$$.8 \quad (p = 0 \vee p = U \rightarrow \nabla(p \wedge c) = (p \wedge \nabla c))$$

$$.9 \quad (A \neq 0 \neq B \rightarrow \nabla \nabla \nabla(A \,, B) = \nabla A \cup \nabla B)$$

Hint. Use .1, 2.24.38, 2.58.1 and the fact that

$$(\forall x(x \in A) = U = \forall y(y \in B)).$$

- .10 (basicrelation is $R \rightarrow$
 $R \subset S \leftrightarrow \bigwedge x \bigwedge y (x, y \in R \rightarrow x, y \in S)$)
- .11 (basicrelation is $R \wedge$ basicrelation is $S \rightarrow$
 $R = S \leftrightarrow \bigwedge x \bigwedge y (x, y \in R \leftrightarrow x, y \in S)$)
- .12 $(A \cup B) \cup C = (A \cup C) \cup (B \cup C)$
- .13 $(A \cap B) \cap C = (A \cap C) \cap (B \cap C)$
- .14 $((A \cap B) \cap C) = A \cap (B \cap C)$
- .15 $(A \cap B = 0 \leftrightarrow A = 0 \vee B = 0)$

2.60 LEMMAS.

- .0 $(0 \in ss\ a)$
- .1 $(\forall ss\ a = a)$
- .2 $(ss\ (a \cup b) = ss\ a \cup ss\ b)$
- .3 $(a \subset b \leftrightarrow ss\ a \subset ss\ b)$
- .4 $(ss\ (a \cap b) = ss\ a \cap ss\ b)$

Hint. First check that

$$(sng\ 0 \cap (x \in a \wedge sng\ sng\ x) = 0)$$

and that

$$\begin{aligned} &(\forall y(x \in a \wedge sng\ sng\ x \wedge y \in b \wedge sng\ sng\ y) \\ &= (x \in a \wedge x \in b \wedge sng\ sng\ x) \\ &= (x \in a \cap b \wedge sng\ sng\ x)). \end{aligned}$$

- .5 $(\forall \forall \forall (a, b) = sng\ 0 \cup a \cup b)$
- .6 $(bsvs\ (a, b)\ 0 = ss\ a \wedge bsvs\ (a, b)\ sng\ 0 = ss\ b)$
- .7 $\sim(0 \in a, b)$
- .8 $(sng\ sng\ 0 \in a, b)$

Proof. Using .0 and 2.59.2 we see

$$(sng\ sng\ 0 = 0, 0 \in sng\ 0, ss\ a \subset a, b).$$

2.61 THE ORDERED PAIR THEOREMS.

- .0 $(a, b \in U \leftrightarrow a \in U \wedge b \in U)$
- .1 $(\text{crd}'(a, b) = a \wedge \text{crd}''(a, b) = b)$
- .2 $(a, b = c, d \leftrightarrow a = c \wedge b = d)$

As developed by Kelley⁵ the ordered pair introduced by 2.56.0 has many pleasant features. However, it is a defect in 2.56.0 that from Theorem 2.61.2, which enhances our theory of notation, a theorem is *not* obtained by replacing ‘,’ by ‘,’.

Of interest to us but unused hereinafter are

2.62 THEOREMS.

- .0 $(a, b \subset c, d \leftrightarrow a \subset c \wedge b \subset d)$
- .1 $(\text{crd}'(p \cup q) = \text{crd}' p \cup \text{crd}' q \wedge \text{crd}''(p \cup q) = \text{crd}'' p \cup \text{crd}'' q)$
- .2 $(p \subset q \rightarrow \text{crd}' p \subset \text{crd}' q \wedge \text{crd}'' p \subset \text{crd}'' q)$

2.63 THEOREMS.

- .0 (orderedpair is $p \rightarrow 0 \sim \in p$)
- .1 (orderedpair is $p \rightarrow \text{sng sng } 0 \in p$)
- .2 $\sim$ orderedpair is 0

SUBSTITUTION**2.64 THEOREMS.**

- .0 (st $yx \underline{ux} = \underline{uy}$)

Proof. After glancing back at 0.57 we check

$$\begin{aligned}
 (\text{st } yx \underline{ux} &= \vee x(y = x \wedge \underline{ux}) \\
 &= \vee x(y = x \wedge \underline{uy}) \\
 &= \underline{uy} \cap \vee x(y = x) \\
 &= \underline{uy} \cap U \\
 &= \underline{uy}).
 \end{aligned}$$

⁵ J. L. Kelley, *General Topology*, p. 259, Van Nostrand, Princeton, New Jersey, 1955, or Springer-Verlag, New York, 1975.

- .1 (st ty $\underline{u'xy} = \underline{u'xt}$)
- .2 (st (s, t) x, y $\underline{u'xy} = \underline{u'st}$)
- .3 ($\sim$ orderedpair is $z \rightarrow$ st z x, y $\underline{u'xy} = 0$)
- .4 ($x = y \wedge \underline{ux} \rightarrow$ st yx $\underline{ux}$)

2.65 LEMMAS.

- .0 ($\sim \wedge z ; \underline{uz} \underline{vz} = \vee z ; \underline{uz} \sim \underline{vz}$)
- .1 ($\wedge z ;$ st z x, y $\underline{u'xy} \sim$ st z x, y $\underline{v'xy} =$
 $\wedge z ;$ st z x, y $\underline{u'xy}$ st z x, y $\sim \underline{v'xy}$)

Hint. (orderedpair is $z \vee \sim$ orderedpair is z)

Theorem 2.61.2 gives rise to 2.64.2 which fortunately gives rise in turn to:

2.66 THEOREM. ($\wedge x, y ; \underline{u'xy} \underline{v'xy} = \wedge x \wedge y ; \underline{u'xy} \underline{v'xy}$)

This follows from our theory of notation and:

Lemma.

$$(\wedge x \wedge y \wedge z (\underline{w'xy} = (0 \in \underline{u'xy} \rightarrow \underline{v'xy}) \wedge \\ \underline{uz} = (0 \in \text{st z x, y } \underline{u'xy} \rightarrow \text{st z x, y } \underline{v'xy})) \rightarrow \\ \wedge x \wedge y \underline{w'xy} = \wedge z \underline{uz})$$

Proof.

- ($c = a, b \rightarrow \wedge z \underline{uz} \subset \underline{uc} = \underline{w'ab}$)
- ($c = a, b \rightarrow \wedge z \underline{uz} \subset \underline{w'ab}$)
- .0 ($\wedge z \underline{uz} \subset \underline{w'ab}$)
- ($c = a, b \rightarrow \wedge x \wedge y \underline{w'xy} \subset \underline{w'ab} = \underline{uc}$)
- ($c = a, b \rightarrow \wedge x \wedge y \underline{w'xy} \subset \underline{uc}$)
- (orderedpair is $c \rightarrow \wedge x \wedge y \underline{w'xy} \subset \underline{uc}$)
- ($\sim$ orderedpair is $c \rightarrow \wedge x \wedge y \underline{w'xy} \subset \underline{U} = \underline{uc}$)
- .1 ($\wedge x \wedge y \underline{w'xy} \subset \underline{uc}$)

Using .0 and .1 we conclude

$$(\wedge z \underline{uz} \subset \wedge a \wedge b \underline{w'ab} = \wedge x \wedge y \underline{w'xy} \subset \wedge c \underline{uc} = \wedge z \underline{uz}).$$

Because of 2.65 we can now easily check

2.67 THEOREM. $(\forall x, y; \underline{u'xy} \underline{v'xy} = \forall x \forall y; \underline{u'xy} \underline{v'xy})$

Illustrative of our theory of notation are

2.68 THEOREMS.

.0 $(\wedge x \cap y; \underline{u'xy} \underline{v'xy} = \wedge x \wedge y; \underline{u'xy} \underline{v'xy})$

.1 $(\wedge x \cap y \underline{u'xy} = \wedge x, y \underline{u'xy} = \wedge x \wedge y \underline{u'xy})$

.2 $(\wedge x \cap y \in A \underline{u'xy} = \wedge x \wedge y(x \cap y \in A \rightarrow \underline{u'xy}))$

.3 $(\forall x \cap y \in A \underline{u'xy} = \wedge x \wedge y(x \cap y \in A \wedge \underline{u'xy}))$

A different favor is found in

2.69 THEOREMS.

.0 $(E x \cup y \underline{u'xy} = E z \forall x \forall y(z = x \cup y \wedge \underline{u'xy}))$

.1 $(E x \cup y \in A \underline{u'xy} = E z \forall x \forall y(z = x \cup y \in A \wedge \underline{u'xy}))$

2.70 THEOREMS.

.0 $(\wedge x; \underline{vx} E y \underline{u'xy} = E y \wedge x; \underline{vx} \underline{u'xy})$

.1 $(\forall x; \underline{vx} E y \underline{u'xy} = E y \forall x; \underline{vx} \underline{u'xy})$

.2 $(y \in U \rightarrow y \in \wedge x; \underline{vx} \underline{ux} \leftrightarrow \wedge x; \underline{vx} (y \in \underline{ux}))$

.3 $(y \in \forall x; \underline{vx} \underline{ux} \leftrightarrow \forall x; \underline{vx} (y \in \underline{ux}))$

.4 $(\prod A = \wedge x \in A x)$

.5 $(\nabla A = \forall x \in A x)$

.6 $(\prod \forall x; \underline{vx} \underline{ux} = \wedge x; \underline{vx} \prod \underline{ux})$

.7 $(\nabla \forall x; \underline{vx} \underline{ux} = \forall x; \underline{vx} \nabla \underline{ux})$

.8 $(\forall x \underline{vx} \rightarrow \wedge x; \underline{vx} a = a = \forall x; \underline{vx} a)$

.9 $(\sim \forall x \underline{vx} \rightarrow \wedge x; \underline{vx} \underline{ux} = U \wedge \forall x; \underline{vx} \underline{ux} = 0)$

.10 $(\sim \wedge x; \underline{vx} \underline{ux} = \forall x; \underline{vx} \sim \underline{ux})$

.11 $(\sim \forall x; \underline{vx} \underline{ux} = \wedge x; \underline{vx} \sim \underline{ux})$

- .12 $(\wedge x; \underline{w}x (\underline{u}x \cap \underline{v}x) = \wedge x; \underline{w}x \underline{u}x \cap \wedge x; \underline{w}x \underline{v}x)$
- .13 $(\vee x; \underline{w}x (\underline{u}x \cup \underline{v}x) = \vee x; \underline{w}x \underline{u}x \cup \vee x; \underline{w}x \underline{v}x)$
- .14 $(\wedge x; \underline{w}x (\underline{u}x \subset \underline{v}x) \rightarrow$
 $\wedge x; \underline{w}x \underline{u}x \subset \wedge x; \underline{w}x \underline{v}x \wedge \vee x; \underline{w}x \underline{u}x \subset \vee x; \underline{w}x \underline{v}x)$
- .15 $(\underline{v}x \rightarrow \wedge x; \underline{v}x \underline{u}x \subset \underline{u}x \subset \vee x; \underline{v}x \underline{u}x)$
- .16 $(\wedge x; (x = y) x = y = \vee x; (x = y) x)$
- .17 $(\wedge x(\underline{u}x \leftrightarrow \underline{v}x) \rightarrow$
 $\wedge x; \underline{u}x \underline{w}x = \wedge x; \underline{v}x \underline{w}x \wedge \vee x; \underline{u}x \underline{w}x = \vee x; \underline{v}x \underline{w}x)$
- .18 $(\wedge x; \underline{u}x (y \cup \underline{v}x) = y \cup \wedge x; \underline{u}x \underline{v}x)$
- .19 $(\vee x; \underline{u}x (y \cap \underline{v}x) = y \cap \vee x; \underline{u}x \underline{v}x)$
- .20 $(\wedge x; \underline{u}x \wedge y; \underline{v}y \underline{u}'xy = \wedge y; \underline{v}y \wedge x; \underline{u}x \underline{u}'xy)$
- .21 $(\vee x; \underline{u}x \vee y; \underline{v}y \underline{u}'xy = \vee y; \underline{v}y \vee x; \underline{u}x \underline{u}'xy)$
- .22 $(\vee x; \underline{u}x \wedge y; \underline{v}y \underline{u}'xy \subset \wedge y; \underline{v}y \vee x; \underline{u}x \underline{u}'xy)$
- .23 $(\nabla \wedge x; \underline{v}x \underline{u}x \subset \wedge x; \underline{v}x \nabla \underline{u}x)$
- .24 $(\top \wedge x; \underline{v}x \underline{u}x \supset \vee x; \underline{v}x \top \underline{u}x)$
- .25 $(y \subset \wedge x; \underline{v}x \underline{u}x \leftrightarrow \wedge x; \underline{v}x(y \subset \underline{u}x))$
- .26 $(sb \wedge x; \underline{v}x \underline{u}x = \wedge x; \underline{v}x sb \underline{u}x)$
- .27 $(\wedge x \in U(\underline{u}x \rightarrow \underline{v}x) \leftrightarrow Ex \underline{u}x \subset Ex \underline{v}x)$
- .28 $(\wedge x \in U(\underline{u}x \leftrightarrow \underline{v}x) \leftrightarrow Ex \underline{u}x = Ex \underline{v}x)$

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Persuaded by R. A. Alps, I have recast 2.71 in such a way that the definiendum in 2.71.0 has, in common with other set-theoretic statementlike definienda, the Kronecker character fundamentally illustrated by 2.24.50 and 2.24.51.

2.71 DEFINITIONS.

- .0 $(\text{One } x \underline{u}x \equiv \vee y \wedge x(0 \in \underline{u}x \leftrightarrow x = y))$
- .1 $(\text{There is just one } x \text{ such that } \underline{u}x \equiv \text{One } x \underline{u}x)$
- .2 $(\text{The } x \underline{u}x \equiv (\text{One } x \underline{u}x \rightarrow \wedge x; \underline{u}x x))$

2.72 LEMMAS.**.0** $(\text{One } x \underline{ux} \wedge \underline{ux} \rightarrow \underline{uz} \leftrightarrow z = x)$ **Proof.**

$$\begin{aligned}
& (\wedge x(\underline{ux} \leftrightarrow x = y) \rightarrow (\underline{ux} \rightarrow x = y) \wedge (\underline{uz} \rightarrow z = y) \\
& \quad \rightarrow (\underline{ux} \wedge \underline{uz} \rightarrow x = y \wedge z = y) \\
& \quad \rightarrow (\underline{ux} \wedge \underline{uz} \rightarrow z = x)) \\
& (\wedge x(\underline{ux} \leftrightarrow x = y) \rightarrow (\underline{ux} \wedge \underline{uz} \rightarrow z = x)) \\
& (\vee y \wedge x(\underline{ux} \leftrightarrow x = y) \rightarrow (\underline{ux} \wedge \underline{uz} \rightarrow z = x)) \\
& (\text{One } x \underline{ux} \rightarrow (\underline{ux} \wedge \underline{uz} \rightarrow z = x)) \\
& (\text{One } x \underline{ux} \wedge \underline{ux} \rightarrow (\underline{uz} \rightarrow z = x))
\end{aligned}$$

Also

$$(\text{One } x \underline{ux} \wedge \underline{ux} \rightarrow (z = x \rightarrow \underline{uz}))$$

and the desired conclusion is at hand.

.1 $(\text{One } x \underline{ux} \wedge \underline{ux} \rightarrow x = \text{The } x \underline{ux})$ **Proof.** From .0 we learn

$$(\underline{uz} \leftrightarrow z = x).$$

Consequently, with the help of 2.70.17 and 2.70.16 we infer

$$(\text{The } x \underline{ux} = \wedge x; \underline{ux} x = \wedge z; \underline{uz} z = \wedge z; (z = x) z = x).$$

.2 $(\text{One } x \underline{ux} \wedge \underline{ux} \rightarrow (\underline{uz} \leftrightarrow z = \text{The } x \underline{ux}))$ **Proof.** $(\underline{uz} \leftrightarrow z = x \leftrightarrow z = \text{The } x \underline{ux})$ **.3** $(\text{One } x \underline{ux} \rightarrow \vee x \underline{ux})$

For the sake of completeness, not utility, we state

2.73 THEOREM.

$$(\text{One } x \underline{ux} \leftrightarrow \vee x \underline{ux} \wedge \wedge x \wedge y(\underline{ux} \wedge \underline{uy} \rightarrow x = y))$$

2.74 THE THEOREM OF UNICITY.

$$(\text{One } x \underline{u}x \rightarrow \underline{u}x \leftrightarrow x = \text{The } x \underline{u}x)$$

Proof. Because of 2.72.2

$$(\text{One } x \underline{u}x \wedge \forall x \underline{u}x \rightarrow \underline{u}z \leftrightarrow z = \text{The } x \underline{u}x).$$

Because of this and 2.72.3

$$(\underline{u}z \leftrightarrow z = \text{The } x \underline{u}x).$$

The desired conclusion is at hand.

2.75. THEOREM. $(\sim \text{One } x \underline{u}x \rightarrow \text{The } x \underline{u}x = U)$

RELATIONS

In connection with 2.76.19, again recall 0.62.

2.76 DEFINITIONS.

- .0** (relation is $R \equiv \wedge x \in R$ orderedpair is x)
- .1** (relation $RS \equiv (\text{relation is } R \wedge \text{relation is } S)$)
- .2** ($\text{dmn } R \equiv \exists x \forall y(x, y \in R)$)
- .3** (The domain of $R \equiv \text{dmn } R$)
- .4** ($\text{rng } R \equiv \exists y \forall x(x, y \in R)$)
- .5** (The range of $R \equiv \text{rng } R$)
- .6** ($\text{fld } R \equiv (\text{dmn } R \cup \text{rng } R)$)
- .7** (The field of $R \equiv \text{fld } R$)
- .8** ($\text{vs } Rx \equiv \exists y(x, y \in R)$)
- .9** (The vertical section of R at $x \equiv \text{vs } Rx$)
- .10** (The set of points which come after x under $R \equiv \text{vs } Rx$)
- .11** ($\text{hs } Ry \equiv \exists x(x, y \in R)$)
- .12** (The horizontal section of R at $y \equiv \text{hs } Ry$)
- .13** (The set of points which come before y under $R \equiv \text{hs } Ry$)

- .14 ($\text{inv } R \equiv \exists x, y (y, x \in R)$)
- .15 (The inverse of $R \equiv \text{inv } R$)
- .16 ($((R : S) \equiv \exists x, y \forall z (x, z \in S \wedge z, y \in R))$)
- .17 ($((R \text{ composed with } S) \equiv (R : S))$)
- .18 ($((R : S) \equiv (S : R))$)
- .19 ($\text{rct } AB \equiv (A \text{ ,, } B)$)
- .20 ($\text{rectangle } AB \equiv \text{rct } AB$)
- .21 ($\text{sqr } A \equiv \text{rct } AA$)
- .22 ($\text{square } A \equiv \text{sqr } A$)
- .23 ($\text{strc } RA \equiv (R \cap \text{rct } AU)$)
- .24 ($\text{strict } RA \equiv \text{strc } RA$)
- .25 (The restriction of R to $A \equiv \text{strc } RA$)
- .26 ($\text{strn } RB \equiv (R \cap \text{rct } UB)$)
- .27 ($\text{strun } RB \equiv \text{strn } RB$)
- .28 (The restriction in range of R to $B \equiv \text{strn } RB$)
- .29 ($*RA \equiv \forall x \in A \text{ vs } Rx$)
- .30 (The image of A under $R \equiv *RA$)
- .31 ($*RB \equiv \forall y \in B \text{ hs } Ry$)
- .32 (The inverse image of B under $R \equiv *RB$)

By replacing ‘,’ by ‘,’ in the proof of 2.59.0 we obtain a proof of

2.77 THEOREM. $(x, y \in E \text{ } x, y \text{ } \underline{u'}xy \leftrightarrow \underline{u'}xy \wedge x, y \in U)$

2.78 THEOREMS.

- .0 relation is $E \text{ } x, y \text{ } \underline{u'}xy$
- .1 relation is 0
- .2 (relation is $R \rightarrow R = E \text{ } x, y (x, y \in R)$)

2.79 THEOREMS.

- .0 (relation is $S \wedge R \subset S \rightarrow$ relation is R)
- .1 (relation is $R \rightarrow R \subset S \leftrightarrow \wedge x, y \in R (x, y \in S)$)
- .2 (relation $RS \rightarrow R = S \leftrightarrow \wedge x, y (x, y \in R \leftrightarrow x, y \in S)$)

- .3 (relation is $R \wedge \text{dmn } R \subset A \rightarrow$
 $R \subset S \leftrightarrow \bigwedge x \in A (\text{vs } Rx \subset \text{vs } Sx))$
- .4 (relation is $R \wedge \text{rng } R \subset B \rightarrow$
 $R \subset S \leftrightarrow \bigwedge y \in B (\text{hs } Ry \subset \text{hs } Sy))$
- .5 ($\text{dmn strc } RA = A \cap \text{dmn } R$)
- .6 ($\text{rng strn } RB = B \cap \text{rng } R$)
- .7 ($\text{dmn } (R \cup S) = \text{dmn } R \cup \text{dmn } S$)
- .8 ($\text{rng } (R \cup S) = \text{rng } R \cup \text{rng } S$)

2.80 THEOREMS.

- .0 ($\text{dmn } R = \text{Ex}(\text{vs } Rx \neq 0) \wedge \text{rng } R = \text{Ey}(\text{hs } Ry \neq 0))$
- .1 (relation is $R \rightarrow \text{dmn } R = \bigvee p \in R \text{ sng crd}' p \wedge$
 $\text{rng } R = \bigvee p \in R \text{ sng crd}'' p$)
- .2 ($\text{dmn } R = \text{dmn } (R \cap \text{sqr } U) \wedge \text{rng } R = \text{rng } (R \cap \text{sqr } U)$)
- .3 ($R \in U \rightarrow \text{dmn } R \in U \wedge \text{rng } R \in U$)
- .4 ($0 \neq A \text{ ,, } B = S \rightarrow \text{dmn } S = A \wedge \text{rng } S = B$)
- .5 (relation is $R \rightarrow \text{dmn } R = 0 \vee \text{rng } R = 0 \leftrightarrow R = 0$)
- .6 ($\text{sqr } A = \text{sqr } B \leftrightarrow A = B$)

2.81 THEOREMS.

- .0 ($\text{dmn inv } R = \text{rng } R \wedge \text{rng inv } R = \text{dmn } R$)
- .1 (relation is $R \rightarrow \text{inv inv } R = R$)
- .2 ($(R : (S : T) = (R : S) : T = R : S : T)$)
- .3 ($\text{inv } (R : S) = \text{inv } S : \text{inv } R$)
- .4 ($A \cup B \text{ ,, } C = (A \text{ ,, } C) \cup (B \text{ ,, } C)$)
- .5 ($A \text{ ,, } B \cup C = (A \text{ ,, } B) \cup (A \text{ ,, } C)$)
- .6 ($((A \text{ ,, } B) \cap (C \text{ ,, } D) = A \cap C \text{ ,, } B \cap D)$)

2.82 THEOREMS.

- .0 (relation is $R \rightarrow \text{strc } R \sim A = R \smile \text{strc } RA$)
- .1 (relation is $R \rightarrow \text{strn } R \sim B = R \smile \text{strn } RB$)
- .2 ($\text{strc } R (A \cap C) = \text{strc } RA \cap \text{strc } RC$)
- .3 ($\text{strn } R (B \cap C) = \text{strn } RB \cap \text{strn } RC$)

- .4 $(x \in A \rightarrow \text{vs strc } RA \ x = \text{vs } Rx)$
- .5 $(y \in B \rightarrow \text{hs strn } RB \ y = \text{hs } Ry)$
- .6 $(\text{relation is } R \wedge \text{dmn } R \subset A \rightarrow \text{strc } RA = R)$
- .7 $(\text{relation is } R \wedge \text{rng } R \subset B \rightarrow \text{strn } RB = R)$
- .8 $(A \subset C \rightarrow * \text{strc } RC \ A = *RA)$
- .9 $(B \subset C \rightarrow * \text{strn } RC \ B = *RB)$
- .10 $(*RA = \text{E}y \vee x \in A \ (x, y \in R) \wedge *RB = \text{E}x \vee y \in B \ (x, y \in R))$
- .11 $(\text{vs } Rx = *R \ \text{sng } x \wedge \text{hs } Ry = *R \ \text{sng } y)$
- .12 $(*RA = * \text{inv } R \ A \wedge *RB = * \text{inv } R \ B)$
- .13 $(*RU = \text{rng } R \wedge *RU = \text{dmn } R \wedge$
 $*R(U \smallfrown \text{dmn } R) = *R(U \smallfrown \text{rng } R) = 0)$
- .14 $(*R0 = *R0 = 0)$
- .15 $(*R(A \cup C) = *RA \cup *RC)$
- .16 $(*R(B \cup C) = *RB \cup *RC)$
- .17 $(*R \vee x; \underline{v}x \ \underline{u}x = \vee x; \underline{v}x \ *R \ \underline{u}x)$
- .18 $(*R \ \vee x; \underline{v}x \ \underline{u}x = \vee x; \underline{v}x \ *R \ \underline{u}x)$
- .19 $(A \subset B \rightarrow *RA \subset *RB \wedge *RA \subset *RB)$
- .20 $(*R(A \cap C) \subset *RA \cap *RC)$
- .21 $(*R(B \cap C) \subset *RB \cap *RC)$
- .22 $(*R \ \wedge x; \underline{v}x \ \underline{u}x \subset \wedge x; \underline{v}x \ *R \ \underline{u}x)$
- .23 $(*R \ \wedge x; \underline{v}x \ \underline{u}x \subset \wedge x; \underline{v}x \ *R \ \underline{u}x)$
- .24 $(*R(A \smallfrown C) \supset *RA \smallfrown *RC)$
- .25 $(*R(B \smallfrown C) \supset *RB \smallfrown *RC)$
- .26 $(B \cap *RA \subset *R(*RB \cap A))$
- .27 $(A \cap *RB \subset *R(*RA \cap B))$
- .28 $(A \cap \text{dmn } R \subset *R*RA)$
- .29 $(B \cap \text{rng } R \subset *R*RB)$
- .30 $(* (R : S) A = *R *S A)$
- .31 $(* (R : S) B = *S *R B)$
- .32 $(\text{vs } (R : S) x = *R \ \text{vs } S x)$
- .33 $(\text{hs } (R : S) y = *S \ \text{hs } R y)$

2.83 THEOREMS.

- .0 $(\text{relation is } R \rightarrow 0 \smallfrown \in R)$

Proof. Because of 2.63.2,

$$(0 \in R \rightarrow \text{orderedpair is } 0 \rightarrow 0).$$

.1 (relation is $R \rightarrow \sim \text{orderedpair is } R$)

Proof. Helped by 2.63.1 and 2.24.32, we infer

$$\begin{aligned} &(\text{orderedpair is } R \rightarrow \text{sng sng } 0 \in R \\ &\quad \rightarrow \text{orderedpair is sng sng } 0 \\ &\quad \rightarrow \text{sng sng } 0 \in \text{sng sng } 0 \\ &\quad \rightarrow \text{sng sng } 0 \subset \text{sng } 0 \\ &\quad \rightarrow \text{sng } 0 \subset 0 \\ &\quad \rightarrow 0 \in 0 \\ &\quad \rightarrow 0). \end{aligned}$$

FUNCTIONS

The notation of 2.84.6 is adapted from the lambda-notation of Alonzo Church.

2.84 DEFINITIONS.

- .0 (function is $f \equiv (\text{relation is } f \wedge \wedge x \in \text{dmn } f \text{ singleton is vs } fx))$
- .1 (function $fg \equiv (\text{function is } f \wedge \text{function is } g))$
- .2 (univalent is $f \equiv \text{function } f \text{ inv } f$)
- .3 ($.fx \equiv \top\top \text{ vs } fx$)
- .4 (The value of f at $x \equiv .fx$)
- .5 (${}_x f \equiv .fx$)
- .6 ($\lambda x \underline{ux} \equiv E \ x, y \ (y = \underline{ux})$)
- .7 (lonzo $x \underline{ux} \equiv \lambda x \underline{ux}$)
- .8 (upon A is $f \equiv (\text{function is } f \wedge \text{dmn } f \subset A)$)
- .9 (on A is $f \equiv (\text{function is } f \wedge \text{dmn } f = A)$)
- .10 (upon A to B is $f \equiv (\text{upon } A \text{ is } f \wedge \text{rng } f \subset B)$)
- .11 (on A to B is $f \equiv (\text{on } A \text{ is } f \wedge \text{rng } f \subset B)$)
- .12 (upon A onto B is $f \equiv (\text{upon } A \text{ is } f \wedge \text{rng } f = B)$)
- .13 (on A onto B is $f \equiv (\text{on } A \text{ is } f \wedge \text{rng } f = B)$)

- .14 (Upon $A \equiv \text{Ef upon } A \text{ is } f$)
- .15 (On $A \equiv \text{Ef on } A \text{ is } f$)
- .16 (To $B \equiv \text{Ef upon } U \text{ to } B \text{ is } f$)
- .17 (Onto $B \equiv \text{Ef upon } U \text{ onto } B \text{ is } f$)
- .18 (Uto $B \equiv \text{Ef}(\text{univalent is } f \wedge \text{rng } f \subset B)$)
- .19 (Uonto $B \equiv \text{Ef}(\text{univalent is } f \wedge \text{rng } f = B)$)

2.85 THEOREM. (function is $f \rightarrow x, y \in f \leftrightarrow y = .fx \wedge x, y \in U$)

Proof. From 2.61.0, 2.84.0, and 2.84.3 we deduce

$$\begin{aligned}
 (x, y \in f &\leftrightarrow x \in U \wedge y \in \text{vs } fx \\
 &\leftrightarrow x \in U \wedge y = .fx \in U \\
 &\leftrightarrow y = .fx \wedge x, y \in U).
 \end{aligned}$$

2.86 THEOREMS.

- .0 (function is $\lambda x \underline{u}x \wedge$ function is $\lambda x ; \underline{v}x \underline{u}x$)
- .1 (function is $f \rightarrow f = \lambda x .fx$)

Proof. (function is $f \rightarrow$ relation is $f \rightarrow f = \text{E } x, y (x, y \in f)$
 $= \text{E } x, y (y = .fx)$
 $= \lambda x .fx$)

- .2 (function is $0 \wedge \text{dmn } 0 = \text{rng } 0 = 0$)
- .3 (function is $g \wedge f \subset g \rightarrow$ function is f)

2.87 THEOREMS.

- .0 (function $fg \rightarrow \wedge x \in \text{dmn } f (.fx = .gx) \leftrightarrow f \subset g$)
- .1 (function $fg \rightarrow \wedge x \in \text{dmn } f \cup \text{dmn } g (.fx = .gx) \leftrightarrow f = g$)
- .2 (function $fg \rightarrow \wedge x (.fx = .gx) \leftrightarrow f = g$)

2.88 THEOREMS.

- .0 (function is $f \rightarrow x \in \text{dmn } f \leftrightarrow .fx \in U$)
- .1 (function is $f \rightarrow x \in \text{dmn } f \leftrightarrow .fx = U$)

- .2 (function is $f \rightarrow y \in \text{rng } f \leftrightarrow \forall x \in \text{dmn } f (y = .fx)$)
- .3 (function is $f \rightarrow f = \forall x \in \text{dmn } f \text{ sng } (x, .fx)$)
- .4 (function is $f \wedge \text{dmn } f \in U \rightarrow f \in U \wedge \text{rng } f \in U$)
- .5 ($f = \lambda x \underline{ux} \rightarrow \text{dmn } f = \text{Ex}(\underline{ux} \in U) \wedge (x \in \text{dmn } f \rightarrow .fx = \underline{ux})$)
- .6 ($f = \lambda x ; \underline{vx} \underline{ux} \rightarrow$
 $\text{dmn } f = \text{Ex}(\underline{vx} \wedge \underline{ux} \in U) \wedge (x \in \text{dmn } f \rightarrow .fx = \underline{ux})$)
- .7 ($f = \lambda x \in A \underline{ux} \rightarrow$
 $\text{dmn } f = \text{Ex} \in A (\underline{ux} \in U) \wedge (x \in \text{dmn } f \rightarrow .fx = \underline{ux})$)
- .8 ($\lambda x \underline{ux} = \lambda x \in U \underline{ux}$)

If to each point x in A there is intuitively correlated a point $\underline{ux}$ then according to .7 there is a function which carries out this correlation.

2.89 THEOREMS.

- .0 (univalent is $f \rightarrow \text{univalent is inv } f$)
- .1 (univalent is $f \wedge x \in \text{dmn } f \rightarrow . \text{inv } f .fx = x$)
- .2 (univalent is $f \wedge y \in \text{rng } f \rightarrow .f . \text{inv } f y = y$)
- .3 (function is $f \rightarrow \text{univalent is } f \leftrightarrow \wedge x \wedge y (fx = .fy \in U \rightarrow x = y)$)
- .4 (univalent is $f \rightarrow \text{rng } f \in U \leftrightarrow \text{dmn } f \in U \leftrightarrow f \in U$)

2.90 THEOREMS.

- .0 (function $fg \rightarrow f: g = \lambda x . f.gx$)
- .1 (function $fg \wedge \wedge x \in A (.fx = .gx) \rightarrow \text{strc } fA = \text{strc } gA$)
- .2 (function is $f \wedge x \in A \rightarrow . \text{strc } fA x = .fx$)
- .3 (function is $f \rightarrow x \in *fB \leftrightarrow .fx \in B$)
- .4 (function is $f \rightarrow *f(B \cap C) = *fB \cap *fC$)
- .5 (function is $f \wedge \forall x \underline{vx} \rightarrow *f \wedge x ; \underline{vx} \underline{ux} = \wedge x ; \underline{vx} *f \underline{ux}$)
- .6 (function is $f \rightarrow *f(B \vee C) = *fB \vee *fC$)
- .7 (function is $f \rightarrow B \cap *fA = *f(*fB \cap A)$)
- .8 (function is $f \rightarrow *f*fA = A \cap \text{rng } f$)
- .9 (function is $f \wedge A \cap *fB \neq 0 \rightarrow *fA \cap B \neq 0$)
- .10 (function is $f \wedge A \subset B \subset \text{rng } f \rightarrow *fA \subset *fB$)

2.91 THEOREMS.

- .0** (upon $\text{sqr } U$ is $\wedge x, y \underline{u'}xy \wedge$ upon $\text{sqr } U$ is $\wedge x, y; \underline{v'}xy \underline{u'}xy$)
- .1** (upon $\text{sqr } U$ is $f \rightarrow f = \wedge x, y . f(x, y)$)
- .2** (upon $\text{sqr } U$ is $f \wedge$ function is $g \rightarrow$
 $\wedge x, y \in \text{dmn } f . (f(x, y) = .g(x, y)) \leftrightarrow f \subset g$)
- .3** (upon $\text{sqr } U$ is $f \wedge$ upon $\text{sqr } U$ is $g \rightarrow$
 $\wedge x, y \in \text{dmn } f \cup \text{dmn } g . (f(x, y) = .g(x, y)) \leftrightarrow f = g$)
- .4** (upon $\text{sqr } U$ is $f \wedge$ upon $\text{sqr } U$ is $g \rightarrow$
 $\wedge x \wedge y . (f(x, y) = .g(x, y)) \leftrightarrow f = g$)
- .5** ($f = \wedge x, y; \underline{v'}xy \underline{u'}xy \rightarrow \text{dmn } f = \exists x, y (\underline{v'}xy \wedge \underline{u'}xy \in U) \wedge$
 $(x, y \in \text{dmn } f \rightarrow .f(x, y) = \underline{u'}xy)$)
- .6** ($f = \wedge x, y \in A \underline{u'}xy \rightarrow \text{dmn } f = \exists x, y \in A (\underline{u'}xy \in U) \wedge$
 $(x, y \in \text{dmn } f \rightarrow .f(x, y) = \underline{u'}xy)$)
- .7** ($\wedge x, y \underline{u'}xy = \wedge x, y \in \text{sqr } U \underline{u'}xy$)

ORDINALS**2.92 DEFINITIONS.**

- .0** (nest is $N \equiv \wedge x \in N \wedge y \in N (x \subset y \vee y \subset x)$)
- .1** (wellordered is $N \equiv \wedge A; (0 \neq A \subset N)(\cap A \in A)$)
- .2** (strung is $N \equiv$
 $(\text{wellordered is } N \wedge \wedge x \in N \wedge y \in N (x \subset y \leftrightarrow x \in y))$)
- .3** (ordinal is $N \equiv (\forall N \subset N \wedge \text{strung is } N)$)
- .4** (ordinal $ab \equiv (\text{ordinal is } a \wedge \text{ordinal is } b)$)
- .5** ($Q \equiv \exists x \text{ ordinal is } x$)
- .6** ($\text{scsr } x \equiv (x \cup \text{sng } x)$)
- .7** (successor $x \equiv \text{scsr } x$)

2.93 THEOREM. (wellordered is $N \rightarrow$ nest is N)

2.94 LEMMAS.

- .0** ($M \subset N \wedge \text{wellordered is } N \rightarrow \text{wellordered is } M$)
- .1** ($M \subset N \wedge \text{strung is } N \rightarrow \text{strung is } M$)
- .2** ($\wedge \beta \in F (\nabla \beta \subset \beta) \rightarrow \nabla \nabla F \subset \nabla F \wedge \nabla \cap F \subset \cap F$)

Hint. For the second conclusion, recall 2.37.21 and 2.70.26.

.3 $(0 \neq F \subset Q \rightarrow \prod F \in Q)$

.4 (strung is $N \rightarrow N \sim \in N$)

Proof. $(x = y = N \in N \rightarrow x \in N \wedge y \in N \wedge x \in y$
 $\rightarrow x \subset \cdot y$
 $\rightarrow N \subset \cdot N)$

.5 (ordinal is $N \wedge x \in N \rightarrow x \subset \cdot N$)

Proof. Because of 2.92.3 and .4

$$(x \in N \rightarrow x \subset \nabla N \subset N$$

$$\rightarrow x \subset \cdot N \vee N = x \in N$$

$$\rightarrow x \subset \cdot N).$$

.6 (ordinal is $N \wedge t \in x \in N \rightarrow t \subset \cdot x$)

.7 (ordinal is $N \wedge x \in N \rightarrow \nabla x \subset x$)

.8 (ordinal is $N \wedge x \in N \rightarrow x \in Q$)

.9 (strung is $b \wedge t \in b \wedge y \in b \rightarrow t \in y \vee t = y \vee y \in t$)

Proof. Use 2.93 and 2.92.2.

.10 (ordinal $ab \wedge a \subset b \wedge y \in b \sim a \rightarrow a \subset y$)

Proof. From .9 and .5 we infer

$$(t \in a \rightarrow t \in b \wedge y \in b$$

$$\rightarrow t \in y \vee y = t \in a \vee y \in t \subset a$$

$$\rightarrow t \in y \vee y \in a$$

$$\rightarrow t \in y \vee y \in a \sim a$$

$$\rightarrow t \in y).$$

.11 (ordinal is $N \wedge A \subset N \rightarrow A \cap \prod A = 0$)

Proof. Because of .6 we learn

$$(t \in A \cap \prod A \rightarrow t \in \prod A \subset t \in A \subset N$$

$$\rightarrow t \in t \in N$$

$$\rightarrow t \subset \cdot t).$$

.12 (ordinal $ab \wedge a \subset b \rightarrow a \subset \prod(b \sim a)$)

Proof. Use .10 and 2.24.64.

.13 (ordinal $ab \wedge a \subset b \rightarrow a \supset \bigcap(b \smallfrown a) \in b$)

Proof. From .5 and .11 we infer

$$\begin{aligned} (A = b \smallfrown a \rightarrow \bigcap A \in A \subset b \wedge \bigcap A \in b \wedge \\ \bigcap A \subset b \subset a \cup A \wedge \\ \bigcap A = (a \cup A) \cap \bigcap A = a \cap \bigcap A \subset a \wedge \\ a \supset \bigcap A \in b). \end{aligned}$$

An immediate consequence of .12 and .13 is our pivotal Theorem 2.95.1 below.

2.95 THEOREMS.

.0 (ordinal is $N \rightarrow N \subset Q$)

.1 (ordinal $ab \wedge a \subset b \rightarrow a = \bigcap(b \smallfrown a) \in b$)

.2 (ordinal $ab \rightarrow a \subset b \leftrightarrow a \in b$)

.3 wellordered is Q

Hint. $(0 \neq F \subset Q \wedge \bigcap F \smallfrown \in F \rightarrow \bigwedge y \in F (\bigcap F \subset y))$

.4 $(\nabla Q \subset Q)$

.5 ordinal is Q

.6 $\sim (Q \in U)$

.7 $(\nabla N \subset N \subset Q \leftrightarrow \text{ordinal is } N)$

.8 $(F \subset Q \rightarrow \text{ordinal is } \nabla F)$

Hint. $(\nabla \nabla F \subset \nabla F \subset \nabla Q \subset Q)$

.9 $(x \in Q \rightarrow \nabla \text{scsr } x = x)$

.10 $(x \in Q \rightarrow x \in \text{scsr } x \in Q)$

Hint. $(x \subset \text{scsr } x \subset Q)$

.11 $(\nabla Q = Q)$

.12 $(0 \in Q)$

.13 $(x \in y \in Q \rightarrow \text{scsr } x \in \text{scsr } y)$

- .14 $(x \in Q \wedge y \in Q \wedge x \subset y \rightarrow \text{scsr } x \subset \text{scsr } y)$
 .15 $(x \subset y \subset \text{scsr } x \rightarrow y = x \vee y = \text{scsr } x)$
 .16 (ordinal is $x \rightarrow \nabla x \subset x \subset \text{scsr } \nabla x$)
 .17 (ordinal is $x \leftrightarrow x \in Q \vee x = Q$)
 .18 $(\text{sng sng } 0 \in \sim Q \neq 0)$
 .19 $(A \subset Q \rightarrow A \cap \nabla A = 0)$
 .20 $(C \subset Q \wedge \wedge x \in C (x \cap C \subset A \rightarrow x \in A) \rightarrow C \subset A)$

Hint. Helped by .18 and .3, check that

$$(\xi = \text{TT}(C \smile A) \wedge C \smile A \neq 0 \rightarrow \xi \cap C \smile A = 0 \wedge \xi \cap C \subset A \\ \rightarrow \xi \in A \rightarrow 0).$$

- .21 (ordinal is $N \wedge \wedge x \in N (x \subset A \rightarrow x \in A) \rightarrow N \subset A$)
 .22 $(C \subset Q \rightarrow C \cap \text{TT}(C \cap B) \subset \sim B)$
 .23 (ordinal is $N \wedge N \cap B \neq 0 \rightarrow \text{TT}(N \cap B) \subset \sim B$)

In .20 above we find a general principle of proof by induction of which .21 is a much used corollary. An obvious consequence of .19 is .22 of which .23 is a corollary much used instinctively without reference.

It is a fact that

$$(\text{ordinal is } N \leftrightarrow \wedge x ; (\nabla x \subset x \subset \cdot N)(x \in N)).$$

Thus a more concise definition of ordinal is certainly possible. See J. R. Isbell, *A Definition of Ordinal Numbers*, Amer. Math. Monthly, January 1960.

DEFINITION BY INDUCTION

2.96 DEFINITIONS

- .0 (Induced $Rxy \underline{u'} xy \equiv (\text{relation is } R \wedge \text{dmn } R \subset Q \wedge \\ \wedge x \in Q (\text{vs } Rx = \text{st strc } Rxy \underline{u'} xy)))$
 .1 $((R \text{ is induced by } \underline{u'} xy \text{ in } x \text{ and } y) \equiv \text{Induced } Rxy \underline{u'} xy)$
 .2 $(\text{Ndc } xy \underline{u'} xy \equiv \text{The } R \text{ Induced } Rxy \underline{u'} xy)$

- .3** (upon α , f is induced by $H \equiv$
 (function is $H \wedge$ ordinal is $\alpha \wedge$
 upon α is $f \wedge \bigwedge x \in \alpha (fx = .H \text{ strc } fx))$)
- .4** ($\text{ndc } H\alpha \equiv \text{Ndc } xy (x \in \alpha \wedge \text{sng } .Hy)$)

Remark. According to 2.97, $\text{vs } Rx$ is determined by x and the *prior* behavior of R .

The initial supposition in the proof of 2.100 is motivated by the following plausible heuristic theorem.

$$\begin{aligned} &(\text{Induced } Rxy \text{ } \underline{u'}xy \wedge \\ &\quad \wedge \alpha (\underline{u}\alpha = \text{Ndc } xy (x \in \alpha \wedge \underline{u'}xy)) \rightarrow \\ &\quad \wedge x \in Q (\text{strc } Rx = \underline{u}x)) \end{aligned}$$

2.97 THEOREM.

$$\begin{aligned} &(\text{Induced } Rxy \text{ } \underline{u'}xy \leftrightarrow \\ &\quad \text{relation is } R \wedge \text{dmn } R \subset Q \wedge \bigwedge x \in Q (\text{vs } Rx = \underline{u'}x \text{ strc } Rx)) \end{aligned}$$

2.98 LEMMA. $(\text{Induced } Rxy \text{ } \underline{u'}xy \wedge \text{Induced } Sxy \text{ } \underline{u'}xy \rightarrow R = S)$

Proof. Suppose on the contrary that

$$(0 \neq A = \text{Ex} \in Q (\text{vs } Rx \neq \text{vs } Sx)).$$

Let

$$(\xi = \bigvee A).$$

Clearly then

$$.0 \quad (\text{vs } R\xi \neq \text{vs } S\xi),$$

but

$$\bigwedge t \in \xi (\text{vs } Rt = \text{vs } St).$$

Hence

$$(t \in \xi \rightarrow \text{vs strc } R\xi t = \text{vs } Rt = \text{vs } St = \text{vs strc } S\xi t).$$

Accordingly,

$$(\text{strc } R\xi = \text{strc } S\xi \wedge \text{vs } R\xi = \underline{u}'\xi \text{ strc } R\xi = \underline{u}'\xi \text{ strc } S\xi = \text{vs } S\xi)$$

in contradiction to .0.

From 2.98 we see at once

2.99 THEOREM. (Induced $Rxy \underline{u}'xy \rightarrow R = \text{Ndc } xy \underline{u}'xy$)

2.100 THEOREM. ($R = \text{Ndc } xy \underline{u}'xy \rightarrow \text{Induced } Rxy \underline{u}'xy$)

Proof. Suppose

$$(\wedge \alpha (\underline{u}\alpha = \text{Ndc } xy (x \in \alpha \wedge \underline{u}'xy)) \wedge S = \text{E } x, y (x \in Q \wedge y \in \underline{u}'x \underline{u}x)).$$

We divide the remainder of the proof into four parts, the first of which is evident.

Part 0. (relation is $S \wedge \text{dmn } S \subset Q \wedge \wedge x \in Q (\text{vs } Sx = \underline{u}'x \underline{u}x)$)

Part 1. ($\alpha \in Q \wedge \wedge x \in \alpha (\text{vs } Sx = \underline{u}'x \text{ strc } Sx) \rightarrow \text{strc } S\alpha = \underline{u}\alpha$)

Proof. Let

$$(T = \text{strc } S\alpha).$$

Now on the one hand,

$$\begin{aligned} (x \in \alpha \rightarrow \text{strc } Tx &= \text{strc strc } S\alpha x \\ &= \text{strc } S(\alpha \cap x) \\ &= \text{strc } Sx \\ \rightarrow \text{vs } Tx &= \text{vs } Sx \\ &= \underline{u}'x \text{ strc } Sx \\ &= \underline{u}'x \text{ strc } Tx \\ &= (x \in \alpha \wedge \underline{u}'x \text{ strc } Tx)). \end{aligned}$$

On the other hand,

$$(x \in Q \sim \alpha \rightarrow \text{vs } Tx = 0 = (x \in \alpha \wedge \underline{u}'x \text{ strc } Tx)).$$

Hence

$$\text{Induced } Txy (x \in \alpha \wedge \underline{u}'xy).$$

From 2.99 we learn

$$(T = \underline{u}\alpha).$$

Part 2. Induced $Sxy \underline{u}'xy$

Proof. Suppose on the contrary that

$$(0 \neq A = \exists x \in Q (\forall Sx \neq \underline{u}'x \text{ strc } Sx))$$

and let

$$(\xi = \prod A).$$

Clearly then

$$.0 \quad (\forall S\xi \neq \underline{u}'\xi \text{ strc } S\xi);$$

but

$$\wedge x \in \xi (\forall Sx = \underline{u}'x \text{ strc } Sx).$$

Because of this, and Parts 0 and 1,

$$(\forall S\xi = \underline{u}'\xi \underline{u}'\xi = \underline{u}'\xi \text{ strc } S\xi)$$

in contradiction to .0.

Part 3. Induced $Rxy \underline{u}'xy$

Proof. Because of Part 2 and 2.99 we conclude

$$(S = \text{Ndc } xy \underline{u}'xy = R \wedge \text{Induced } Rxy \underline{u}'xy).$$

The general principle of definition by induction is summed up in:

2.101 THE GENERAL INDUCTION THEOREM.

$$(\text{Induced } Rxy \underline{u}'xy \leftrightarrow R = \text{Ndc } xy \underline{u}'xy)$$

An easy application of the General Induction Theorem is 2.102.0 below. Since the vertical sections may not be points, the General Induction Theorem is not a special case of Theorem 2.102.0.

More demanding applications in which the Ordinary Induction Theorems are of little use can arise. See, for example, 2.143.2, 2.182, and 2.183 wherein many vertical sections are not points.

2.102 THE ORDINARY INDUCTION THEOREMS.

- .0 (function is $H \wedge$ ordinal is $\alpha \rightarrow$
upon α , f is induced by $H \leftrightarrow f = \text{ndc } H\alpha$)
- .1 (function is $H \wedge$ ordinal is $\alpha \wedge f = \text{ndc } H\alpha \rightarrow$
 $\text{dmn } f = \alpha \cap \prod(\alpha \prec \text{dmn } f)$)

REGULARITY AND CHOICE

by T. J. McMinin and A. P. Morse

The set U_0 is constructed in 2.103.3. Now it is this *construction*, made earlier in a slightly different way by von Neumann, that enables the replacement of the original chooser *mel* by a better chooser *Mel*. This *Mel* led one of us certainly to 2.126.0 and now leads us both fortuitously to 2.126.1. Partly because of this, we presently study in some detail the built-in structure of U_0 .

Our success in checking 2.126.1 stems from 2.107.1 and 2.106.5.

Since we rely a lot on ordinal theory, we find it interesting that, in checking 2.104, we prove nothing by induction and that we never check the univalence of *sbb*.

Because of 2.104.7, 2.104.10, and 2.103.6, it is fairly easy to check the exercises below. In connection with these exercises, because of .2 the Cantorian plausibility of 2.105.1, which is our theorem of size,⁶ is greatly enhanced by the realization of 2.105.1 for the subuniverse U_0 ; because of .3 it follows that, without transfinite machinery, U_0 could be defined and the exercises could be checked.

EXERCISES.

- .0 (sb $A \subset A \rightarrow U_0 \subset A$)

Hint. ($\sim A \cap U_0 \neq 0 \rightarrow 0 \neq \sim A \cap \text{nub } \sim A \subset \sim A \cap \text{sb } \sim \sim A$)

- .1 (sb $U_0 = U_0$)
- .2 (sb $A \subset A \subset U_0 \rightarrow A = U_0$)

⁶ See also Quine, W. V., *Set Theory and its Logic*, Belknap Press, Harvard, 1969, p. 286.

.3 $(U_0 = \bigwedge A ; (sb\ A \subset A)A)$

.4 regular is U_0

2.103 DEFINITIONS.

.0 $(2C \equiv \langle g\ sb\ \nabla\ rng\ g)$

.1 $(sbb \equiv ndc\ 2C\ Q)$

.2 $(subsetnest \equiv rng\ sbb)$

.3 $(U_0 \equiv \nabla\ rng\ sbb)$

.4 $(nick\ A \equiv \exists x \in Q\ (A \cap .sbb\ x \neq 0))$

.5 $(nub\ A \equiv .sbb\ \prod\ nick\ A)$

.6 $(regular\ is\ S \equiv \bigwedge A \subset S\ (A \neq 0 \rightarrow A \cap sb \sim A \neq 0))$

.7 $(Mel\ A \equiv (A \neq 0 \rightarrow mel\ (A \cap nub\ A)))$

2.104 LEMMA. $(f = sbb \rightarrow$

.0 $(x \in Q \rightarrow .fx = sb\ \nabla\ rng\ strc\ fx) \wedge$

.1 $dmn\ f = Q \wedge$

.2 $(x \in Q \wedge y \in Q \wedge x \subset y \rightarrow .fx \subset .fy) \wedge$

.3 $U_0 = \nabla\ rng\ f = \bigvee x \in Q\ .fx \wedge$

.4 $nest\ is\ subsetnest \wedge \nabla\ subsetnest = U_0 \wedge$

.5 $(x \in Q \rightarrow \nabla\ .fx = \bigvee t \in x\ .ft) \wedge$

.6 $(A \cap U_0 \neq 0$
 $\rightarrow A \cap nub\ A \neq 0 \wedge nub\ A \in sb\ U_0 \wedge nub\ A \subset sb \sim A)$

.7 $(A \cap U_0 \neq 0 \rightarrow 0 \neq A \cap nub\ A \subset A \cap sb \sim A) \wedge$

.8 $(A \cap U_0 = 0 \rightarrow nub\ A = U) \wedge$

.9 $(\alpha \in Q \wedge y \subset .f\alpha \rightarrow y \in .f\ scsr\ \alpha \subset U_0) \wedge$

.10 $sb\ U_0 \subset U_0 \wedge$

.11 regular is $U_0 \wedge$

.12 $(regular\ is\ U \rightarrow U = U_0) \wedge$

.13 $(regular\ is\ U \leftrightarrow U = U_0) \wedge$

.14 $(B \subset A \rightarrow nub\ A \subset nub\ B))$

Proof .0. Use 2.102.0.

Proof .1. Use .0.

Proof .2. Use .0.

Proof .3. Use 2.103.3 and .1.

Proof .4. First use 2.103.2 and .2. Next use 2.103.2 and 2.103.3.

Proof .5. Because of .0, .1, and 2.37.16,

$$(\nabla .fx = \nabla \text{rng strc } fx = \forall t \in x .ft).$$

Proof .6. We let

$$(\xi = \top\top \text{ nick } A),$$

make the remainder of the proof in nine steps, and infer the three desired conclusions from Step 4, Step 8, and Step 7.

Step 0. $(\xi = \top\top \text{ nick } A \wedge \text{nick } A = \text{Ex} \in Q (A \cap .fx \neq 0))$

Proof. Use 2.103.4.

Step 1. $(0 \neq \text{nick } A \subset Q)$

Proof. Use .3.

Step 2. $(\xi \in \text{nick } A \subset Q)$

Proof. Use Step 1 and 2.95.3.

Step 3. $(\xi \in Q \wedge \text{nub } A = .f\xi = \text{sb } \nabla \text{rng strc } f\xi)$

Proof. Because of Step 2, 2.103.5, and .0,

$$(\text{nub } A = .f\xi = \text{sb } \nabla \text{rng strc } f\xi).$$

Step 4. $(A \cap \text{nub } A \neq 0)$

Proof. Because of Step 2, Step 0, and Step 3,

$$(A \cap .f\xi \neq 0 \wedge A \cap \text{nub } A \neq 0).$$

Step 5. $(t \in \xi \rightarrow A \cap .ft = 0)$

Proof. Use Step 2 and Step 0.

Step 6. $(\nabla \text{ nub } A \subset \sim A)$

Proof. Because of Step 3 and .5,

$$(A \cap \nabla \text{ nub } A = A \cap \forall t \in \xi .ft = \forall t \in \xi (A \cap .ft) = \forall t \in 0 = 0).$$

Step 7. $(\text{nub } A \subset \text{sb } \sim A)$

Proof. Because of 2.37.16 and Step 6,

$$(\text{nub } A \subset \text{sb } \nabla \text{ nub } A \subset \text{sb } \sim A).$$

Step 8. $(\text{nub } A \in \text{sb } U_0)$

Proof. Use Step 3, .1, and .3.

Proof .7. Use .6.

Proof .8. Because of .3 and 2.103.4,

$$(\text{nick } A = 0).$$

Hence, because of 2.103.5,

$$(\text{TT } \text{nick } A = U \wedge .f \text{TT } \text{nick } A = U \wedge \text{nub } A = .f \text{TT } \text{nick } A = U).$$

Proof .9.

$$\begin{aligned} (\text{scsr } \alpha \in Q \wedge \alpha \in \text{scsr } \alpha &= \text{dmn } \text{strc } f \text{scsr } \alpha \wedge \\ & .f\alpha \in \text{rng } \text{strc } f \text{scsr } \alpha \wedge \\ & .f\alpha \subset \nabla \text{rng } \text{strc } f \text{scsr } \alpha \wedge \\ y \in \text{sb } .f\alpha &\subset \text{sb } \nabla \text{rng } \text{strc } f \text{scsr } \alpha \\ &= .f \text{scsr } \alpha \subset U_0) \end{aligned}$$

Proof .10. The desired conclusion follows from the

Statement. $(y \in \text{sb } U_0 \rightarrow y \in U_0)$

Proof. Let

$$\begin{aligned} (g &= \lambda s \in y \prod \text{Ex} \in Q (s \in .fx) \wedge \\ A &= \text{rng } g \wedge \\ \alpha &= \nabla A) \end{aligned}$$

and note that

$$(A \in U \wedge \alpha = \nabla A \in Q)$$

and use 2.95.3 in checking

$$\begin{aligned} (g \in \text{On } y \cap \text{To } Q \wedge \\ (s \in y \rightarrow s \in .f.gs \wedge \text{sng } s \subset .f.gs \wedge \\ .gs \in A \wedge .gs \subset \nabla A = \alpha)). \end{aligned}$$

Hence notice, because of .2, that

$$(y = \bigvee s \in y \text{ sng } s \subset \bigvee s \in y .f.gs \subset \bigvee s \in y .f\alpha \subset .f\alpha).$$

Consequently, because of .9,

$$(y \in U_0).$$

Proof .11. Use .7.

Proof .12. Let

$$(B = \sim U_0)$$

and note, because of .10,

$$(0 = \sim U_0 \cap U_0 = B \cap U_0 \supset B \cap \text{sb } U_0 = B \cap \text{sb } \sim B).$$

Since we now know

$$(B \subset U \wedge B \cap \text{sb } \sim B = 0),$$

we infer contrapositively from 2.103.6 that $(B = 0)$ and hence that $(U = U_0)$.

Proof .13. Use .11 and .12.

Proof .14. Clearly, because of 2.103.4,

$$(\prod \text{nick } C \in Q \vee \prod \text{nick } C = U)$$

and hence, because of 2.103.4, 2.24.39, .2, and 2.103.5,

$$\begin{aligned} &(\text{nick } B \subset \text{nick } A \wedge \\ &\prod \text{nick } A \subset \prod \text{nick } B \wedge \\ &.f \prod \text{nick } A \subset .f \prod \text{nick } B \wedge \\ &\text{nub } A \subset \text{nub } B). \end{aligned}$$

Remarks. In spite of 2.104 our treatment of sbb is far from exhaustive. Helped in part by induction, we could also have checked the more or less well-known results

- .0 strung is subsetnest
- .1 $(x \in Q \rightarrow \nabla .\text{sbb } x \subset .\text{sbb } x)$
- .2 $(\text{nub } A \subset \text{sb } \text{nub } A)$
- .3 $(x \in Q \rightarrow x \in .\text{sbb } x)$
- .4 $(x \in y \in U_0 \rightarrow y \sim \in \text{nub } \text{sng } x)$

Because of .3 we know $(0 \in .\text{sbb } 0)$ and hence that our construction is a bit different from that of von Neumann.

Remarks. Theorem 2.105.1 is our rather plausible theorem of size.

Notice in 2.105 that .4 is not only an immediate consequence of .2 and .0 but also an immediate consequence of 2.103.6 and the pleasantly structured .3. Noteworthy here is that

$$(\text{regular is } U \rightarrow \text{sb } A \subset A \leftrightarrow A = U).$$

The only use made of 2.5.9 is in checking 2.105.1. By starting with but not using 2.5.9 we notice

$$\begin{aligned} &((\sim \vee y((y \in \sim A) \wedge (y \subset A)) \rightarrow (A = U)) \\ &\leftrightarrow (\sim \vee y(y \in \sim A \wedge y \subset A) \rightarrow A = U) \\ &\leftrightarrow (\sim A \cap \text{sb } A = 0 \rightarrow A = U) \\ &\leftrightarrow (\text{sb } A \subset A \rightarrow A = U)). \end{aligned}$$

Accordingly 2.105.1 is equivalent to 2.5.9.

Also in 2.105 we can infer .0 from 2.104.13, infer .2 from 2.104.10 and .1, infer .3 from .2 and 2.104.7, infer .4 from .3 and 2.103.6, and infer .5 from .2 and 2.104.4.

2.105 THEOREMS.

- .0 (regular is $U \leftrightarrow U = U_0$)
- .1 (sb $A \subset A \rightarrow A = U$)
- .2 ($U = U_0$)
- .3 ($0 \neq A \rightarrow 0 \neq A \cap \text{nub } A \subset A \cap \text{sb } \sim A$)
- .4 regular is U
- .5 (nest is subsetnest $\wedge \nabla$ subsetnest = U)

Bearing in mind 2.105.2, we can in 2.106 below infer .0 and .1 from 2.104.6, infer .2 from 2.104.8, infer .3 from 2.104.14, infer .4 from .3 and .1, and infer .5 from 2.104.6.

2.106 THEOREMS.

- .0 ($A \neq 0 \rightarrow A \cap \text{nub } A \neq 0$)
- .1 ($A \neq 0 \rightarrow \text{nub } A \in U$)
- .2 ($A = 0 \rightarrow \text{nub } A = U$)
- .3 ($B \subset A \rightarrow \text{nub } A \subset \text{nub } B$)
- .4 ($z \in A \rightarrow \text{nub } A \subset \text{nub sng } z \in U$)
- .5 ($\text{nub } A \subset \text{sb } \sim A$)

In checking 2.107.0 below, we use our Zermelo choice axiom for the first time.

Bearing in mind 2.103.7, we can in 2.107 infer .0 from 2.5.8, infer .1 from .0 and 2.106.0, and infer .2 directly from 2.103.7.

2.107 THEOREMS.

- .0 ($A \neq 0 \rightarrow \text{mel } A \in A$)
- .1 ($A \neq 0 \rightarrow \text{Mel } A \in A \cap \text{nub } A$)
- .2 ($A = 0 \rightarrow \text{Mel } A = U$)

2.108 THE THEOREMS OF CHOICE.

- .0 ($z \in A \wedge P = \text{nub sng } z \rightarrow P \in U \wedge \text{nub } A \subset P \wedge \text{Mel } A \in A \cap P$)
- .1 ($\nabla P \in U \wedge A; (z \in A)(\text{Mel } A \in A \cap P)$)

MAXIMALITY

2.109 DEFINITIONS.

- .0 (hereditary is $F \equiv \wedge \alpha \wedge \beta ; (\alpha \subset \beta \in F)(\alpha \in F)$)
- .1 (nestling is $F \equiv$
(hereditary is $F \wedge \wedge N \in \text{sb } F ; \text{ nest is } N(\forall N \in F)$))

2.110 DEFINITIONS.

- .0 (adjoinable $Fg \equiv \exists t \in \sim \text{rng } g (\text{rng } g \cup \text{sng } t \in F)$)
- .1 (adjoiner $F \equiv \wedge g \text{ Mel adjoinable } Fg$)
- .2 (nestbuilder $F \equiv \text{ndc adjoiner } F Q$)
- .3 (nester $F \equiv$
 $\forall x \in \text{dmn nestbuilder } F \text{ sng } * \text{ nestbuilder } F \text{ scsr } x$)

2.111 DEFINITION.

(capped is $F \equiv \wedge N ; (\text{nest is } N \wedge N \subset F) \vee z(\forall N \subset z \in F)$)

2.112 DEFINITIONS.

- .0 (nested R is $\beta \equiv (\text{sqr } \beta \subset \wedge x \in \text{fld } R \ x \cup R \cup \text{inv } R)$)
- .1 ($\text{cap } RA \equiv \exists x \in \text{fld } R (A \subset \text{hs } Rx \cap \wedge y \in \text{vs } Rx \text{ hs } Ry)$)
- .2 (capped $R \equiv \wedge \beta ; \text{ nested } R \text{ is } \beta (\text{cap } R\beta \neq 0)$)
- .3 (maximal $R \equiv \exists x \in \text{fld } R (\text{vs } Rx \subset \text{hs } Rx)$)

We shall not use 2.111 and 2.112 until 2.116.
It is easy to check

2.113 LEMMA.

- ($K = \text{nestbuilder } F \rightarrow$
- .0 ordinal is $\text{dmn } K \wedge$
- .1 ($x \in Q \rightarrow .Kx = \text{Mel adjoinable } F \text{ strc } Kx$) $\wedge$
($x \in \text{dmn } K \rightarrow$
- .2 $.Kx \in \text{adjoinable } F \text{ strc } Kx \wedge$

- .3 $.Kx \sim \in \text{rng strc } Kx \wedge$
 .4 $\text{rng strc } Kx \cup \text{sng } .Kx \in F)$

2.114 LEMMA.

$(N = \text{nester } F \wedge K = \text{nestbuilder } F \rightarrow$
 $\text{univalent is } K \wedge \text{nest is } N \wedge N \subset F \wedge \nabla N = \text{rng } K)$

Proof. Using 2.113.3,

$$(y \in x \in \text{dmn } K \rightarrow .Ky \in \text{rng strc } Kx \rightarrow .Ky \neq .Kx).$$

Consequently univalent is K .

Since

$$(N = \bigvee x \in \text{dmn } K \text{ sng } *K \text{ scsr } x),$$

it is evident that nest is N . Also

$$\begin{aligned} (N &= \bigvee x \in \text{dmn } K \text{ sng } *K (x \cup \text{sng } x) \\ &= \bigvee x \in \text{dmn } K \text{ sng } (*Kx \cup \text{sng } .Kx) \\ &= \bigvee x \in \text{dmn } K \text{ sng } (\text{rng strc } Kx \cup \text{sng } .Kx)). \end{aligned}$$

Because of 2.113.4 we now know $(N \subset F)$. Also

$$\begin{aligned} (\text{rng } K &\supset \bigvee x \in \text{dmn } K (\text{rng strc } Kx \cup \text{sng } .Kx) \\ &= \nabla N \\ &\supset \bigvee x \in \text{dmn } K \text{ sng } .Kx \\ &= \text{rng } K \wedge \\ &\text{rng } K = \nabla N). \end{aligned}$$

We shall now establish the very powerful⁷

2.115 MAXIMAL PRINCIPLE.

$(\text{nestling is } F \wedge N = \text{nester } F \rightarrow$
 $\text{nest is } N \wedge N \subset F \wedge \sim \bigvee \eta \subset F (\text{nest is } \eta \wedge \nabla N \subset \cdot \nabla \eta))$

⁷ Note that N is maximal in the strong sense that F includes no other nest which covers more. Because of 2.120 we feel that the existence of such a maximal N is not an easy consequence of Hausdorff's Maximal Principle.

Proof. The first two conclusions follow from 2.114. Let

$$(K = \text{nestbuilder } F).$$

In order to complete the proof by showing

$$\sim \forall \eta \subset F \text{ (nest is } \eta \wedge \nabla N \subset \nabla \eta)$$

we assume the contrary, so choose η that

$$(\eta \subset F \wedge \text{nest is } \eta \wedge \nabla N \subset \nabla \eta),$$

and so determine z that

$$(z \in \nabla \eta \sim \nabla N).$$

In accordance with Theorem of Choice 2.108.1 we so choose P that

$$.0 \quad (P \in U \wedge \bigwedge A ; (z \in A)(\text{Mel } A \in A \cap P))$$

and in a four step argument infer in contradiction to 2.95.6 that $(Q \in U)$.

Step 0. $(x \in Q \rightarrow \text{rng strc } Kx \cup \text{sng } z \in F)$

Proof. Let

$$\begin{aligned} (B &= \text{rng strc } Kx \cup \text{sng } z \wedge \\ \eta' &= \bigvee \beta \in \eta \text{ sng } (B \cap \beta)). \end{aligned}$$

Since

$$(\text{rng strc } Kx \subset \text{rng } K = \nabla N \subset \nabla \eta \wedge z \in \nabla \eta),$$

we are sure

$$(B \subset \nabla \eta).$$

Evidently,

$$\begin{aligned} (\text{nest is } \eta' \wedge \eta' \subset F \wedge \\ \nabla \eta' = \bigvee \beta \in \eta (B \cap \beta) = B \cap \bigvee \beta \in \eta \beta = B \cap \nabla \eta = B \in U). \end{aligned}$$

Consequently,

$$(\eta' \in U \wedge \eta' \in \text{sb } F \wedge B = \nabla \eta' \in F).$$

Step 1. $(x \in Q \rightarrow z \in \text{adjoinable } F \text{ strc } Kx)$

Proof. Since

$$(z \in \sim \nabla N = \sim \text{rng } K \subset \sim \text{rng strc } Kx),$$

we infer from Step 0 and 2.110.0 that

$$(z \in \text{adjoinable } F \text{ strc } Kx).$$

Step 2. $(x \in Q \rightarrow .Kx \in P)$

Proof. From .0, Step 1, and 2.113.1, we infer

$$(.Kx = \text{Mel adjoinable } F \text{ strc } Kx \in P).$$

Step 3. $(Q \in U)$

Proof. From Step 2, we infer

$$(\text{dmn } K = Q \wedge \text{rng } K \subset P \in U \wedge \text{rng } K \in U).$$

Since according to 2.114 univalent is K , we now conclude from 2.89.4

$$(Q = \text{dmn } K \in U).$$

Obviously, we have

2.116 THEOREM.

$$(\wedge N ; (\text{nest is } N \wedge N \subset F)(\nabla N \in F) \rightarrow \text{capped is } F)$$

It should be fairly clear that

$$\begin{aligned} (\text{nested } R \text{ is } \beta \leftrightarrow \\ \beta \subset \text{fld } R \wedge \wedge x \in \beta \wedge y \in \beta (x = y \vee x, y \in R \vee y, x \in R)). \end{aligned}$$

We now establish a modification of Hausdorff's maximal principle.

2.117 THEOREM.

$$\begin{aligned} (F = \text{E}\alpha \text{ nested } R \text{ is } \alpha \wedge B = \nabla \text{nester } F \rightarrow \\ \text{nested } R \text{ is } B \wedge \sim \vee \beta (\text{nested } R \text{ is } \beta \wedge B \subset \cdot \beta)) \end{aligned}$$

Proof. Let

$$(N = \text{nester } F)$$

and note

$$.0 \quad (\text{nest is } \eta \wedge \eta \subset F \rightarrow \text{nested } R \text{ is } \nabla \eta)$$

and also

$$.1 \quad (a \subset b \wedge \text{nested } R \text{ is } b \rightarrow \text{nested } R \text{ is } a).$$

Thus nestling is F and our Maximal Principle assures us

$$.2 \quad (\text{nest is } N \wedge N \subset F \wedge \sim \nabla \eta \subset F (\text{nest is } \eta \wedge \nabla N \subset \cdot \nabla \eta)).$$

From .2 and .0 we infer nested R is B . In order to show

$$\sim \nabla \beta (\text{nested } R \text{ is } \beta \wedge B \subset \cdot \beta)$$

we assume the contrary, so choose β that

$$(\text{nested } R \text{ is } \beta \wedge B \subset \cdot \beta),$$

let

$$(\eta' = \nabla \alpha \in \text{subsetnest sng } (\beta \cap \alpha)),$$

and in contradiction to .2 infer from .1 and 2.105.5 that

$$(\text{nest is } \eta' \wedge \eta' \subset F \wedge \nabla N = B \subset \cdot \beta = \nabla \eta').$$

An immediate consequence of 2.117 is the following modification of a lemma due to Kuratowski.

2.118 THEOREM.

$$\nabla B (\text{nested } R \text{ is } B \wedge \sim \nabla \beta (\text{nested } R \text{ is } \beta \wedge B \subset \cdot \beta))$$

By letting

$$(R = \text{sqr } F \text{ E } x, y (x \subset y))$$

we see that a special case of 2.118 is

2.119 HAUSDORFF'S MAXIMAL PRINCIPLE.

$$\nabla N (\text{nest is } N \wedge N \subset F \wedge \sim \nabla \eta (\text{nest is } \eta \wedge \eta \subset F \wedge N \subset \cdot \eta))$$

That 2.118 is not a simple consequence of Hausdorff's Maximal Principle is, we think, brought out by the easily checked

2.120 THEOREM.

$$\begin{aligned}
 (R = \text{sqr } U \wedge F = U \wedge B = N = Q \wedge \beta = U \rightarrow \\
 F = \exists \alpha \text{ nested } R \text{ is } \alpha \wedge \text{nest is } N \wedge N \subset F \wedge \\
 \sim \forall \eta (\text{nest is } \eta \wedge \eta \subset F \wedge N \subset \eta) \wedge \\
 B = \forall N \wedge \text{nested } R \text{ is } B \wedge \text{nested } R \text{ is } \beta \wedge B \subset \beta)
 \end{aligned}$$

Often it is existence, rather than maximality, which is of prime importance; therefore interesting and efficient is the following inductive⁸ modification of Zorn's lemma.

2.121 THEOREM.

$$\begin{aligned}
 (A = \text{fld } R \wedge \text{capped } R \wedge \bigwedge x \in A \sim C \vee y(x, y \in R \sim \text{inv } R) \\
 \rightarrow A \cap C \neq 0)
 \end{aligned}$$

Proof. With the help of 2.118 we so choose B that

$$\begin{aligned}
 &(\text{nested } R \text{ is } B \wedge \\
 .0 \quad &\sim \forall \beta (\text{nested } R \text{ is } \beta \wedge B \subset \beta)).
 \end{aligned}$$

Since $\text{capped } R$, we choose

$$.1 \quad (x \in \text{cap } RB)$$

and complete the proof by verifying that

$$(x \in A \cap C).$$

Suppose on the contrary

$$\sim (x \in A \cap C).$$

Since .1 and 2.112.1 guarantee that $(x \in A)$ we now know that

$$(x \in A \sim C).$$

In accordance with our premise we so choose y that

$$.2 \quad (x, y \in R \sim \text{inv } R).$$

Because of .1 and 2.112.1,

$$.3 \quad (B \subset \text{hs } Rx).$$

⁸ In 2.121 the last part of the premise assures us that if we have not yet found what we are looking for, then at least we can advance a step and look again.

Because of .2, .1, and 2.112.1,

$$(B \subset \text{hs } Ry).$$

Hence

$$\text{nested } R \text{ is } (B \cup \text{sng } y).$$

According to this and .0,

$$.4 \quad (y \in B).$$

Using .4, .3, and .2 we conclude

$$(y, x \in R \wedge x, y \in \text{inv } R \sim \text{inv } R = 0).$$

Next we have a noninductive modification of Zorn's lemma.

2.122 THEOREM. (capped $R \rightarrow \text{maximal } R \neq 0$)

Proof. Let

$$\begin{aligned} (A &= \text{fld } R \wedge \\ C &= \text{maximal } R) \end{aligned}$$

and check

$$\wedge x \in A \sim C \vee y(x, y \in R \sim \text{inv } R).$$

Now apply 2.121 to learn

$$(0 \neq A \cap C = C).$$

Very useful is

2.123 THE INDUCTIVE PRINCIPLE OF INCLUSION.

$$(\text{capped is } F \wedge \wedge x \in F \sim K \vee y(x \subset y \in F) \rightarrow F \cap K \neq 0)$$

Proof. Let

$$\begin{aligned} (R &= \text{sqr } F \cap E x, y (x \subset y) \wedge \\ A &= F \wedge \\ C &= K). \end{aligned}$$

Notice

$$(A = \text{fld } R \wedge \text{capped } R \wedge \wedge x \in A \sim C \vee y(x, y \in R \sim \text{inv } R))$$

and then apply 2.121 to learn

$$(0 \neq A \cap C = F \cap K).$$

The Inductive Principle of Inclusion can be proved in many ways. It follows easily, for example, from Hausdorff's Maximal Principle.

Remark. In connection with 2.115 it is sometimes possible to pinpoint in advance ∇N and even F . We illustrate this with

$$\begin{aligned} &(\text{nestling is } F \wedge N = \text{nester } F \wedge \\ &\wedge \alpha \in F \wedge \beta \in F (\alpha \cup \beta \in F) \rightarrow \\ &\nabla N = \nabla F \wedge \\ &(F \subset \nabla F \rightarrow \nabla N = F = U)). \end{aligned}$$

The first conclusion follows rather easily from 2.115 and the second from 2.105.3. Thus in the proof of 2.126.0 we have an alternative assurance that

$$(\nabla N = U).$$

WELL ORDERING

2.124 DEFINITIONS.

- .0 (The Ω inception of $\beta \equiv E\xi \in \beta$ ($\beta \subset \text{vs } \Omega\xi$))
- .1 ((Ω wellorders A) $\equiv$ (relation is $\Omega \wedge$
 $\wedge \beta; (0 \neq \beta \subset A)$ singleton is The Ω inception of β))
- .2 (The Ω start of $\beta \equiv \top\top$ The Ω inception of β)
- .3 ((Ω orders A) $\equiv \wedge x \in A \wedge y \in A \wedge z \in A$ (Ω wellorders $\{xyz\}$))
- .4 ($\text{Mel}' \equiv \text{nestbuilder } U$)
- .5 ($\text{Melorder} \equiv E x, y \zeta, \text{inv Mel}' x \subset . \text{inv Mel}' y$)

2.125 THEOREMS.

- .0 ($A \subset B \wedge \Omega$ wellorders $B \rightarrow \Omega$ wellorders A)
- .1 ($x \in \beta \subset A \wedge \Omega$ wellorders $A \wedge \xi = \text{The } \Omega \text{ start of } \beta \rightarrow$
 $\beta \subset \text{vs } \Omega x \leftrightarrow x = \xi$)

- .2 $(0 \neq \beta \subset A \wedge \Omega \text{ wellorders } A \wedge \xi = \text{The } \Omega \text{ start of } \beta \rightarrow \xi \in \beta \subset \text{vs } \Omega\xi)$
- .3 $(\Omega = E \ x, y \ (x \subset y) \rightarrow \text{wellordered is } A \leftrightarrow \Omega \text{ wellorders } A)$
- .4 $(\Omega \text{ wellorders } A \rightarrow \Omega \text{ orders } A)$
- .5 $(\Omega \text{ orders } A \rightarrow (x \in A \wedge y \in A \rightarrow x, y \in \Omega \vee y, x \in \Omega))$
- .6 $(\Omega \text{ orders } A \rightarrow (x \in A \rightarrow x, x \in \Omega))$
- .7 $(\Omega \text{ orders } A \rightarrow (x \in A \wedge y \in A \wedge x, y \in \Omega \wedge y, x \in \Omega \rightarrow x = y))$
- .8 $(\Omega \text{ orders } A \rightarrow (x \in A \wedge y \in A \wedge z \in A \wedge x, y \in \Omega \wedge y, z \in \Omega \rightarrow x, z \in \Omega))$

Remarks. Although we did not check it earlier, Theorem 2.126.1 was valid in the setting of *A Theory of Sets*, Academic Press, 1965.

It interests us that 2.105.4 is an easy consequence of 2.95.5 and 2.126. According to 2.126

$$(\psi = \text{inv Mel}' \rightarrow \text{on } U \text{ onto } Q \text{ is } \psi \wedge \bigwedge x \bigwedge y (x \in y \in U \rightarrow .\psi x \in .\psi y)).$$

K. Gödel (see footnote 4, p. 66) obtained a similar membership-preserving function, Od , which is on L to On , where L is his class of constructible sets and On is his class of ordinals. In his constructible setting Gödel's reasoning involved neither choice nor regularity whereas in our larger setting our reasoning involves both.

2.126 THEOREMS.

- .0 $(\text{univalent is Mel}' \wedge \text{dmn Mel}' = Q \wedge \text{rng Mel}' = U)$
- .1 $(. \text{Mel}' x \in . \text{Mel}' y \in U \rightarrow x \in y \in Q)$

Proofs. Let

$$(F = U \wedge K = \text{nestbuilder } F \wedge N = \text{nester } F)$$

and from 2.124.4 learn that

$$(K = \text{Mel}').$$

Proof .0. From 2.114 we learn

$$(\text{univalent is } K \wedge \text{rng } K = \nabla N).$$

Now let ($\eta = \text{subsetnest}$). According to 2.105.5,

$$(\text{nest is } \eta \wedge \nabla \eta = U).$$

Since

$$(\text{nestling is } F \wedge \eta \subset F),$$

it follows from our Maximal Principle that

$$(\nabla N \sim \subset \cdot \nabla \eta \wedge \nabla N \sim \subset \cdot U \wedge \text{rng } K = \nabla N = U).$$

Since 2.88.4 assures us

$$(\text{dmn } K \in Q \rightarrow U = \text{rng } K \in U),$$

we conclude from 2.113.0 that

$$(\text{dmn } K = Q).$$

Proof .1. Helped by 2.110.0, .0, 2.113.2, 2.113.1, 2.107.1, and 2.106.5, we infer

$$\begin{aligned} (x \in Q \rightarrow 0 \neq \sim \text{rng strc } Kx &= \text{adjoinable } F \text{ strc } Kx \wedge \\ &\quad .Kx = \text{Mel} \sim \text{rng strc } Kx \in \text{sb} \sim \sim \text{rng strc } Kx \\ &\rightarrow .Kx \in \text{sb rng strc } Kx \\ &\rightarrow .Kx \subset \text{rng strc } Kx). \end{aligned}$$

Thus

$$(y \in Q \rightarrow .Ky \subset \text{rng strc } Ky)$$

and, because of 2.113.3,

$$\begin{aligned} (.Kx \in .Ky \in U \wedge y \subset x \\ \rightarrow x \in Q \wedge y \in Q \wedge .Kx \in .Ky \subset \text{rng strc } Ky \subset \text{rng strc } Kx \wedge \\ .Kx \in \text{rng strc } Kx \wedge .Kx \sim \in \text{rng strc } Kx \\ \rightarrow 0). \end{aligned}$$

Accordingly,

$$(.Kx \in .Ky \in U \rightarrow x \in Q \wedge y \in Q \wedge y \sim \subset x \rightarrow x \in y \in Q).$$

2.127 THE WELL ORDERING THEOREMS.

.0 (Melorder wellorders U)

.1 ($y \in U \rightarrow \text{hs Melorder } y \in U$)

2.128 THEOREM.

$(\Omega \text{ wellorders } A \wedge \wedge y \in A (\text{hs } \Omega y \in U) \rightarrow$

One f (univalent is $f \wedge$ ordinal is $\text{dmn } f \wedge A = \text{rng } f \wedge$
 $\wedge \alpha \in \text{dmn } f \wedge \beta \in \text{dmn } f (\alpha < \beta \rightarrow .f\alpha, .f\beta \in \Omega))$

2.129 THEOREM.

(ordinal $ab \wedge$ on a to b is $f \wedge \wedge \alpha \in a \wedge \beta \in \alpha (.f\beta \in .f\alpha) \rightarrow$

.0 $\wedge \alpha \in a (\alpha < .f\alpha) \wedge$

.1 $a < b$)

NATURAL NUMBERS**2.130 DEFINITIONS.**

.0 (naturalnumberclass is $A \equiv (0 \in A \wedge \wedge x \in A (\text{scsr } x \in A)))$

.1 ($\omega \equiv \wedge A$; naturalnumberclass is A A)

.2 (naturalnumber is $n \equiv (n \in \omega)$)

2.131 DEFINITIONS.

.0 ($1 \equiv \text{scsr } 0$)

.1 ($2 \equiv \text{scsr } 1$)

.2 ($3 \equiv \text{scsr } 2$)

.3 ($4 \equiv \text{scsr } 3$)

.4 ($5 \equiv \text{scsr } 4$)

.5 ($6 \equiv \text{scsr } 5$)

.6 ($7 \equiv \text{scsr } 6$)

.7 ($8 \equiv \text{scsr } 7$)

.8 ($9 \equiv \text{scsr } 8$)

2.132 THEOREMS.

.0 naturalnumberclass is ω

.1 ($\omega \subset Q$)

Theorems 2.133 are the Peano axioms for natural numbers.

2.133 THEOREMS.

- .0** $(0 \in \omega)$
- .1** $(x \in \omega \rightarrow \text{scsr } x \in \omega)$
- .2** $(x \in \omega \rightarrow \text{scsr } x \neq 0)$
- .3** $(x \in \omega \wedge y \in \omega \wedge \text{scsr } x = \text{scsr } y \rightarrow x = y)$
- .4** $(0 \in S \subset \omega \wedge \wedge x \in S (\text{scsr } x \in S) \rightarrow S = \omega)$

2.134 THEOREM. $(\omega = \nabla \omega)$

Proof. Let

$$(S = \omega \cap \text{sb } \omega).$$

We have

$$\begin{aligned} (x \in S &\rightarrow x \in \omega \wedge x \subset \omega \\ &\rightarrow \text{sng } x \subset \omega \wedge x \subset \omega \\ &\rightarrow \text{scsr } x = x \cup \text{sng } x \subset \omega \\ &\rightarrow \text{scsr } x \in \omega \wedge \text{scsr } x \subset \omega \\ &\rightarrow \text{scsr } x \in S). \end{aligned}$$

Since $(0 \in S)$, we are assured by 2.133.4 that $(S = \omega)$. Thus

$$(x \in \omega \rightarrow x \subset \omega).$$

Consequently,

$$(\nabla \omega \subset \omega).$$

On the other hand,

$$(x \in \omega \rightarrow x \in \text{scsr } x \in \omega \rightarrow x \in \nabla \omega).$$

Accordingly,

$$(\omega = \nabla \omega).$$

2.135 THEOREM. ordinal is ω

Proof. Use 2.95.8, 2.132.1, and 2.134.

2.136 THEOREM. $(0 \neq x \in \omega \rightarrow \text{scsr } \nabla x = x)$

Hint. Let

$$(S = \omega \cap \text{Ex}(0 \neq x \rightarrow \text{scsr } \nabla x = x))$$

and show that $(S = \omega)$ by virtue of 2.133.4.

SEQUENCES

2.137 DEFINITIONS.

- .0 (sequence is $S \equiv \text{on } \omega \text{ is } S)$
- .1 ($\text{sqnc } A \equiv \text{ES on } \omega \text{ to } A \text{ is } S)$
- .2 (sequence $A \equiv \text{sqnc } A)$
- .3 ($\text{ndc}''' Ta \equiv \text{ndc } \wedge g (g = 0 \wedge a \vee g \neq 0 \wedge$
 $\quad .T(\nabla \text{dmn } g, .g \nabla \text{dmn } g)) \omega)$
- .4 ($\text{ndc}'' Sa \equiv \text{ndc}''' \wedge x, y ..Sxya)$
- .5 ($\text{ndc}' ha \equiv \text{ndc}''' \wedge x, y .hya)$

2.138 THEOREMS.

- .0 ($a \in U \rightarrow f = \text{ndc}''' Ta \leftrightarrow \text{function is } f \wedge$
 $(\text{dmn } f \in \omega \vee \text{dmn } f = \omega) \wedge .f0 = a \wedge$
 $\wedge n \in \omega (.f \text{scsr } n = .T(n, .fn))$)

Proof. Recall the Ordinary Induction Theorems.

- .1 ($a \in U \rightarrow f = \text{ndc}'' Sa \leftrightarrow \text{function is } f \wedge$
 $(\text{dmn } f \in \omega \vee \text{dmn } f = \omega) \wedge .f0 = a \wedge$
 $\wedge n \in \omega (.f \text{scsr } n = ..Sn.fn))$)
- .2 ($a \in U \rightarrow f = \text{ndc}' ha \leftrightarrow \text{function is } f \wedge$
 $(\text{dmn } f \in \omega \vee \text{dmn } f = \omega) \wedge .f0 = a \wedge$
 $\wedge n \in \omega (.f \text{scsr } n = .h.fn))$)
- .3 ($a \in G \wedge \text{on } G \text{ to } G \text{ is } h \rightarrow \text{on } \omega \text{ to } G \text{ is } \text{ndc}' ha)$
- .4 ($a \in G \wedge S \in \text{sqnc} (\text{On } G \cap \text{To } G) \rightarrow \text{on } \omega \text{ to } G \text{ is } \text{ndc}'' Sa)$
- .5 ($a \in G \wedge \text{on } \text{rct } \omega G \text{ to } G \text{ is } T \rightarrow \text{on } \omega \text{ to } G \text{ is } \text{ndc}''' Ta)$

2.139 THEOREM. ($\omega \in Q$)

Proof. Use 2.5.6 to so secure ($C \in U$) that

$$(0 \in C \wedge \bigwedge x \in C (\text{sng } x \in C)).$$

Let

$$\begin{aligned} (\xi &= \bigwedge x \in C \text{ sng } x \wedge \\ f &= \text{ndc}' \xi 0) \end{aligned}$$

and divide the remainder of the proof into six parts.

$$\text{Part 0. } (S = \omega \cap \text{En}(.fn = .f0 \rightarrow n = 0) \rightarrow S = \omega)$$

Proof. Clearly ($0 \in S$). Also

$$\begin{aligned} (n \in S \rightarrow .f \text{ scsr } n = \text{sng } .fn \neq 0 = .f0 \\ \rightarrow .f \text{ scsr } n \neq .f0 \\ \rightarrow (.f \text{ scsr } n = .f0 \rightarrow \text{scsr } n = 0) \wedge \text{scsr } n \in \omega \\ \rightarrow \text{scsr } n \in S). \end{aligned}$$

Consequently, because of 2.133.4, ($S = \omega$).

$$\text{Part 1. } \bigwedge n \in \omega (.fn = .f0 \rightarrow n = 0)$$

Proof. Use Part 0.

$$\text{Part 2. } (S = \omega \cap \text{Em} \wedge n \in \omega (.fn = .fm \rightarrow n = m) \rightarrow S = \omega)$$

Proof. From Part 1 we infer ($0 \in S$). Now

$$\begin{aligned} (m \in \omega \wedge 0 \in n \in \omega \rightarrow \\ .f \text{ scsr } m = \text{sng } .fm \wedge \\ .fn = .f \text{ scsr } \nabla n = \text{sng } .f \nabla n) \end{aligned}$$

and hence

$$\begin{aligned} (m \in S \wedge 0 \neq n \in \omega \\ \rightarrow (.f \nabla n = .fm \rightarrow \nabla n = m) \\ \rightarrow (\text{sng } .f \nabla n = \text{sng } .fm \rightarrow n = \text{scsr } \nabla n = \text{scsr } m) \\ \rightarrow (.fn = .f \text{ scsr } m \rightarrow n = \text{scsr } m)). \end{aligned}$$

Using Part 1 again

$$(m \in S \wedge 0 = n \in \omega \rightarrow (.fn = .f \text{ scsr } m \rightarrow n = \text{scsr } m)).$$

Consequently

$$(m \in S \wedge n \in \omega \rightarrow (.fn = .f \text{ scsr } m \rightarrow n = \text{scsr } m)).$$

Thus $(S = \omega)$.

Part 3. $(m \in \omega \wedge n \in \omega \rightarrow (.fn = .fm \rightarrow n = m))$

Proof. Use Part 2.

Part 4. $(\omega \in U)$

Proof. We see univalent is f by Part 3. Since according to 2.138.3
on ω to C is f

we conclude from 2.89.4

$$(\text{rng } f \subset C \in U \wedge \text{rng } f \in U \wedge \omega = \text{dmn } f \in U).$$

Part 5. $(\omega \in Q)$

Proof. Use Part 4 and 2.135.

REITERATION

2.140 DEFINITION. $(\text{reit } Tn \equiv \wedge x . \text{ndc}' Tx n)$

2.141 THEOREMS.

.0 (function is $T \rightarrow$

$$\wedge x \in U \wedge n \in \omega (. \text{reit } T0 x = x \wedge . \text{reit } T1 x = .Tx \wedge . \text{reit } T2 x = .T.Tx \wedge . \text{reit } T \text{ scsr } n x = .T. \text{reit } Tn x))$$

.1 (function is $T \wedge n \in \omega \rightarrow$

$$\text{reit } T0 = \wedge x x \wedge \text{reit } T1 = T \wedge \text{reit } T \text{ scsr } n = T : \text{reit } Tn)$$

.2 (function is $T \rightarrow \forall n \in \omega . \text{reit } Tn 0 = \forall n \in \omega . \text{reit } T \text{ scsr } n 0)$

Hint. $(. \text{reit } T0 0 = 0)$

FIXED SETS AND BIPARTITION

Herein we follow the lead of Tarski and Knaster.

For those readers uninterested in 2.149 but interested in the Cantor–Bernstein Theorem, 2.154, we follow 2.154 with Remarks wherein we directly attack the Cantor–Bernstein Theorem with a manageable sequence of five lemmas which are influenced in *formulation* by 2.147 and 2.148.4. In the last of these lemmas we strive to capture the content of 2.148.4 by using functionally inductive 2.140 and 2.141. Our failure impresses us with the utility of *schematically* inductive 2.101.

In order to analyze effective proofs of 2.154 we suppose, for a bit, that

$$\begin{aligned} &(\text{univalent is } f \wedge \text{univalent is } g \wedge \\ &\text{on } A \text{ to } B \text{ is } f \wedge \text{upon } A \text{ onto } B \text{ is } g). \end{aligned}$$

In spite of their great variety all effective proofs seem to have in common one central, but sometimes hidden, thread:

C is so constructed that

$$\begin{aligned} &(C \subset A \wedge A \smile C \subset \text{dmn } g \wedge \\ &_*g(A \smile C) = B \smile *_fC). \end{aligned}$$

Put equivalently another way:

C is so constructed that

$$(C \subset A \wedge *_fg(B \smile *_fC) = A \smile C).$$

Put equivalently still another way:

C is so constructed that

$$(A \smile *_fg(B \smile *_fC) = C).$$

This last rendition seems the most fruitful and efficient.

If now and then we agree that

$$\wedge \alpha (\underline{u}\alpha = A \smile *_fg(B \smile *_f\alpha)),$$

then we hope to so construct C that

$$(\underline{u}C = C);$$

moreover here

$$\wedge \beta \wedge \alpha \subset \beta (\underline{u}\alpha \subset \underline{u}\beta).$$

Interested thus in fixed sets we conjecture that if

$$\wedge \beta \wedge \alpha \subset \beta(\underline{u}\alpha \subset \underline{u}\beta),$$

then C can always be so constructed that

$$(\underline{u}C = C).$$

Our most elementary construction is given by 2.142.0. Under a stronger premise a countable iterative construction appears in 2.145 by way of 2.142.2 and 2.143.2.

The notation used here suggests applications but does not even require, in 2.147 for example, that

$$(\text{relation is } f \wedge \text{relation is } g).$$

Bipartite effectiveness stems from the next two definitions. The Cantor–Bernstein Theorem is an immediate consequence of readily grasped 2.149. The key that unlocks 2.148.0–2.148.3 is our very brief 2.147. In the quite lengthy 2.148 we regard with interest the *countable* resolution of C described by 2.148.4; but in checking 2.148.0–2.148.3 and 2.149 we use neither 2.142.2 nor 2.143.1 nor 2.143.2 nor 2.145 nor 2.146.1. Accordingly, no recursive device need be used in checking the Cantor–Bernstein Theorem.

In 2.147 the sets A and B are disentangled from f and g ; in 2.148 and 2.149, on the contrary, the sets A and B are completely determined by f and g . Thus 2.148 and 2.149 are essentially theorems about two intertwined functions. In contrast, however, the Cantor–Bernstein Theorem is essentially a theorem about two intertwined *univalent* functions.

2.142 DEFINITIONS.

- .0 (Fix $\alpha \underline{u}\alpha \equiv \wedge \alpha (\underline{u}\alpha \subset \alpha \rightarrow \alpha)$)
- .1 (Bip $fgAB \equiv \text{Fix } \alpha (A \curvearrowright *g(B \curvearrowright *f\alpha))$)
- .2 (Reitr $\alpha \underline{u}\alpha \equiv \text{Ndc } xy (0 \in x \in \omega \wedge \text{st vs } y \nabla x \alpha \underline{u}\alpha)$)

In 2.143 below, .0 springs from 2.24.63, .1 merely requires attention.

2.143 LEMMAS.

- .0 ($\underline{u}\alpha \subset \alpha \wedge C = \text{Fix } \alpha \underline{u}\alpha \rightarrow C \subset \alpha$)
- .1 ($(\nabla R(\underline{u} \nabla n \in \omega \text{ vs } Rn = \wedge n \in \omega \underline{u} \text{ vs } Rn) \rightarrow \underline{u}(\alpha \cup \beta) = \underline{u}\alpha \cup \underline{u}\beta)$)

Hints. Suppose that

$$(R = Ex, y(x = 0 \wedge y \in \alpha \vee 0 \in x \in \omega \wedge y \in \beta)).$$

Independently of our lemma's premise check first in some detail that

$$(\underline{u}\alpha \cup \underline{u}\beta = \bigvee n \in \omega \underline{u} \text{ vs } Rn),$$

consequently that

$$(\wedge x(\underline{u}x = x) \rightarrow \alpha \cup \beta = \bigvee n \in \omega \text{ vs } Rn)$$

and thereby that

$$(\alpha \cup \beta = \bigvee n \in \omega \text{ vs } Rn).$$

Now notice that

$$(\underline{u}(\alpha \cup \beta) = \underline{u} \bigvee n \in \omega \text{ vs } Rn = \bigvee n \in \omega \underline{u} \text{ vs } Rn = \underline{u}\alpha \cup \underline{u}\beta).$$

In .2 below it is intuitively clear that there is precisely one P which obeys the conclusion and that the premise exhibits this unique P .

$$\begin{aligned} \textbf{.2} \quad & (P = \text{Reitr } \alpha \underline{u}\alpha \rightarrow \text{relation is } P \wedge \text{dmn } P \subset \omega \wedge \\ & \text{vs } P0 = 0 \wedge \wedge n \in \omega (\text{vs } P \text{ scsr } n = \underline{u} \text{ vs } Pn)) \end{aligned}$$

Hint. First use 2.64.0; then use 2.101 and 2.97.

2.144 THEOREM.

$$(\wedge \beta \wedge \alpha \subset \beta(\underline{u}\alpha \subset \underline{u}\beta) \wedge C = \text{Fix } \alpha \underline{u}\alpha \rightarrow \underline{u}C = C)$$

Proof. In the reasoning below, .0 comes from 2.143.0 and .2 follows from .1 and .0.

$$\begin{aligned} & ((\alpha \subset \beta \rightarrow \underline{u}\alpha \subset \underline{u}\beta) \wedge \\ \textbf{.0} \quad & (\underline{u}\alpha \subset \alpha \rightarrow C \subset \alpha) \wedge \\ & (\underline{u}\alpha \subset \alpha \rightarrow \underline{u}C \subset \underline{u}\alpha \subset \alpha) \wedge \\ & (\underline{u}\alpha \subset \alpha \rightarrow \underline{u}C \subset \alpha) \wedge \\ & \underline{u}C \subset C \wedge \\ \textbf{.1} \quad & \underline{u}\underline{u}C \subset \underline{u}C \wedge \\ \textbf{.2} \quad & C \subset \underline{u}C \wedge \\ & \underline{u}C = C) \end{aligned}$$

2.145 THEOREM.

$$(\wedge R(\underline{u} \vee n \in \omega \text{ vs } Rn = \vee n \in \omega \underline{u} \text{ vs } Rn) \wedge \\ C = \text{Fix } \alpha \underline{u}\alpha \wedge P = \text{Reitr } \alpha \underline{u}\alpha \rightarrow \\ \underline{u}C = C = \vee n \in \omega \text{ vs } Pn)$$

Proof. Let

$$(K = \vee n \in \omega \text{ vs } Pn)$$

and

$$(S = \exists n \in \omega (\text{vs } Pn \subset C)).$$

Because of 2.143.1

$$(\alpha \subset \beta \rightarrow \underline{u}\alpha \subset \underline{u}\beta);$$

hence by 2.144

$$(\underline{u}C = C).$$

Accordingly a check that

$$(C = K)$$

will complete the proof.

In the reasoning below 2.143.2 is used thrice and 2.143.0 is used penultimately.

$$\begin{aligned} & ((x \subset C \rightarrow \underline{u}x \subset \underline{u}C = C \rightarrow \underline{u}x \subset C) \wedge \\ & (n \in S \rightarrow \text{vs } Pn \subset C \rightarrow \text{vs } P \text{ scsr } n = \underline{u} \text{ vs } Pn \subset C \rightarrow \text{scsr } n \in S) \wedge \\ & 0 \in S \wedge \wedge n \in S (\text{scsr } n \in S) \wedge \\ & S = \omega \wedge \\ & \wedge n \in \omega (\text{vs } Pn \subset C) \wedge \\ & K \subset C \wedge \\ & \underline{u}K = \vee n \in \omega \underline{u} \text{ vs } Pn = \vee n \in \omega \text{ vs } P \text{ scsr } n \subset K \wedge \\ & C \subset K \wedge \\ & C = K) \end{aligned}$$

2.146 THEOREM.

$$(\wedge \alpha(\underline{u}\alpha = A \smile *g(B \smile *f\alpha)) \rightarrow$$

$$.0 \quad (\alpha \subset \beta \rightarrow \underline{u}\alpha \subset \underline{u}\beta) \wedge$$

$$.1 \quad (\text{function is } g \rightarrow \underline{u} \vee n \in \omega \text{ vs } Rn = \vee n \in \omega \underline{u} \text{ vs } Rn))$$

Proof. After checking .0, we momentarily agree that

$$(D = A \smile *gB \wedge h = \text{inv } g \wedge k = h : f).$$

Helped by 2.90.6, 2.82.12, and 2.82.30, we infer that

$$(\text{function is } g \rightarrow \underline{u}\alpha = D \cup A \cap *k\alpha)$$

and now notice that .1 is at hand.

2.147 THEOREM.

$$(C = \text{Bip } fgAB \rightarrow A \smile *g(B \smile *fC) = C)$$

Proof. Suppose

$$\wedge \alpha (\underline{u}\alpha = A \smile *g(B \smile *f\alpha)).$$

Because of 2.142.1,

$$(C = \text{Fix } \alpha \underline{u}\alpha).$$

According to 2.146.0 and 2.144,

$$(\underline{u}C = C)$$

and the desired conclusion is at hand.

2.148 THE THEOREM OF BIPARTITION.

(on A to B is $f \wedge$ upon A onto B is $g \wedge$

$$C = \text{Bip } fgAB \wedge$$

$$h = \text{strc } fC \cup \text{strc } g(A \smile C) \rightarrow$$

$$.0 \quad C \subset A \wedge *g(B \smile *fC) = A \smile C \wedge$$

$$.1 \quad A \smile C \subset \text{dmn } g \wedge *g(A \smile C) = B \smile *fC \wedge$$

$$.2 \quad \text{on } A \text{ onto } B \text{ is } h \wedge$$

$$.3 \quad (\text{univalent is } f \wedge \text{univalent is } g \rightarrow \text{univalent is } h) \wedge$$

$$.4 \quad (\wedge \alpha (\underline{u}\alpha = A \smile *g(B \smile f\alpha)) \wedge P = \text{Reitr } \alpha \underline{u}\alpha \rightarrow \\ C = \bigvee_{n \in \omega} \text{vs } Pn))$$

Proof .0. Because of 2.147

$$(C = A \smile *g(B \smile *fC) \wedge C \subset A \wedge \\ A \smile C = *g(B \smile *fC)).$$

Proof .1. Because of .0 and 2.90.8

$$(A \smile C \subset \text{dmn } g \wedge {}_*g(A \smile C) = {}_*g{}^*g(B \smile {}_*fC) = B \smile {}_*fC).$$

Proof .2. Because of .0 and .1, we see from the left that

on A to B is h

and then from the right that

on A onto B is h .

Proof .3. Because of .1,

$$({}_*fC \cap {}_*g(A \smile C) = 0)$$

and hence the univalence of h follows from that of f and g .

Proof .4. Use 2.142.1, 2.146.1, 2.145.

2.149 THEOREM.

(on A to B is $f \wedge$ upon A onto B is $g \wedge$
 $C = \text{Bip } fgAB \wedge$
 $h = \text{strc } fC \cup \text{strc } g(A \smile C) \rightarrow$
 on A onto B is $h \wedge$
 (univalent is $f \wedge$ univalent is $g \rightarrow$ univalent is h))

Proof. Use 2.148.2 and 2.148.3.

EQUINUMEROSITY

2.150 DEFINITIONS.

.0 $((A \text{ eq } B) \equiv \forall f; \text{univalent is } f (\text{dmn } f = A \wedge \text{rng } f = B))$

.1 $((A \text{ is equinumerous with } B) \equiv (A \text{ eq } B))$

2.151 THEOREMS.

.0 $(A \text{ eq } A)$

.1 $(A \text{ eq } B \rightarrow B \text{ eq } A)$

- .2 $(A \text{ eq } B \text{ eq } C \rightarrow A \text{ eq } C)$
- .3 $(A \text{ eq } 0 \leftrightarrow A = 0)$

2.152 THEOREMS.

- .0 $(B \text{ eq } B' \wedge C \text{ eq } C' \wedge B \cap C = 0 = B' \cap C' \rightarrow B \cup C \text{ eq } B' \cup C')$
- .1 $(B \text{ eq } B' \wedge C \text{ eq } C' \rightarrow \text{rct } BC \text{ eq } \text{rct } B'C')$
- .2 $(\text{univalent is } f \wedge A \subset \text{dmn } f \rightarrow A \text{ eq } *fA)$
- .3 $(\text{function is } f \rightarrow \text{dmn } f \text{ eq } f)$

2.153 THEOREM. $(A \text{ eq } B \in U \rightarrow A \in U)$

2.154 THE CANTOR-BERNSTEIN THEOREM.

$$(A \text{ eq } D' \subset A' \text{ eq } D \subset A \rightarrow A \text{ eq } A')$$

Proof. Use 2.149.

Remarks. The Cantor-Bernstein Theorem is, of course, simpler in statement than either 2.149 or the Theorem of Bipartition. However the latter reveals in 2.148.4 a simple explicit mechanism which might be useful in other contexts such as when f and g are continuous functions.

The Cantor-Bernstein Theorem is clearly a direct consequence of our fifth lemma below.

Some motivation for what follows is to be found in the third paragraph of Fixed Sets and Bipartition.

It will appear later that an immediate consequence of our fourth lemma is our unused

First Lemma.

$$(\text{univalent is } g \wedge \text{dmn } g \subset A \wedge \text{function is } f \rightarrow \\ \forall C(A \smile *g(B \smile *fC) = C))$$

Since for a number of reasons we strongly suspect that C could be of the form

$$\forall n \in \omega * \text{reit } TnL$$

where $\text{function is } T$ we are led at once to fashion with the help of 2.82.30 and 2.141.1 a simple quite general exploratory

Second Lemma.

$$\begin{aligned} &(\text{function is } T \wedge \\ &C = \bigvee n \in \omega \text{reit } TnL \rightarrow \\ &C = L \cup \text{*}TC) \end{aligned}$$

Now in connection with our first lemma it is easy with the help of 2.90.6, 2.82.12, and 2.82.30 to check our

Third Lemma.

$$\begin{aligned} &(\text{upon } A \text{ is } g \wedge k = \text{inv } g \rightarrow \\ &A \smile \text{*}g(B \smile \text{*}f\alpha) = A \smile \text{*}gB \cup \text{*}(k : f)\alpha \wedge \\ &(C = A \smile \text{*}g(B \smile \text{*}fC) \leftrightarrow C = A \smile \text{*}gB \cup \text{*}(k : f)C)) \end{aligned}$$

Now in the hope that

$$(C = A \smile \text{*}gB \cup \text{*}(k : f)C)$$

in our third lemma, we take

$$(L = A \smile \text{*}gB \wedge T = k : f)$$

in our second lemma and then completely understand the source of C in our

Fourth Lemma.

$$\begin{aligned} &(\text{univalent is } g \wedge k = \text{inv } g \wedge \\ &\text{dmn } g \subset A \wedge \text{function is } f \wedge \\ &C = \bigvee n \in \omega \text{*reit } (k : f)n(A \smile \text{*}gB) \rightarrow \\ &A \smile \text{*}g(B \smile \text{*}fC) = C) \end{aligned}$$

Proof. Assume

$$(L = A \smile \text{*}gB \wedge T = k : f)$$

so that by our second lemma

$$(C = L \cup \text{*}TC = A \smile \text{*}gB \cup \text{*}(k : f)C).$$

Because of this and our third lemma, the desired conclusion is at hand.

Fifth Lemma.

(univalent is $g \wedge k = \text{inv } g \wedge$
 on A to B is $f \wedge \text{upon } A \text{ onto } B \text{ is } g \wedge$
 $C = \forall n \in \omega \star \text{reit } (k : f)n(A \smallfrown \star gB) \wedge$
 $h = \text{strc } fC \cup \text{strc } g(A \smallfrown C) \rightarrow$
 on A onto B is $h \wedge$
 (univalent is $f \rightarrow \text{univalent is } h)$)

Proof. Because of our fourth lemma we infer, with the penultimate help of 2.90.8, that

$(C \subset A \wedge$
 $A \smallfrown C = \star g(B \smallfrown \star fC) \subset \text{dmn } g \wedge$
 on A to B is $h \wedge$
 $\star g(A \smallfrown C) = \star g \star g(B \smallfrown \star fC) = B \smallfrown \star fC \wedge$
 on A onto B is $h)$.

Thus we know

on A onto B is h

and since we also now know

$$(\star fC \cap \star g(A \smallfrown C) = 0),$$

we finally conclude

(univalent is $f \rightarrow \text{univalent is } f \wedge \text{univalent is } g$
 $\rightarrow \text{univalent is } h)$.

In connection with 2.149 and our fourth lemma there arises a natural question which is answered by

Statement.

(univalent is $g \wedge k = \text{inv } g \wedge$
 $\text{dmn } g \subset A \wedge \text{function is } f \rightarrow$
 $\text{Bip } fgAB = \forall n \in \omega \star \text{reit } (k : f)n(A \smallfrown \star gB))$

Hints. Let

$(C = \text{Bip } fgAB \wedge C' = \forall n \in \omega \star \text{reit } (k : f)n(A \smallfrown \star gB) \wedge$
 $L = A \smallfrown \star gB \wedge T = k : f \wedge \wedge \alpha(\underline{u}\alpha = L \cup \star T\alpha))$

so that

$$(\underline{u}\alpha = A \smile *g(B \smile *f\alpha) \wedge C = \text{Fix } \alpha \underline{u}\alpha \wedge \\ \underline{u}C' = C' = \bigvee n \in \omega * \text{reit } TnL).$$

Next check with the help of 2.145 that

$$(L = \underline{u}0 \subset C),$$

by induction that

$$(n \in \omega \rightarrow * \text{reit } TnL \subset C),$$

and finally that

$$(C' \subset C \subset C').$$

Although our fifth lemma has a simple and almost traditional recursive charm, it nevertheless fails to embrace 2.149.

2.155 LEMMA. $(s \in A \text{ eq } B \ni t \rightarrow A \smile \text{sng } s \text{ eq } B \smile \text{sng } t)$

CARDINALS

2.156 DEFINITIONS.

- .0** (cardinal is $\alpha \equiv (\text{ordinal is } \alpha \wedge \bigwedge \beta \in \alpha \sim (\alpha \text{ eq } \beta))$)
- .1** (cardinal $\alpha\beta \equiv (\text{cardinal is } \alpha \wedge \text{cardinal is } \beta)$)
- .2** ($C \equiv E\alpha \text{ cardinal is } \alpha$)

2.157 LEMMAS.

- .0** (cardinal $\alpha\beta \wedge \alpha \text{ eq } \beta \rightarrow \alpha = \beta$)
- .1** (cardinal $\alpha\beta \wedge \alpha \subset \gamma \text{ eq } \beta \rightarrow \alpha \subset \beta$)

Proof. Recalling the Cantor–Bernstein Theorem we see

$$(\beta \in \alpha \rightarrow \bigvee \alpha' \subset \alpha (\beta \text{ eq } \alpha') \wedge \bigvee \beta' \subset \beta (\alpha \text{ eq } \beta') \\ \rightarrow \alpha \text{ eq } \beta \\ \rightarrow \alpha = \beta \wedge \alpha \subset \beta).$$

But ordinal $\alpha\beta$ and therefore

$$(\alpha < \beta \vee \beta \in \alpha).$$

.2 $(\gamma \text{ eq } \beta \in \alpha \wedge \text{cardinal } \alpha\beta \wedge \text{ordinal is } \gamma \rightarrow \gamma \in \alpha)$

Proof. According to 2.156.0 and .1,

$$(\text{ordinal } \alpha\gamma \wedge \alpha \smile \subset \gamma);$$

hence $(\gamma \in \alpha)$.

2.158 THEOREMS.

.0 $(\omega \subset C)$

Hint. 2.155.

.1 $(\omega \in C)$

Proof. Using 2.157.2 we see

$$\begin{aligned} (\sim \text{cardinal is } \omega \rightarrow \\ \forall n(\omega \text{ eq } n \in \omega \wedge \omega \text{ eq } n \in \text{scsr } n \wedge \\ \text{cardinal } n \text{ scsr } n \wedge \text{ordinal is } \omega \wedge \\ \omega \in \text{scsr } n \subset \omega \wedge \omega \in \omega \wedge 0)). \end{aligned}$$

.2 $(\omega \subset \beta \in Q \rightarrow \text{scsr } \beta \in \sim C)$

Proof.

$$\begin{aligned} (f = \wedge x \in \beta (x = 0 \wedge \beta \vee 0 \in x \in \omega \wedge \nabla x \vee \omega \subset x \wedge x) \rightarrow \\ \text{univalent is } f \wedge \text{dmn } f = \beta \wedge \text{rng } f = \text{scsr } \beta) \end{aligned}$$

.3 cardinal is Q

CARDINALITY

2.159 DEFINITIONS.

.0 $(\text{pwr } A \equiv (Q \cap \text{TTE}\alpha(A \text{ eq } \alpha \in Q)))$

.1 $(\text{power } A \equiv \text{pwr } A)$

- .2 (The cardinality of $A \equiv \text{pwr } A$)
- .3 (finite is $A \equiv (\text{pwr } A \in \omega)$)
- .4 ($\text{fnt} \equiv \exists A \text{ finite is } A$)
- .5 (infinite is $A \equiv (\text{pwr } A \supset \omega)$)
- .6 (countable is $A \equiv (\text{pwr } A \subset \omega)$)
- .7 ($\text{cbl} \equiv \exists A \text{ countable is } A$)
- .8 (denumerable is $A \equiv (\text{pwr } A = \omega)$)

2.160 DEFINITIONS.

- .0 ($2D \ A \equiv (\forall \text{ dmn } A, \neg A \ \forall \text{ dmn } A)$)
- .1 ($2E \ A \equiv 2D \ (A \cap \text{sqr } \forall t \in Q \ (A \cap \text{sqr } t \neq 0))$)
- .2 ($2F \equiv \wedge f \ 2E \ (\text{sqr } Q \sim \text{rng } f)$)
- .3 ($\text{Sq} \equiv \text{ndc } 2F \ Q$)

We shall not use 2.160 until 2.167.

2.161 THEOREMS.

- .0 cardinal is $\text{pwr } A$
- .1 ($A \text{ eq } B \rightarrow \text{pwr } A = \text{pwr } B$)
- .2 (ordinal is $\alpha \rightarrow \alpha \text{ eq } \text{pwr } \alpha \subset \alpha$)
- .3 (cardinal is $\alpha \rightarrow \text{pwr } \alpha = \alpha$)
- .4 ($A \text{ eq } \alpha \wedge \text{cardinal is } \alpha \rightarrow \text{pwr } A = \alpha$)
- .5 ($\text{pwr } A \in Q \rightarrow A \text{ eq } \text{pwr } A$)
- .6 ($A \subset b \wedge \text{ordinal is } b \rightarrow A \text{ eq } \text{pwr } A \subset b$)

Hint. Use 2.125.3, 2.128, 2.129.1.

- .7 ($A \text{ eq } C \subset Q \rightarrow A \text{ eq } \text{pwr } A$)
- .8 ($A \subset B \rightarrow \text{pwr } A \subset \text{pwr } B$)

Hint. ($\text{pwr } B = Q \vee \text{pwr } B \in Q$)

- .9 ($A \subset C \text{ eq } B \rightarrow \text{pwr } A \subset \text{pwr } B$)
- .10 ($A \text{ eq } C \subset B \rightarrow \text{pwr } A \subset \text{pwr } B$)
- .11 (upon Q is $f \rightarrow \text{pwr } \text{rng } f \subset \text{pwr } \text{dmn } f$)

Hint. Let

$$(g = \lambda y \top \top \text{hs } fy).$$

.12 (function is $f \rightarrow \text{pwr rng } f \subset \text{pwr dmn } f = \text{pwr } f$)

Among the above .6 and .11 are incidental.

The principle of choice is involved in 2.162–2.165.

2.162 THEOREM. $\forall C \subset Q (A \text{ eq } C)$

Proof. Use 2.126.

2.163 THE THEOREMS OF CARDINALITY.

.0 ($A \text{ eq pwr } A$)

.1 ($A \text{ eq } B \leftrightarrow \text{pwr } A = \text{pwr } B$)

2.164 THEOREMS.

.0 ($0 \neq A \wedge \text{ordinal is } \alpha \rightarrow \text{pwr } A \subset \alpha \leftrightarrow \forall f \text{ on } \alpha \text{ onto } A \text{ is } f$)

.1 ($\text{ordinal is } \alpha \rightarrow \text{pwr } A \subset \alpha \leftrightarrow \forall f (\text{on } \alpha \text{ is } f \wedge A \subset \text{rng } f)$)

2.165 THEOREM.

$$(\wedge x \in A (\text{pwr } A \subset \text{pwr } \underline{u}x) \rightarrow \\ \vee g (\text{univalent is } g \wedge \wedge x \in A (.gx \in \underline{u}x)))$$

2.166 THEOREMS.

.0 ($\text{pwr } A = n \in \omega \wedge x \in \sim A \rightarrow \text{pwr } (A \cup \text{sng } x) = \text{scsr } n$)

.1 ($\text{pwr } A = n \in \omega \wedge x \in A \rightarrow \text{pwr } (A \smile \text{sng } x) = \nabla n$)

.2 ($A \in \text{fnt} \wedge B \in \text{fnt} \rightarrow A \cup B \in \text{fnt}$)

.3 ($F \in \text{fnt} \cap \text{sb fnt} \rightarrow \nabla F \in \text{fnt}$)

2.167 LEMMA. ($0 \neq A \subset \text{sqr } Q \rightarrow 2D A \in A$)

Hint. Recall 2.84.3.

2.168 THEOREMS.

.0 (univalent is $\text{Sq} \wedge \text{dmn Sq} = Q \wedge \text{rng Sq} = \text{sqr } Q \wedge \bigwedge x \in Q (*\text{Sq} \text{ sqr } x \in Q)$)

.1 (ordinal is $\alpha \rightarrow \alpha \subset * \text{Sq} \text{ sqr } \alpha$)

Hint. Use 2.90.10, 2.129.0, and .0.

2.169 THEOREM. ($\text{pwr} \text{ sqr } \omega = \omega$)

Proof. Let

$$(y = * \text{Sq} \text{ sqr } \omega).$$

Notice that

$$(\text{sqr } \omega = \bigvee n \in \omega \text{ sqr } n \wedge y = \bigvee n \in \omega * \text{Sq} \text{ sqr } n).$$

We know from 2.166.3 that

$$(n \in \omega \rightarrow \text{sqr } n \in \text{fnt}).$$

Hence, because of 2.168.0,

$$(n \in \omega \rightarrow * \text{Sq} \text{ sqr } n \in \omega).$$

Consequently, $(y \subset \omega)$ and, because of 2.168.1,

$$(y = \omega \wedge \text{sqr } \omega = * \text{Sq } \omega \wedge \text{pwr} \text{ sqr } \omega = \omega).$$

Remark. It can be seen without much trouble that

$$(\text{cardinal is } \alpha \wedge \omega \subset \alpha \rightarrow \text{sqr } \alpha = * \text{Sq } \alpha \wedge \text{pwr} \text{ sqr } \alpha = \alpha).$$

We again use the principle of choice in

2.170 THEOREM. ($F \in \text{cbl} \cap \text{sb cbl} \rightarrow \nabla F \in \text{cbl}$)

Proof. Let

$$(H = F \smallfrown 1)$$

and recall that

$$(1 = \text{sng } 0).$$

Clearly

$$.0 \quad (\nabla F = \nabla H \cup \nabla(F \smile H) = \nabla H \cup 0 = \nabla H).$$

Since the desired conclusion is evident in the event $(H = 0)$, assume $(H \neq 0)$. Use 2.164.0 to choose

$$(h \in \text{On } \omega \cap \text{Onto } H).$$

Aided by 2.164.0 and the principle of choice, select such a function g that

$$(g \in \text{On } \omega \wedge \wedge n \in \omega (.gn \in \text{On } \omega \cap \text{Onto } .hn)).$$

Let

$$(f = \lambda m, n \in \text{sqr } \omega ..gmn).$$

Evidently

$$(f \in \text{On } \text{sqr } \omega \cap \text{Onto } \nabla H).$$

According to .0, 2.161.12, and 2.169,

$$(\text{pwr } \nabla H \subset \text{pwr } \text{sqr } \omega = \omega).$$

The next three theorems, which are of interest in themselves, lead up to 2.174.

2.171 THEOREM. $(y \in U \rightarrow \text{On } y \cap \text{To } 2 \text{ eq sb } y)$

Taking

$$(f = \lambda xx \wedge y = U),$$

we see that 2.172 generalizes 2.38.

2.172 THEOREM.

$$(\text{on } y \text{ is } f \wedge \alpha = E x (x \in \sim .fx) \rightarrow y \supset \alpha \smile \in \text{rng } f)$$

Proof. $(t \in \alpha \leftrightarrow t \in \sim .ft \leftrightarrow t \in y \wedge t \smile \in \sim .ft)$

Hence $(\alpha \subset y)$ and

$$(\sim .ft = \alpha \rightarrow t \in \alpha \leftrightarrow t \smile \in \alpha \rightarrow 0).$$

Accordingly,

$$(\alpha \in \text{rng } f \rightarrow \forall t(.ft = \alpha) \rightarrow \forall t0 \rightarrow 0)$$

and the desired conclusion is at hand.

According to 2.172, α is such a subset of the domain of f that α does not belong to the range of f .

2.173 THEOREM. $\sim \forall f \in \text{On } y \text{ (sb } y \subset \text{rng } f)$

The principle of choice is involved in proving:

2.174 CANTOR'S POWER THEOREM.

$$(y \in Q \rightarrow y \in \text{pwr sb } y)$$

Proof. Use 2.164.1 and 2.173.

In 2.175 we formulate two special cases of the principle of choice. 2.175.0 is the finite principle of choice and 2.175.1 is the countable principle of choice. 2.175.0 is to be proved without the principle of choice; however 2.175.1 cannot be proved without the principle of choice.

2.175 THEOREMS.

$$.0 \quad (F \in \text{fnt} \cap \text{sb} \sim 1 \rightarrow \forall f \in \text{On } F \wedge \beta \in F (.f\beta \in \beta))$$

$$.1 \quad (F \in \text{cbl} \cap \text{sb} \sim 1 \rightarrow \forall f \in \text{On } F \wedge \beta \in F (.f\beta \in \beta))$$

CARDINAL ARITHMETIC

2.176 DEFINITIONS.

$$.0 \quad ((\alpha \oplus \beta) \equiv \text{pwr } \exists x, y (x \in \alpha \wedge y = 0 \vee x \in \beta \wedge y = 1))$$

$$.1 \quad ((\alpha \odot \beta) \equiv \text{pwr } \text{rct } \alpha\beta)$$

- .2 $({}^\odot\alpha\beta \equiv \text{pwr } (\beta \sim \in U \wedge (\text{pwr } (\alpha \subset 1 \rightarrow \alpha) \vee \text{On } \beta \cap \text{To } \alpha)))$
 .3 (exponential α to the $\beta \equiv {}^\odot\alpha\beta$)

2.177 THEOREMS.

- .0 $(\alpha \oplus \beta = \beta \oplus \alpha)$
 .1 $((\alpha \oplus \beta) \oplus \gamma = \alpha \oplus (\beta \oplus \gamma) = \alpha \oplus \beta \oplus \gamma)$
 .2 $(\alpha \odot \beta = \beta \odot \alpha)$
 .3 $((\alpha \odot \beta) \odot \gamma = \alpha \odot (\beta \odot \gamma) = \alpha \odot \beta \odot \gamma)$
 .4 $((\alpha \oplus \beta) \odot \gamma = \alpha \odot \gamma \oplus \beta \odot \gamma)$

2.178 THEOREMS.

- .0 $({}^\odot\alpha\beta \odot {}^\odot\alpha\gamma = {}^\odot\alpha(\beta \oplus \gamma))$
 .1 $({}^\odot\alpha\gamma \odot {}^\odot\beta\gamma = {}^\odot(\alpha \odot \beta) \gamma)$
 .2 $({}^\odot\odot\alpha\beta\gamma = {}^\odot\alpha(\beta \odot \gamma))$

2.179 THEOREM. $(\omega \subset \alpha \in C \rightarrow \alpha \oplus \alpha = \alpha \odot \alpha = \alpha \in {}^\odot 2\alpha = {}^\odot\alpha\alpha)$

The famous hypothesis of the continuum asserts

$$({}^\odot 2\omega = \text{TT}(C \sim \text{scsr } \omega)),$$

that is, ${}^\odot 2\omega$ is the smallest cardinal larger than ω . We are not at all sure that Gödel's approach⁹ can be used here to show the consistency of the hypothesis of the continuum.

2.180 THEOREMS.

- .0 $(A \cap B = 0 \rightarrow \text{pwr } (A \cup B) = \text{pwr } A \oplus \text{pwr } B)$
 .1 $(\text{pwr } (A \cup B) \subset \text{pwr } A \oplus \text{pwr } B)$
 .2 $(n \in \omega \rightarrow n \oplus 1 = \text{scsr } n)$

⁹ See the sentences following .7' on p. 66.

DIRECT EXTENSIONS

Herein is found a partial background for R. S. Arnold and A. P. Morse, *Plus and Times*, Pacific J. Math. **64** (2), 1976.

Suppose

$$(\text{on } A \text{ is } f \wedge \\ B = \text{To } A \smallfrown A \smallfrown 1).$$

If we wish to extend the domain of f so that it becomes $(A \cup B)$, then it is natural to so determine F that

$$(\text{on } (A \cup B) \text{ is } F \wedge f \subset F \wedge \wedge z \in B (.Fz = f: z)).$$

And there is no need to stop here! Our interest in direct extensions of this sort motivated 2.181–2.183. Definition 2.181.5 is of value, for example, in extending addition and multiplication functions. Theorems 2.182 and 2.183 help us evaluate the result.

In checking 2.182 and 2.183 the Ordinary Induction Theorems fail us but the General Induction Theorem does not.

2.181 DEFINITIONS.

- .0 (noz $R \equiv (R = 0 \vee R)$)
- .1 (domonde $A\theta \equiv$
Ndc $xy (A \cup Ez \in \text{dmn } \theta (\text{rng } .\theta z \subset \vee t \in x \text{ vs } yt)))$
- .2 (domo $A\theta x \equiv \text{vs domonde } A\theta x$)
- .3 ($\text{dom } A\theta \equiv \vee x \in Q \text{ domo } A\theta x$)
- .4 ($\text{drexndc } fA\theta \equiv \text{Ndc } xy (f \cup$
 $\wedge z \in \text{domo } A\theta x \smallfrown \vee t \in x \text{ domo } A\theta t \text{ noz } (\vee t \in x \text{ vs } yt : .\theta z)))$
- .5 ($\text{drex } fA\theta \equiv \vee x \in Q \text{ vs drexndc } fA\theta x$)
- .6 (The direct extension of f beyond A via $\theta \equiv \text{drex } fA\theta$)

2.182 THEOREM.

$$(D = \text{dom } A\theta \rightarrow \\ A \subset D = A \cup Ez \in \text{dmn } \theta (\text{rng } .\theta z \subset D))$$

2.183 THEOREM.

(function is $f \wedge \text{dmn } f \subset A \wedge D = \text{dom } A\theta \wedge F = \text{drex } fA\theta \rightarrow$
 function is $F \wedge f \subset F \wedge \text{dmn } F \subset D \wedge$
 $\wedge z \in A (.fz = .Fz) \wedge \wedge z \in D \sim A (.Fz = \text{noz } (F : .\theta z)))$)

FAMILIES OF SETS**2.184 DEFINITIONS.**

- .0** (disjointed is $F \equiv \wedge \alpha \in F \wedge \beta \in F (\alpha = \beta \vee \alpha \cap \beta = 0)$)
- .1** (dsjn $\equiv \exists F$ disjointed is F)
- .2** (dsn' $F \equiv \forall G \in \text{fnt} \cap \text{dsjn} \cap \text{sb } F \text{ sng } \nabla G$)
- .3** ($\nabla' F \equiv \forall G \in \text{fnt} \cap \text{sb } F \text{ sng } \nabla G$)
- .4** ($\Pi' F \equiv \forall G \in \text{fnt} \cap \text{sb } F \text{ sng } (\nabla F \cap \Pi G)$)
- .5** ($\sim' F \equiv \forall \beta \in F \text{ sng } (\nabla F \sim \beta)$)
- .6** (dsn'' $F \equiv \forall G \in \text{cbl} \cap \text{dsjn} \cap \text{sb } F \text{ sng } \nabla G$)
- .7** ($\nabla'' F \equiv \forall G \in \text{cbl} \cap \text{sb } F \text{ sng } \nabla G$)
- .8** ($\Pi'' F \equiv \forall G \in \text{cbl} \cap \text{sb } F \text{ sng } (\nabla F \cap \Pi G)$)
- .9** ($\nabla^- F \equiv \forall G \in \text{sb } F \text{ sng } \nabla G$)
- .10** ($\Pi^- F \equiv \forall G \in \text{sb } F \text{ sng } (\nabla F \cap \Pi G)$)
- .11** (borel $F \equiv (\text{sb } \nabla F \cap \Pi \text{EG} \supset F (\nabla'' G = \Pi'' G = G \subset \text{sb } \nabla F)))$)
- .12** (Borel $F \equiv (\text{sb } \nabla F \cap \Pi \text{EG} \supset F (\Pi'' G = \sim' G = G \subset \text{sb } \nabla F)))$)
- .13** (topology $\equiv \exists T (T = \nabla^- T = \Pi' T)$)
- .14** ($((T \text{ topologizes } A) \equiv (T \in \text{topology} \wedge \nabla T = A))$)
- .15** (diskompact $\equiv \exists F \wedge H \in \text{fnt} \cap \text{sb } F (\nabla H \neq \nabla F)$)
- .16** (kompact $\equiv \exists F \wedge H \subset F ; (\nabla H = \nabla F)(H \in \sim \text{diskompact}))$)
- .17** ($((G \subset \subset F) \equiv \wedge \alpha \in G \vee \beta \in F (\alpha \subset \beta))$)
- .18** ($((G \text{ is a refinement of } F) \equiv (G \subset \subset F))$)
- .19** (refined is $F \equiv$

$$\wedge H \in \text{cbl} \cap \text{sb } F \vee G \in \text{cbl} \cap \text{dsjn} \cap \text{sb } F (G \subset \subset H \wedge \nabla G = \nabla H))$$

2.185 THEOREMS.

- .0** $(F \subset \nabla' F \subset \nabla'' F \subset \nabla^- F)$
- .1** $(F \subset \Pi' F \subset \Pi'' F \subset \Pi^- F)$
- .2** $(0 \in \nabla' F)$
- .3** $(F \in U \rightarrow \nabla F \in \Pi' F)$
- .4** $(F \subset G \rightarrow \nabla' F \subset \nabla' G \wedge \nabla'' F \subset \nabla'' G \wedge \nabla^- F \subset \nabla^- G)$
- .5** $(F \subset G \wedge \nabla F = \nabla G \rightarrow$
 $\Pi' F \subset \Pi' G \wedge \Pi'' F \subset \Pi'' G \wedge \Pi^- F \subset \Pi^- G)$
- .6** $(F \subset G \wedge \nabla F = \nabla G \rightarrow \sim' F \subset \sim' G)$
- .7** $(\nabla' F = \text{sng } 0 \cup \bigvee G \in \text{fnt} \cap \text{sb } F \sim 1 \text{ sng } \nabla G \wedge$
 $\nabla'' F = \text{sng } 0 \cup \bigvee G \in \text{cbl} \cap \text{sb } F \sim 1 \text{ sng } \nabla G \wedge$
 $\nabla^- F = \text{sng } 0 \cup \bigvee G \in \text{sb } F \sim 1 \text{ sng } \nabla G)$
- .8** $(\Pi' F = \text{sng } \nabla F \cup \bigvee G \in \text{fnt} \cap \text{sb } F \sim 1 \text{ sng } \Pi G \wedge$
 $\Pi'' F = \text{sng } \nabla F \cup \bigvee G \in \text{cbl} \cap \text{sb } F \sim 1 \text{ sng } \Pi G \wedge$
 $\Pi^- F = \text{sng } \nabla F \cup \bigvee G \in \text{sb } F \sim 1 \text{ sng } \Pi G)$
- .9** $(\nabla^- F \cup \Pi^- F \cup \sim' F \subset \text{sb } \nabla F)$
- .10** $(\nabla \nabla' F = \nabla \nabla'' F = \nabla \nabla^- F = \nabla F)$
- .11** $(\nabla \Pi' F = \nabla \Pi'' F = \nabla \Pi^- F = \nabla F)$
- .12** $(\nabla \sim' F \subset \nabla F)$
- .13** $(\nabla' \nabla' F = \nabla' F \wedge \nabla'' \nabla'' F = \nabla'' F \wedge \nabla^- \nabla^- F = \nabla^- F)$
- .14** $(\Pi' \Pi' F = \Pi' F \wedge \Pi'' \Pi'' F = \Pi'' F \wedge \Pi^- \Pi^- F = \Pi^- F)$
- .15** $(\wedge \alpha \in F (\nabla F \sim \alpha \in G) \rightarrow \sim' F \subset G)$
- .16** $(\alpha \in F \in U \rightarrow \nabla F \sim \alpha \in \sim' F)$
- .17** $(\alpha \in \sim' F \rightarrow \nabla F \sim \alpha \in F \in U)$
- .18** $(\sim' F \subset F \in U \leftrightarrow \sim' F = F)$
- .19** $(F \in U \rightarrow \sim' (F \cup \text{sng } 0) = \sim' F \cup \text{sng } \nabla F \wedge$
 $\sim' (F \cup \text{sng } \nabla F) = \sim' F \cup \text{sng } 0)$
- .20** $(F \in U \rightarrow \nabla \sim' F = \nabla F \sim \Pi F)$
- .21** $(0 \neq F \in U \rightarrow \Pi \sim' F = 0)$
- .22** $(\sim' \sim' F = F \leftrightarrow \nabla \sim' F = \nabla F)$
- .23** $(\sim' \sim' F = F \leftrightarrow F \in U \wedge \nabla F \cap \Pi F = 0)$
- .24** $(0 \in F \in U \rightarrow \sim' \sim' F = F)$
- .25** $(F \in U \rightarrow \sim' (F \cup \sim' F) = F \cup \sim' F)$
- .26** $(\sim' \sim' \sim' F = \sim' F)$

- .27 $(0 \neq G \subset F \wedge \nabla G \in F \cup \text{sng } 0 \rightarrow \nabla G \in F)$
- .28 $(0 \neq G \subset F \wedge \prod G \in F \cup \text{sng } \nabla F \rightarrow \prod G \in F)$
- .29 $(\nabla' F = F \cup \text{sng } 0 \leftrightarrow \wedge \alpha \in F \wedge \beta \in F (\alpha \cup \beta \in F))$
- .30 $(\prod' F = F \cup \text{sng } \nabla F \leftrightarrow \wedge \alpha \in F \wedge \beta \in F (\alpha \cap \beta \in F))$
- .31 $(\nabla'' F = F \cup \text{sng } 0 \leftrightarrow \wedge G \in \text{cbl} \cap \text{sb } F \sim 1 (\nabla G \in F))$
- .32 $(\prod'' F = F \cup \text{sng } \nabla F \leftrightarrow \wedge G \in \text{cbl} \cap \text{sb } F \sim 1 (\prod G \in F))$
- .33 $(\sim' \sim' F = F \rightarrow \sim' \nabla' F = \prod' \sim' F \wedge \sim' \nabla'' F = \prod'' \sim' F \wedge \sim' \nabla^- F = \prod^- \sim' F)$
- .34 $(\sim' \sim' F = F \rightarrow \sim' \prod' F = \nabla' \sim' F \wedge \sim' \prod'' F = \nabla'' \sim' F \wedge \sim' \prod^- F = \nabla^- \sim' F)$
- .35 $(F \sim \in U \rightarrow \text{borel } F = \text{Borel } F = \text{sb } \nabla F)$
- .36 $(F \subset \text{borel } F \subset \text{Borel } F)$
- .37 $(\nabla \text{ borel } F = \nabla \text{ Borel } F = \nabla F)$
- .38 $(\nabla'' \text{ borel } F = \prod'' \text{ borel } F = \text{borel } F)$
- .39 $(\sim' \text{ Borel } F \subset \nabla'' \text{ Borel } F = \prod'' \text{ Borel } F = \text{Borel } F)$
- .40 $(F \subset G \wedge \nabla F = \nabla G \rightarrow \text{borel } F \subset \text{borel } G)$
- .41 $(F \subset G \wedge \nabla F = \nabla G \rightarrow \text{Borel } F \subset \text{Borel } G)$
- .42 $(\text{borel borel } F = \text{borel } F)$
- .43 $(\text{Borel Borel } F = \text{Borel } F)$
- .44 $(\sim' \sim' F = F \rightarrow \sim' \text{Borel } F = \text{Borel } \sim' F)$
- .45 $(\sim' \sim' F = F \rightarrow \sim' \text{borel } F = \text{borel } \sim' F)$
- .46 $(\sim' F \subset \text{borel } F \leftrightarrow \text{borel } F = \text{Borel } F)$
- .47 $(F \in U \rightarrow \text{Borel } F = \nabla G \in \text{cbl} \cap \text{sb } F \text{ Borel } (G \cup \text{sng } \nabla F))$
- .48 $(F \in U \rightarrow \text{borel } F = \nabla G \in \text{cbl} \cap \text{sb } F \text{ borel } (G \cup \text{sng } \nabla F))$
- .49 $(\nabla G \subset A \rightarrow \text{borel } (G \cup \text{sng } A) \subset \text{sb } \nabla G \cup \text{sng } A)$
- .50 $(\sim' F \subset \text{borel } F \in U \rightarrow \nabla F \in \nabla'' F)$

2.186 THEOREMS.

- .0 $(\prod' \nabla^- \prod' F = \nabla^- \prod' F)$
- .1 $(F \in U \rightarrow \nabla^- \prod' F \text{ topologizes } \nabla F)$

2.187 THEOREM.

$(F \in \text{topology} \leftrightarrow \wedge G \subset F \wedge \alpha \in F \wedge \beta \in F (\nabla G \in F \wedge \alpha \cap \beta \in F))$

2.188 THEOREMS.

- .0 (diskompact $\subset \sim$ kompact)
- .1 $(\nabla^- F \in \text{kompact} \leftrightarrow F \in \text{kompact})$

Hint. Use 2.175.0

A certain amount of effort may be required to check the beautiful:

2.189 THEOREM. $(\nabla^- \prod' F \in \text{kompact} \leftrightarrow F \in \text{kompact})$

In connection with 2.190 recall 0.62.

2.190 THEOREM.

$$(F \sim \sim F \subset \text{dsn}' F = K \rightarrow$$

- .0 $K \sim \sim K \subset K \wedge$
- .1 $K \cap \cap K \subset K \wedge$
- .2 $K \cup \cup K \subset K \wedge$
- .3 refined is F)

Hint. Notice first that

$$(K \sim \sim F \subset K)$$

TUPLES

We shall use 2.56 and 2.57.

2.191 DEFINITIONS.

- .0 (bsdmn $x \equiv \text{Et} (\text{bsvs } xt \neq 0)$)
- .1 (tuple is $x \equiv (\text{basicrelation is } x \wedge$
 $\wedge t \in \text{bsdmn } x (\text{bsvs } xt = \text{ss } \nabla \text{bsvs } xt)))$
- .2 (tuple a is $x \equiv (\text{tuple is } x \wedge \text{bsdmn } x = a)$)
- .3 (crd $tx \equiv \nabla \text{bsvs } xt$)
- .4 (The t coordinate of $x \equiv \text{crd } tx$)

2.192 DEFINITIONAL SCHEMAS.

.0 We accept as a definition each expression obtained from

$$'((x, < y) \equiv (\text{tuple is } x \wedge \wedge t \in \text{bsdmn } x (\text{crd } tx < y)))'$$

by replacing '<' by a nexus which is not a comma.

.1 We accept as a definition each expression obtained from

$$'(\text{Each coordinate of } x < y \equiv (x, < y))'$$

by replacing '<' by a nexus which is not a comma.

2.193 DEFINITIONS.

$$.0 \quad ((x, x', x'') \equiv ((x, x') \cup (\text{sng } 2, \text{ss } x'')))$$

$$.1 \quad ((x, x', x'', x''') \equiv ((x, x', x'') \cup (\text{sng } 3, \text{ss } x''')))$$

$$.2 \quad ((x, x', x'', x''', x''') \equiv ((x, x', x'', x''') \cup (\text{sng } 4, \text{ss } x'''')))$$

etc.

2.194 THEOREMS.

$$.0 \quad (\text{tuple } a \text{ is } x \wedge \text{tuple } a \text{ is } y \rightarrow x = y \leftrightarrow \wedge t \in a (\text{crd } tx = \text{crd } ty))$$

$$.1 \quad (x = \vee t \in a (\text{sng } t, \text{ss } \underline{u}t) \rightarrow \text{tuple } \alpha \text{ is } x \wedge \wedge t \in a (\text{crd } tx = \underline{u}t))$$

$$.2 \quad (\text{tuple } a \text{ is } x \rightarrow \nabla \nabla \nabla x = \nabla a \cup \vee t \in a \text{ crd } tx)$$

$$.3 \quad (\text{crd } t (x \cup y) = \text{crd } tx \cup \text{crd } ty)$$

$$.4 \quad (\text{tuple is } x \wedge \text{tuple is } y \rightarrow \text{tuple is } (x \cup y) \wedge \text{tuple is } (x \cap y) \wedge \text{crd } t (x \cap y) = \text{crd } tx \cap \text{crd } ty)$$

$$.5 \quad (\text{crd}' x = \text{crd } 0x \wedge \text{crd}'' x = \text{crd } 1x)$$

$$.6 \quad (p = 0 \rightarrow \text{tuple } 0 \text{ is } p \wedge \text{crd } tp = 0)$$

$$.7 \quad (p = \text{sng } 0, \text{ss } x \rightarrow \text{tuple } 1 \text{ is } p \wedge \text{crd } 0p = x)$$

$$.8 \quad (p = x, x' \rightarrow \text{tuple } 2 \text{ is } p \wedge \text{crd } 0p = x \wedge \text{crd } 1p = x')$$

$$.9 \quad (p = x, x', x'' \rightarrow \text{tuple } 3 \text{ is } p \wedge \text{crd } 0p = x \wedge \text{crd } 1p = x' \wedge \text{crd } 2p = x'')$$

$$.10 \quad (p = x, x', x'', x''' \rightarrow \text{tuple } 4 \text{ is } p \wedge \text{crd } 0p = x \wedge \text{crd } 1p = x' \wedge \text{crd } 2p = x'' \wedge \text{crd } 3p = x''')$$

etc.

The usefulness of 2.192 is illustrated by

2.195 THEOREMS.

.0 $(x, y, \in A \leftrightarrow x \in A \wedge y \in A)$

.1 $(x, y + z, t, < a \leftrightarrow x < a \wedge y + z < a \wedge t < a)$

.2 $(\wedge x, y, z, \in A \underline{u}''xyz \leftrightarrow \wedge x \in A \wedge y \in A \wedge z \in A \underline{u}''xyz)$

APPENDIX A

THE CONSTRUCTION OF DEFINITIONS

Definitions are, as we have said before, more than mere shorthand devices. Since we accept¹⁰ formulas of this sort as theorems, we should use care in making them. For example, if to our present definienda we were in some way to add ' $f(x)$ ', then many resulting formulas would be unreadable. In particular, if

$$'(f(x) \equiv .fx)'$$

were added to our list of definitions, then we could reach a contradiction by reasoning as follows.

First because of the proposed definition,

$$(f(x) = .fx).$$

From this, first by substitution and next by equality,

$$(\Pi f(x) = .\Pi fx)$$

and

$$(\Pi f(x) = \Pi .fx).$$

Accordingly,

$$(. \Pi fx = \Pi .fx).$$

By substitution

$$(. \Pi UU = \Pi .UU).$$

¹⁰ If, however, a definition, or even an axiom, is not a formula, then it is *not* accepted as a theorem.

Next by direct appeal to Chapter 2

$$(.xU = U \wedge .\Pi UU = U \wedge \Pi.UU = 0).$$

Consequently, in conclusion,

$$(U = 0).$$

Our definitions are constructed in accordance with the agreements and rules below; we suggest that the reader bear these in mind when he formulates new definitions of his own.

THE STRUCTURE OF BASIC FORMS

In this section we presuppose terminological familiarity with only the Introduction which begins Chapter 0.

A.0 AGREEMENT. *A* is *basic* if and only if *A* is either ' $(x \equiv y)$ ' or a definiendum.

As in 0.7 we make

A.1 AGREEMENT. α is an *introducer* if and only if α is a constant which is the initial symbol of some definiendum.

A.2 AGREEMENT. α is a *connector* if and only if α is a constant which is not a schemator and some definiendum can be obtained from ' xy ' by replacing ' x ' by a symbol and ' y ' by an expression in which α appears.

A.3 AGREEMENT. α is a *flanker* if and only if α is both an introducer and a connector.

A.4 AGREEMENT. *S* is a *signature* of *A* if and only if there is such an expression *Q*, devoid of schemators, that *A* can be obtained from *Q* by replacing variables by schematic expressions, and *S* can be obtained from *Q* by replacing each variable of *Q* by ' x '.

Thus ' $(x \rightarrow x)$ ' is both a signature of ' $(x \rightarrow A)$ ' and of ' $(z \rightarrow y)$ ', ' x ' is a signature of both ' $\underline{u}y$ ' and ' $\underline{v}''xyz$ ', ' $\wedge xx$ ' is a signature of ' $\wedge x \underline{v}'yy$ ', and the expression ' $\underline{u}\underline{v}x$ ' has no signature.

It should be fairly clear that an expression can have at most one signature.

A.5 RULE. A is a basic form if and only if A either is a primal expression or is a definiendum corresponding to which there is such a simple expression F that no schemator appears in F , every variable which appears in A also appears in F , and A can be obtained from F by replacing variables which do *not* appear in A by schematic expressions.

A.6 RULE. If A is a basic form, then in A no schemator appears more than once.

A.7 RULE. If α is a flanker and A is a basic form in which α appears, then A can be obtained from ' xyx ' by replacing ' x ' by α and ' y ' by some expression B for which α does not appear in B , the terminal symbol of B is a variable, and in some signature of B , ' x ' is not adjacent to ' x '.

A.8 RULE. If A and A' are diverse definienda, S and S' are such expressions different from ' x ' that S is a signature of A and S' is a signature of A' , then S is *not* an initial segment of S' .

A.9 RULE. If A is a basic form, B and C are schematic expressions which are segments of A , and α is a variable, then α appears in B if and only if α appears in C .

Conformity of our definitions to A.0–A.8 assures the variability of primal symbols, the constancy of the opening parenthesis, the unique readability of formulas, the mechanical soundness of our language, the harmony of all our rules and agreements, and the precision of our descriptions of formulas and theorems. These assertions have been confirmed by R. C. Neveln in his dissertation, *Basic Theory of Morse Languages*, Northwestern University, 1975.

If, in nonconformity with A.5, our whole set of definitions were to consist of the expressions enumerated in 0.0 together with the seemingly innocent expression

$$‘(\wedge * \underline{u}'xx \equiv \wedge x \underline{u}'xx)’,$$

then, as the reader may easily check, ‘*x*’ would be both free and not free in ‘ $\wedge * \underline{u}'xx$ ’.

We shall comment later on the rather interesting role played by A.9.

THE STRUCTURE OF DEFINITIONS

In this section we shall give rules of definition which though sometimes complicated in formulation are for the most part followed instinctively by any cautious constructor of definitions.

Throughout the remainder of this appendix we presuppose terminological familiarity with the material preceding and following the Theory of Notation, and with 0.31.0 and 0.51. Here we are primarily interested in mathematical, rather than mechanical, soundness of our language. We want to make sure, for example, that unwanted theorems must arise from axiomatic rather than linguistic defects.

A.10 RULE. Parentheses, the semicolon, binarians, and notarians are constant; in particular, ‘ $\rightarrow$ ’ and ‘ $\wedge$ ’ are constants.

A.11 RULE. The lightface italic Latin letters together with the superscripted and subscripted symbols derived therefrom are variables.

A.12 RULE. Every primal expression is a primitive form.

A.13 RULE. Among the forms are ‘ $(x \rightarrow x')$ ’ and ‘ $\wedge x \underline{u}x$ ’.

At *this* stage it seems pleasanter to endow A.14 with a symmetry which not only smooths our present path but also makes each formula parenthetical.

A.14 RULE. If D is a definition then D can be obtained from ' $(x \equiv y)$ ' by replacing ' x ' and ' y ' by formative parenthetical formulas.

A.15 RULE. If D is a definition and α is a schemator then some formula in which α does not appear can be obtained from D by schematically replacing some schematic expression by ' x '.

A.16 AGREEMENT. S is a *child* of D if and only if D raises an expression of which S is a signature.

A.17 RULE. If S is a child of D and D' , then S differs from ' x ' and D is the same as D' .

In other words, ' x ' is not a child and an expression is the child of at most one expression.

A.18 AGREEMENT. α is *bound* in A if and only if α is a variable, A is a formula, and there is such a formula B that α is absent from B and A can be obtained from B by simultaneously replacing variables free in B by formulas in which α is indicial.

Thus in each of the expressions

$$\begin{aligned} & \text{'}\forall x\ x\text{'}, \quad \text{'}(y \rightarrow \wedge yy)\text{'}, \quad \text{'}(y \rightarrow \forall x\ \underline{u}x)\text{'}, \\ & \text{'}(\wedge x\ \underline{u}x \rightarrow \wedge xy)\text{'}, \quad \text{'}\forall x \in y\ x\text{'}, \quad \text{'}\forall y \in y\ y\text{'}, \end{aligned}$$

' x ' is bound and, except in the first, ' y ' is not. Also ' x ' is bound in

$$\text{'}\wedge x \in 0\ \underline{u}x\text{'}$$

but not in

$$\text{'}\wedge x \in \wedge xx\ \underline{u}x\text{'}$$

It is easy to check that a variable absent from a formula is bound therein.

A.19 RULE. If a definition is obtained from ' $(x \equiv y)$ ' by replacing ' x ' by A and ' y ' by B , then every variable is either free in B or bound in B , a schemator appears in A if and only if it appears in B , a variable

appears precisely once in A if and only if it appears in B and is free therein, and if a variable is indicial in A , then it is bound in B .

A.20 AGREEMENT. B is a *constituent* of A if and only if B is a strict formula and there are such a formula Q and such a variable α that α appears precisely once in Q and A is obtained from Q by replacing α by B .

A.21 AGREEMENT. T is a *progenitor* of A if and only if there are such formulas B and F that B is a constituent of A , F is a fundamental formula from which B can be obtained by replacing free variables of F by formulas, and T is a signature of F .

A.22 AGREEMENT. T is a *forebear* of D if and only if D is a definition and there is such an expression B that D can be obtained from ' $(x \equiv y)$ ' by replacing ' x ' by an expression and ' y ' by B , and T is a progenitor of B .

A.23 AGREEMENT. T is an *ancestor* of S if and only if there is a definition of which S is a child and T is a forbear.

A.24 AGREEMENT. D is an *antecedent* of A if and only if D has a child which is a progenitor of A .

We now use 0.76 and 0.9.

A.25 AGREEMENT. R is a *tree* if and only if R is a chain and corresponding to each link A of R , each ancestor T of A , and each such subchain P of R that P ends with A , there is such a chain Q that R is (PQ) and T is a link of Q .

If D is the definition

$$'(\sim p \equiv (p \rightarrow 0))',$$

then D is the only definition of which ' $\sim x$ ' is a child, the ancestors of ' $\sim x$ ' are the forebears of D , which in turn are the progenitors of ' $(p \rightarrow 0)$ ', the constituents of ' $(p \rightarrow 0)$ ' are ' $(p \rightarrow 0)$ ' and ' 0 ', the progenitors of ' $(p \rightarrow 0)$ ' are ' $(x \rightarrow x)$ ' and ' 0 ', the ancestors of ' $\sim x$ ' are ' $(x \rightarrow x)$ '

and '0', of these, the first has no ancestor and the second has the one ancestor ' $\wedge xx$ ', ' $\wedge xx$ ' has no ancestor, and finally, ' $\sim x$ ' is a link of the tree

$$' \sim x \mid (x \rightarrow x) \mid 0 \mid \wedge xx '.$$

A.26 RULE. If S is a child of D , then there is a tree of which S is a link.

A.27 RULE. Every antecedent of an axiom is also an axiom.

A.28 REMARK. Systems which fail to comply with A.27 have a spurious simplicity. If two systems are to be compared, it is highly desirable that they both comply with A.27. Noncompliance has been the fundamental cause underlying certain anomalies.

ADHERENCE AND TRANSLATABILITY

A.29 AGREEMENT. F is a *formulaclass* if and only if F is a class of formulas, each variable belongs to F , each schematic form belongs to F , and C is in F whenever A and B are such members of F and C is such a formula that either C is a variant of A , or C is a constituent of A , or C is obtained from A by replacing some free variable of A by B , or C is obtained from A by schematically replacing some schematic expression by B .

A.30 AGREEMENT. (formulas G) consists of those formulas A such that A belongs to each formulaclass of which G is a subfamily.

A.31 AGREEMENT. F is a *consequenceclass* if and only if F is such a class of formulas that C is in F whenever A and B are such members of F and C is such a member of (formulas F) that C is entailed by A and by B .

A.32 AGREEMENT. (consequences G) consists of those formulas A such that A belongs to each consequenceclass of which G is a subfamily.

If G is any family of formulas, then (formulas G) is a formulaclass, (consequences G) is a consequenceclass, G is a subfamily of (consequences G), and (consequences G) is a subfamily of (formulas G).

A.33 AGREEMENT. We agree that F adheres to G if and only if each member of (consequences F) which is a member of (formulas G) is also a member of (consequences G).

Thus F adheres to G if and only if

$$((\text{formulas } G) \cap (\text{consequences } F) \subset (\text{consequences } G)).$$

A.34 AGREEMENT. A is *proper* if and only if A is a form with a signature different from 'x'.

A.35 AGREEMENTS.

.0 A is *salient* if and only if A is such a proper primitive form that some progenitor of an axiom is a signature of A .

.1 A is *pristine* if and only if A is a proper primitive form which is not salient.

A pristine form awaits, so to speak, either definitional clarification or axiomatic elucidation.

A.36 AGREEMENT. We agree that A is *annexed* if and only if A can be obtained from

$$'(y \equiv \wedge xx)'$$

by replacing 'y' by a pristine form.

Although annexed formulas are not to be thought of as definitions we nevertheless could easily, in conformity with our rules, so additionally designate as definitions certain of our *present* formulas that no form would be pristine and each *presently* annexed formula would *become* a theorem.

Our salient forms *actually* are

$$'(x \equiv y)', \quad '(x \rightarrow x')', \quad '(\wedge x \underline{u}x)', \quad '(x \in x')', \quad 'mel x'.$$

Of these the second is lifted by 0.0.2, the third by 0.0.3, and the fourth by 2.0.1. It might be noticed in this connection that 0.0.2, 0.0.3, and 2.0.1 are also made into definitions by the Theory of Notation.

A.37 AGREEMENTS.

- .0 **A** is our set of axioms.
- .1 **X** consists of those formulas which are either axioms or definitions.
- .2 **W** consists of those formulas which are either axioms or definitions or annexed formulas.
- .3 **S** is our set of salient forms.

A.38 AGREEMENT.

We agree that *A translates* into *G* if and only if a member of (consequences **W**) can be obtained from ' $(x \equiv y)$ ' by replacing ' x ' by *A* and ' y ' by a member of (formulas *G*).

Remark. We have tried to so formulate the above rules that conformity of our definitions thereto will ensure, among other things, that **X** adheres to **A** and each formula translates into **S**.

If, in defiance of A.9, we were to include

$$'(\wedge * xy \underline{u}x \underline{v}y \equiv \wedge x \wedge y(\underline{u}x \rightarrow \underline{v}y))'$$

among our definitions, then

$$'(\wedge * xy \underline{u}'xy \underline{v}'xy)'$$

would become a formula which would not translate into **S**.

If on the other hand, in conformity with our rules, we were to include

$$((\wedge * xy \underline{u}'xy \underline{v}'xy \equiv \wedge x \wedge y(\underline{u}'xy \rightarrow \underline{v}'xy)))'$$

among our definitions, then

$$'(\wedge * xy \underline{u}x \underline{v}y \equiv \wedge x \wedge y(\underline{u}x \rightarrow \underline{v}y))',$$

would clearly become a theorem.

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APPENDIX B

THE CONSISTENCY OF THE AXIOM OF SIZE

BY T. J. MCMINN AND A. P. MORSE

B.0 AGREEMENT. We agree again that our *revised* axioms are those obtained by striking out 2.5.9, 2.5.8, 2.5.5, and then replacing 2.5.7 by the following axiom of heredity:

$$.7' \quad (((A \subset B) \wedge (B \in U)) \rightarrow (A \in U))$$

We shall show that 2.5.9 is consistent with our revised axioms.

B.1 REMARKS. Axiom 2.5.7 is used for the first time in proving 2.40.0. Although we first used 2.5.5 in checking 2.31, we used neither 2.31 nor 2.32 nor 2.33 in checking 2.34; accordingly we are sure 2.34 is based on our revised axioms.

Also based on our revised axioms are 2.143.0 and 2.144.

Definition B.2.0 below is introduced to avoid dependence on 2.5.5 by way of 2.33.3, 2.33.0, and 2.35.

In this appendix our mathematical reasoning is based on our revised axioms.

B.2 DEFINITIONS.

$$.0 \quad (\text{sb}_0 A \equiv \forall t(t \subset A \wedge \text{sng } t))$$

$$.1 \quad (U^0 \equiv \text{Fix } \alpha \text{ sb}_0 \alpha)$$

$$\mathbf{B.3 THEOREM.} \quad (y \in \text{sb}_0 A \leftrightarrow y \subset A \wedge y \in U)$$

With the help of B.2 and 2.144 and 2.143.0 we infer

B.4 THEOREMS.

- .0 $(A \subset B \rightarrow \text{sb}_0 A \subset \text{sb}_0 B)$
- .1 $(\text{sb}_0 U^0 = U^0)$
- .2 $(\text{sb}_0 A \subset A \subset U^0 \rightarrow A = U^0)$

To clarify and shorten our formulas we regard as of the same type as '∈' the verbs in the middle of the definienda in the

B.5 DEFINITIONS.

- .0 $((x \in^0 y) \equiv (x \cap U^0 \in y))$
- .1 $((x \subset^0 y) \equiv (x \cap U^0 \subset y))$
- .2 $((x =^0 y) \equiv (x \cap U^0 = y \cap U^0))$
- .3 $((x \equiv^0 y) \equiv (x \cap U^0 = y \cap U^0))$
- .4 $(\text{sng}^0 x \equiv \text{sng}(x \cap U^0))$

B.6 LEMMAS.

- .0 $(x \in^0 y \leftrightarrow x \cap U^0 \in y)$
- .1 $(x = y \rightarrow x =^0 y)$
- .2 $(x =^0 y \rightarrow x \leftrightarrow y)$

Proof. Since $(0 \in U^0)$, we infer the desired conclusion from B.5.2 and 2.5.0.

- .3 $((x \equiv^0 y) = (x =^0 y))$
- .4 $((x \equiv^0 y) \rightarrow (x \leftrightarrow y))$
- .5 $((x \equiv^0 y) =^0 (x =^0 y))$
- .6 $(t \in U^0 \rightarrow t \in x \leftrightarrow t \in^0 x)$
- .7 $(y \subset U^0 \rightarrow y \subset A \leftrightarrow y \subset^0 A)$
- .8 $\wedge t(t \in x \cap U^0 \rightarrow t \in y) \wedge t \cap U^0 \in x \rightarrow t \cap U^0 \in x \cap U^0$
 $\rightarrow t \cap U^0 \in y$
- .9 $(\wedge t(t \cap U^0 \in x \rightarrow t \cap U^0 \in y) \wedge t \in x \cap U^0 \rightarrow t \cap U^0 = t \in x$
 $\rightarrow t = t \cap U^0 \in y)$
- .10 $((x \in c \rightarrow \text{sng } x \in c) \rightarrow (x \in c \cap U^0 \rightarrow \text{sng } x \in c \cap U^0))$

B.7 AGREEMENT. The *shadow* of A is the expression obtained from A by replacing: ' $\equiv$ ' by ' $\equiv^0$ '; ' $\in$ ' by ' $\in^0$ '; ' $\subset$ ' by ' $\subset^0$ '; ' $=$ ' by ' $=^0$ '; ' sng ' by ' sng^0 '.

B.8 REMARKS. If Γ and Γ' are subclasses of the augmentation of our axioms with B.0.7' and if Γ and Γ' include all our axioms through 2.4 and if the shadow of each member of Γ is a theorem based on Γ' , then the shadow of each theorem based on Γ is a theorem based on Γ' ; consequently, if Γ' is consistent, then Γ is also consistent.

B.9 THEOREM.

$$(U = \bigvee c \wedge x(c \wedge c \in^0 U \wedge (x \in^0 c \rightarrow \text{sng}^0 x \in^0 c)))$$

Proof. Clearly

$$(x \cap U^0 \in y \cap U^0 \leftrightarrow x \in^0 y)$$

and

$$(x \in y \cap U^0 \rightarrow x \in^0 y).$$

Because of these and B.6.10,

$$\begin{aligned} & (\wedge x(x \in c \rightarrow \text{sng } x \in c) \\ & \rightarrow (x \in c \cap U^0 \rightarrow \text{sng } x \in c \cap U^0) \\ & \rightarrow (x \cap U^0 \in c \cap U^0 \rightarrow \text{sng } (x \cap U^0) \in c \cap U^0) \\ & \rightarrow (x \in^0 c \rightarrow \text{sng}^0 x \in c \cap U^0 \rightarrow \text{sng}^0 x \in^0 c)). \end{aligned}$$

Hence

$$(\wedge x(x \in c \rightarrow \text{sng } x \in c) \rightarrow \wedge x(x \in^0 c \rightarrow \text{sng}^0 x \in^0 c))$$

and because of the Kronecker character of the intersectands and the consequent Kronecker character of the intersects,

$$(\wedge x(x \in c \rightarrow \text{sng } x \in c) \subset \wedge x(x \in^0 c \rightarrow \text{sng}^0 x \in^0 c)).$$

Accordingly, because of B.0.7',

$$\begin{aligned} & ((c \wedge c \in U \wedge \wedge x(x \in c \rightarrow \text{sng } x \in c)) \subset \\ & (c \wedge c \in^0 U \wedge \wedge x(x \in^0 c \rightarrow \text{sng}^0 x \in^0 c))) \end{aligned}$$

and thus

$$\begin{aligned} & (\wedge x(c \wedge c \in U \wedge (x \in c \rightarrow \text{sng } x \in c)) \subset \\ & \wedge x(c \wedge c \in^0 U \wedge (x \in^0 c \rightarrow \text{sng}^0 x \in^0 c))). \end{aligned}$$

Consequently,

$$\begin{aligned} (U &= \forall c \wedge x (c \wedge c \in U \wedge (x \in c \rightarrow \text{sng } x \in c)) \\ &\subset \forall c \wedge x (c \wedge c \in U^0 \wedge (x \in^0 c \rightarrow \text{sng}^0 x \in^0 c) \subset U)). \end{aligned}$$

B.10 THEOREM. $(\sim \forall y (y \in^0 \sim A \wedge y \subset^0 A) \rightarrow A =^0 U)$

Proof. Clearly, because of B.3,

$$(y \in \sim A \wedge y \in \text{sb}_0(A \cap U^0) \rightarrow y \in^0 \sim A \wedge y \subset^0 A).$$

Accordingly,

$$(\forall y (y \in \sim A \wedge y \in \text{sb}_0(A \cap U^0)) \rightarrow \forall y (y \in^0 \sim A \wedge y \subset^0 A)).$$

Because of this and B.4

$$\begin{aligned} &(\sim \forall y (y \in^0 \sim A \wedge y \subset^0 A) \\ &\rightarrow \sim \forall y (y \in \sim A \wedge y \in \text{sb}_0(A \cap U^0)) \\ &\rightarrow \wedge y (y \in \text{sb}_0(A \cap U^0) \rightarrow y \not\in \sim A \rightarrow y \in A) \\ &\rightarrow \text{sb}_0(A \cap U^0) \subset A \\ &\rightarrow \text{sb}_0(A \cap U^0) \subset A \cap U^0 \\ &\rightarrow A \cap U^0 = U^0 = U \cap U^0 \\ &\rightarrow A =^0 U). \end{aligned}$$

B.11 REMARKS. With the help of B.6 and B.9 we can rather easily see that the shadow of each of our revised axioms is a theorem based on our revised axioms.

Also because of B.10, we crucially infer that the shadow of discarded 2.5.9 is also a theorem based on our revised axioms.

In the light of B.8 we are thus persuaded that 2.5.9 is consistent with our revised axioms.

B.12 REMARK. We could, with added effort, show that 2.5.9 is consistent with our revised axioms augmented by axioms inspired by 2.44, 2.45, and 2.48.

AXIOMATIC EQUIVALENCE

Based on our 1965 edition is

C.0 THEOREM. $(\sim \forall y(y \in \sim A \wedge y \subset A) \rightarrow A = U)$

Proof.

$$\begin{aligned} & (z \in A \rightarrow \forall y(y \in A \wedge y \subset \sim A)) \\ & \wedge z(z \in A \rightarrow \forall y(y \in A \wedge y \subset \sim A)) \\ & (A \neq 0 \rightarrow \forall z(z \in A) \rightarrow \forall y(y \in A \wedge y \subset \sim A)) \\ & (\sim \forall y(y \in A \wedge y \subset \sim A) \rightarrow A = 0) \\ & (\sim \forall y(y \in \sim A \wedge y \subset A) \rightarrow \sim A = 0 \rightarrow A = U) \end{aligned}$$

Based on our present edition is:

C.1 THEOREM. $(z \in A \rightarrow \forall y(y \in A \wedge y \subset \sim A))$

Proof.

$$\begin{aligned} & (\sim \forall y(y \in \sim A \wedge y \subset A) \rightarrow A = U) \\ & (\sim \forall y(y \in A \wedge y \subset \sim A) \rightarrow \sim A = U \rightarrow A = 0) \\ & (z \in A \rightarrow A \neq 0 \rightarrow \forall y(y \in A \wedge y \subset \sim A)) \end{aligned}$$

C.2 REMARK. In our two editions only the last axioms differ. Hence, because of C.0 and C.1, it is clear that our 1965 edition is axiomatically equivalent to our present edition.

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An unindexed usage found in a section and unexplained earlier therein is probably explained by the Theory of Notation. Constants fixed by some definition but not subsequently used are not listed.

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